1. Anne-Maria Ernvall-hytönen, *Lattices and modular forms in coset coding*

Wyner (1975). Wiretap channel (Alice, Bob, Eve). Gaussian noise with variance σ^2 (between Alice and Bob) and σ_e^2 (for Eve). We assume $\sigma_e > \sigma$. We would like to do this whole scheme in such a way that we can use the noise so that Bob can still receive the image, but Eve cannot. We aim to do this with lattices $\Lambda \geq \Lambda_e$, where each coset corresponds to a code-letter.

Belfiore and Oggier (2010) (maybe [21]?): secrecy gain. Transmit codeword $x \in \mathbb{R}^n$.

$$\frac{1}{(\sigma_\Lambda \sqrt{2\pi})^n} \int_{V_{\Lambda(x)}} e^{-\|y-x\|^2/2\sigma^2}\, dy.$$

$$\frac{1}{(\sigma_\Lambda \sqrt{2\pi})^n} \sum_{t \in \Lambda_e \cap R} \int_{V_{\Lambda(x+t)}} e^{-\|y-x\|^2/2\sigma^2}\, dy.$$

End up trying to minimize

$$\theta_{\Lambda_e}\left(-\frac{1}{2\pi i \sigma_e^2}\right).$$

Secrecy function

$$\Xi(y) := \frac{\theta_{\lambda\mathbb{Z}^n}(yi)}{\theta_\Lambda(yi)}.$$

Belfiore and Sole: For unimodular Λ, maximum at $y = 1$. Even unimodular: polynomials in

$$E_4 = \frac{1}{2}\left(\vartheta_2^8 + \vartheta_3^8 + \vartheta_4^8\right), \qquad \Lambda = \frac{1}{256}\vartheta_2^8\vartheta_3^8\vartheta_4^8.$$

Inverse of $\Xi(y)$ a polynomial in $\frac{\vartheta_4^4\vartheta_2^4}{\vartheta_3^8}$.

(Maybe [9] is a reference.)

ℓ-modular: $1/\sqrt{\ell}$, $\mathbb{Z} \oplus \sqrt{2}\mathbb{Z} \oplus 2\mathbb{Z}$.

The connection to deeper mathematics is, what kinds of representations as polynomials do you have for various theta functions?

2. Jolanta Marzec-ballesteros, *Doubling method for self-dual linear codes*

Garrett, Piatetski-Shapiro–Rallis (1980's). Integral of cusp form against restriction of Siegel-type Eisenstein series equals L-function attached to cusp form times cusp form or Eisenstein series attached to cusp form:

$$\left\langle E\left(\begin{pmatrix} g & \\ & g' \end{pmatrix}, s\right), f(g) \right\rangle = L(f,s) f(g').$$

Done for G symplectic, orthogonal, unitary over global field, also for congreunce subgroups.

Let's start with an overview. Let f be a cusp form on G, and H a subgroup for which $G \times G \hookrightarrow H$. Then form an Eisenstein series on H

$$E(h,s) = \sum_{\gamma \in P\backslash H} \phi(\gamma h, s).$$

Restrict to $h = \mathrm{diag}(g, g')$ and take inner product with $F(g)$. This leads to an unfolding involving a sum over $\gamma \in P\backslash H/(G \times G)$. In favorable cases, only one

representative γ_0 contributes, leaving us with

$$\begin{aligned}&\sum_{(k,k')\in G\times G}\left\langle \phi\left(\gamma_0\begin{pmatrix}kg & \\ & k'g'\end{pmatrix},s\right),f(g)\right\rangle\\&=\sum_{\beta\in\gamma_0(G\times 1)}\psi(f)f|_\beta(g')\\&=L(f,s)f(g').\end{aligned}$$

We would like to do something similar, but now over finite fields.

A *linear code of length* $2n$ over a finite field $\mathbb{F}$ is a linear subspace $C\subseteq\mathbb{F}^{2n}$. We denote by $\langle,\rangle : C\times C\to\mathbb{F}$ the Euclidean inner product. We say that C is *self-dual* if

$$C=C^\perp:=\left\{v\in\mathbb{F}^{2n}:\langle v,C\rangle=0\right\}.$$

Then (the length $2n$ is even and) $\dim C=n$.

The *weight* of a codeword $c=(c_1,\dots,c_{2n})\in C$ is

$$\operatorname{wt} c=\#\left\{i\in\{1,\dots,2n\}:c_i\neq 0\right\}.$$

Weight enumerators are certain homogeneous polynomials of degree $2n$ in variables from the set $V=\{x_\alpha:\alpha\in\mathbb{F}^g\}$, where $g\in\mathbb{N}$ is the genus.

The *genus one weight enumerator* of a code $C\subseteq\mathbb{F}_2^{2n}$ is a polynomial

$$W_1(C,(x_0,x_1))=\sum_{c\in C}x_0^{2n-\operatorname{wt}c}x_1^{\operatorname{wt}c}.$$

The *genus g weight enumerator* of a code $C\subseteq\mathbb{F}^{2n}$ is a polynomial

$$W_g(C,x)=\sum_{(c^1,\dots,c^g)\in C^g}\prod_{\alpha\in\mathbb{F}^g}x_\alpha^{w_\alpha(c^1,\dots,c^g)}$$

of degree $2n$, where $x=(x_\alpha)_{\alpha\in\mathbb{F}^g}$ and

$$w_\alpha(c^1,\dots,c^g)=\#\left\{\text{rows } r \text{ in } (c_i^j)_{i=1..2n}^{j=1..g}:r=\alpha\right\}.$$

As an example, we give a basis for a Hamming code H_8 5nof weight 8, the span over $\mathbb{F}_2$ inside $\mathbb{F}_2^8$ of the vectors

$$\begin{pmatrix}1\\0\\0\\1\\0\\1\\1\\0\end{pmatrix}\quad\begin{pmatrix}0\\1\\0\\1\\0\\1\\0\\1\end{pmatrix}\quad\begin{pmatrix}0\\0\\1\\1\\0\\0\\1\\1\end{pmatrix}\quad\begin{pmatrix}0\\0\\0\\0\\1\\1\\1\\1\end{pmatrix}.$$

Then

$$\begin{aligned}W_2(H_8,(x_{00},x_{01},x_{10},x_{11}))&=\sum_{\alpha\in\mathbb{F}_2^2}x_\alpha^8+14\sum_{\substack{\alpha_1,\alpha_2\in\mathbb{F}_2^2\\ \alpha_1<\alpha_2}}x_{\alpha_1}^4x_{\alpha_2}^4+168x_{00}^2x_{01}^2x_{10}^2x_{11}^2\\&=(8)+14(4,4)+168(2,2,2,2).\end{aligned}$$

In general,

$$W_g(C)=\sum_A b_A\cdot(A),$$

where

$$A \in \left\{ (a_0, \dots, a_{2^g-1}) : \text{admissible tuples, } \sum_{i=0}^{2^g-1} a_i = 2n \right\}.$$

Some analogies with modular forms:

- $W_g(C)$ is like a modular form f of genus g,
- $\sum_A b_A \cdot (A)$ is like a Fourier expansion,
- $(2n)$ is like a constant term $a(0)$.

EXamples of cusp forms:

- $W_1(G_{24}) - W_1(H_8 \times H_8 \times H_8) = -42(20,4) + 168(16,8) - 252(12,12)$ is a cusp form of genus one.
- $W_3(E_{16}) - W_3(H_8 \times H_8) = -2688(9,1,1,1,1,1,1,1) + \cdots$ is a cusp form of genus 3.

Theorem 1 (Runge, 1996; Nebe, Rains, Sloane, 2006)**.** *We have*

$$\langle W_g(C) : C \text{ self-dual, over } \mathbb{F} \rangle = (\mathbb{C}[x_\alpha : \alpha \in \mathbb{F}^g])^{\mathcal{C}_g},$$

where

$$\mathcal{C}_g := \langle m_r, d_\phi, h_{\iota,u_\iota,v_\iota} : r \in \mathrm{GL}_g(\mathbb{F}), \phi, \iota \rangle$$

with

$$m_r : x_\alpha \mapsto x_{r\alpha},$$
$$d_\phi : x_\alpha \mapsto e^{2\pi i \phi(\alpha)} x_\alpha,$$
$$h_{\iota,u_\iota,v_\iota} : x_\alpha \mapsto (\#\iota\mathbb{F}^g)^{-1/2} \sum_{w \in \iota\mathbb{F}^g} e^{\frac{2\pi i}{p}\langle w, v_\iota \alpha \rangle} x_w + \cdots.$$

Consider the mean polynomial ("Siegel-type Eisenstein series")

$$\begin{aligned} M_{2g}((2n)) &= \sum_{\gamma \in P_{2g} \backslash \mathcal{C}_{2g}} (2n)^\gamma \\ &= \sum_{\gamma \in P_{2g} \backslash \mathcal{C}_{2g}} \sum_{\alpha \in \mathbb{F}^{2g}} ((x_\alpha)^\gamma)^{2n} = \mathrm{const} \sum_{\substack{C \subset \mathbb{F}^{2n} \\ \text{fixed type}}} W_{2g}(C, x), \end{aligned}$$

and an inner product defined on monomials.

What we prove with Bourganis in 2024 is the following:

Theorem 2. *Let $\mathcal{T}$ be a family of self-dual codes of length $2n$ over a field $\mathbb{F}$. Assume either that $\mathbb{F}$ has odd characteristic or is equal to $\mathbb{F}_2$. Let $C \in \mathcal{T}$ be doubly-even, and fix $g \in \mathbb{N}$. Then there exists an (explicit) constant C such that for a cusp form $f \in \mathcal{T}$ of genus r, with $\deg f = 2n$, we have*

$$\langle M_{2g}((2n))(xy), f(x) \rangle = \begin{cases} 0 & \text{if } r < g, \\ C \cdot f(y) & \text{if } r = g \end{cases}.$$

3. Petru Constantinescu, *Non-vanishing of geodesic periods of automorphic forms*

Preprint: [6], joint with Asbhjørn Nordentoft.

Class groups:

- $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$, $K = \mathbb{Q}(\sqrt{D})$,
- Cl_K class group, $h(D) = h_K = |\mathrm{Cl}_K|$ class number,

- $\mathcal{Q}_D$: set of primitive integral binary quadratic forms of discriminant D,
$$\mathcal{Q}_D = \left\{Q(x,y) = ax^2 + bxy + cy^2 : (a,b,c) = 1,\, b^2 - 4ac = D\right\}.$$
- Gauss: $\Gamma \circlearrowright \mathcal{Q}_D$,
$$(Q.\gamma)\begin{pmatrix} x \\ y \end{pmatrix} = Q\left(\gamma \begin{pmatrix} x \\ y \end{pmatrix}\right),$$
- isomorphism
$$\begin{aligned} \mathrm{Cl}_K &\xrightarrow{\cong} \Gamma\backslash\mathcal{Q}_D \\ A &\mapsto [a,b,c]. \end{aligned}$$

Heegner points and closed geodesics:

- $D < 0$: $A \in \mathrm{Cl}_K \rightsquigarrow$ Heegner point $z_A \in \Gamma\backslash\mathbb{H}$,
$$[a,b,c] \rightsquigarrow \frac{-b - i\sqrt{|D|}}{2a}.$$
$$h(D) = |\mathrm{Cl}_K| = |D|^{1/2+o(1)}.$$
- $D > 0$: $A \in \mathrm{Cl}_K^+ \rightsquigarrow$ closed geodesic $C_A \subset \Gamma\backslash\mathbb{H}$, $[a,b,c] \rightsquigarrow$ semicircle with endpoints $\frac{-b\pm\sqrt{D}}{2a}$.
$$h(D)\log\varepsilon_D = D^{1/2+o(1)},$$
$$I(C_A) = 2\log\varepsilon_D.$$

Theorem 3 (Duke '88)**.** *Fix $\Omega \subset \mathrm{PSL}_2(\mathbb{Z})\backslash\mathbb{H}$.*

- *$D < 0$: equidistribution of z_A, $A \in \mathrm{Cl}_{\mathbb{Q}(\sqrt{D})}$.*
- *$D > 0$: similar for closed geodesics.*

Waldspurger formulas. Let f be a nonzero Maass form. Our goal is to study closed geodesic periods $\int_{C_A} f(z)\,\frac{|dz|}{y}$.

$\chi \in \widehat{\mathrm{Cl}_K} \rightsquigarrow \theta_\chi$, the associated theta series (weight one on $\Gamma_0(D)$, nebentypus χ_D).

Theorem 4 (Waldspurger/Zhang/Popa)**.** *Let D be a fundamental discriminant. For $D < 0$,*
$$L(f \times \theta_\chi, \tfrac{1}{2}) = \frac{C_f}{D^{1/2}} \left| \sum_{A \in \mathrm{Cl}_K} \chi(A) f(z_A) \right|^2.$$
For $D > 0$,
$$L(f \times \theta_\chi, \tfrac{1}{2}) = \frac{C_f}{D^{1/2}} \left| \sum_{A \in \mathrm{Cl}_K^+} \chi(A) \int_{C_A} f(z)\,\frac{|dz|}{y} \right|^2.$$

Theorem 5 (Michel–Venkatesh '05)**.** *Let $\delta = 1/2700$. For $D < 0$:*
$$\left|\left\{\chi \in \widehat{\mathrm{Cl}_K} : L(f \times \theta_\chi, \tfrac{1}{2}) \neq 0\right\}\right| \gg D^\delta.$$

Sketch of proof. By orthogonality of characters,
$$\frac{1}{h(D)} \sum_{\chi \in \widehat{\mathrm{Cl}_K}} L(f \times \theta_\chi, \tfrac{1}{2}) = \frac{c_f}{D^{1/2}} \sum_{A \in \mathrm{Cl}_K} |\cdots|^2.$$
This converges by Duke's equidistribution theorem. The conclusion then follows from subconvexity for Rankin–Selberg L-functions, due to Harcos–Michel. □

Same proof does not work for geodesics, cannot apply equidistribution and relate to subconvexity (square is outside integral).

Question 6 (Michel, Oberwolfach 2020)**.** Let K be a real quadratic field of discriminant $D > 0$, and assume that $h_K \gg D^\delta$. Does there exist $A \in \mathrm{Cl}_K^+$ such that

$$\int_{C_A} f \frac{|dz|}{y} \neq 0?$$

Equivalently, does there exist $\chi \in \widehat{\mathrm{Cl}_K^+}$ such that $L(f \times \theta_\chi, \frac{1}{2}) \neq 0$?

The prime geodesic theorem. Let $D > 0, A \in \mathrm{Cl}_K^+$. This gives rise to a closed geodesic C_A, with $I(C_A) = 2 \log \varepsilon_D$. Sound–Young have the best estimate for their count.

Theorem 7 (C–Nordentoft 2024)**.** *Let f be a nonzero Maass form for $\mathrm{SL}_2(\mathbb{Z})$. Then*

$$\# \left\{ C \in \mathcal{C}(X) : \int_C f(z) \frac{|dz|}{y} = 0 \right\} \ll \frac{X}{(\log X)^{5/4}}.$$

Remark 8. We also obtain 100% non-vanishing for periods of weight k holomorphic cusp forms, for any Fuchsian group Γ.

Theorem 9 (C–Nordentoft 2024)**.** *For a positive proportion of positive discriminants $D > 0$ with $\varepsilon_\Delta \leq X$, we get that there exists $\chi \in \widehat{\mathrm{Cl}_{\mathbb{Q}(\sqrt{D})}}$ with $L(f \times \theta_\chi, \frac{1}{2} \neq 0)$.*

We construct a bipartite graph on X_N (double cosets in $\Gamma_\infty \backslash \Gamma / \Gamma_\infty$ with $c \leq N$) times Y_N (conjugacy classes with trace bounded in magnitude by N). This graph relates closed geodesic and vertical geodesics.

4. An excised orthogonal model for families of cusp forms

Talk by Zoe Batterman (Abstract), Akash Narayanan, Christopher Yao. Joint with Owen Barrett, Aditya Jambhale, and Kishan Sharma. Preprint: [3].

Conjecture (Montgomery–Dyson, 1970's): zeros of of zeta vs. GUE.

2005: S.J. Miller noticed a repulsion of the lowest-lying zeros near the central point of a family of even twists of a fixed elliptic curve L-function with finite conductor.

2011: Duenez, Huynh, Keating, Miller, Snaith: proposed an excised orthogonal model to capture the behavior of this repulsion.

Question 10. How accurately do egienvalues of random matrices from classical compact groups model the lowest-lying zeros of families of L-functions associated to a cuspidal newform?

Let

$$S_k^{\mathrm{new}}(M, \chi_f) \ni f(z) = \sum_{n=1}^{\infty} a_f(n) e^{2\pi i n z},$$

$\lambda_f(n) = a_f(n)/n^{(k-1)/2}$.

$$L(s, f) = \sum_{n \geq 1} \lambda_f(n) n^{-s},$$

Various specific families of twists $L(f \otimes \psi_d, s)$, match with classical compact groups:

- principal character, even twists vs. SO(even)
- principal character, odd twists vs. SO(odd)
- non-principal character, self-dual vs. Sp
- generic vs. U

Pictures.

5. On an extension of the Rohrlich-Jensen formula, *Lejla Smajlović*

Joint work with James Cogdell, Jay Jorgensen. Preprint: [5].

What is a Poisson–Jensen formula? We will view it as a way to characterize meromorphic functions in terms of their divisors. Some notation:

- $D_R = \{z = x + iy \in \mathbb{C} : |z| < R\}$
- F: a non-constant meromorphic function on $\overline{D_R}$,

$$F(z) = c_F z^m + \mathrm{O}(z^{m+1}), \quad z \to 0.$$

Then

$$\int_0^{2\pi} \log|F(Re^{i\theta})| \, \frac{d\theta}{2\pi} + \sum_{D_R} \cdots = F(0).$$

Rohrlich, 1980's: a *modular* generalization, characterizing modular forms via divisors. Given f, a meromorphic function on $\mathbb{H}$ that is invariant by $\mathrm{PSL}(2,\mathbb{Z})$. Assume f is holomorphic at the cusp and that the Fourier expansion of f at ∞ has constant term equal to one. Then

$$\int_{\mathrm{PSL}(2,\mathbb{Z})\backslash\mathbb{H}} \log|f(z)| \, \frac{d\mu(z)}{2\pi} + \sum_{w \in \mathcal{F}} \frac{\mathrm{ord}_w(f)}{\mathrm{ord}(w)} P(w) = 0.$$

Here

- $\mathrm{ord}_w(f)$ is the order of f at w as a meromorphic function,
- $\mathrm{ord}(w)$ denotes the order of the *point* w with respect to the action of $\mathrm{PSL}(2,\mathbb{Z})$ on $\mathbb{H}$, and
- $P(w) = \log\left(|\eta(w)|^4 \cdot \Im(w)\right)$ is the Kronecker limit function associated to the parabolic Eisenstein series on $\mathrm{PSL}(2,\mathbb{Z})$. This is the function appearing as the next-order term in the expansion of the Eisenstein series as $s \to 1$.

Another way to interpret this formula is as follows. We have

$$\langle 1, \log|f(z)| \rangle = \lim_{Y \to \infty} \int_{\mathcal{F}(Y)} 1 \cdot \log|f(z)| \, d\mu(z),$$

hence the formula reads

$$\langle 1, \log|f(z)| \rangle = -2\pi \sum_{w \in \mathcal{F}} \frac{\mathrm{ord}_w(f)}{\mathrm{ord}(w)} P(w).$$

Here $\log|f(z)|$ can be replaced by

$$\log\|f(z)\| = \log\left(\Im z^k |f(z)|\right)$$

for weight $2k$ meromorphic modular forms. Generalizations:

- to other Fuchsian groups of the first kind, by Rohrlich.
- to hyperbolic 3-space, by Herrero, Imamoglu, von Pippich, Toth.

Further modular generalization by Bringman and Kane. Keep the modular group setting, but evaluate more general inner products.

- $j(z) = q^{-1} + 744 + \mathrm{O}(q)$: Hauptmodul

- $j_1(z) := j(z) - 744$
- $j_n(z) := j_1|T_n(z)$, for $n \geq 2$
- f: weight $2k$ meromorphic modular form with respect to $\mathrm{PSL}(2,\mathbb{Z})$, normalized so that $f(z) = 1 + \mathrm{O}(q)$

They evaluated the regularized scalar product in terms of the divisor of f, proving that

$$\left\langle j_n(z), \log\left((\Im(z))^k |f(z)|\right)\right\rangle = -2\pi \sum_{w\in\mathcal{F}} \frac{\mathrm{ord}_w(f)}{\mathrm{ord}(w)} \mathbf{j}_n(w) + \frac{k}{6} c_n,$$

where $\mathbf{j}_n$ is characterized in terms of differential operators by Bringman and Kane; application of our results yields a different expression for the same function, namely

$$\mathbf{j}_n(w) = 2\pi\sqrt{n}\partial_s F_{-n}^{\mathrm{PSL}(2,\mathbb{Z})}(w,s)|_{s=1} - 24\sigma(n)P(w).$$

What is our main goal?

- To extend the point of view that the Rohrlich–Jensen formula is the evaluation of a particular type of inner product.
- To prove the extension of this formula in the setting of an arbitrary, not necessarily arithmetic, Fuchsian group of the first kind with one cusp.

We start with $j_n(z) = j_1|T_n(z)$, which is the unique (up to constants) holomorphic function that is $\mathrm{PSL}(2,\mathbb{Z})$-invariant on $\mathbb{H}$ and whose expansion near ∞ is $q^{-n} + o(q^{-1})$.

These properties hold for the special value $s = 1$ of the Niebur–Poincaré series $F_{-n}^{\Gamma}(z,s)$, defined for any Fuchsian group Γ of the first kind with oen cusp by

$$F_m^{\Gamma}(z,s) = \sum_{\gamma_\infty\backslash\Gamma} e\left(m\Re(\gamma z)\right)\left(\Im(\gamma z)\right)^{1/2} I_{s-1/2}\left(2\pi|m|\Im(\gamma z)\right),$$

for $m \neq 0$.

It is an eigenfucntion of the hyperbolic Laplacian, and may be expressed in terms of j_m.

The term $\log\left(\|f(z)\|\right)$ is a bit complicated, involving findings by Jorgensen, von Pippich and the speaker, plus some additional work done in our paper.

Proposition 11. *Let Γ be a cofinite Fuchsian group with one cusp at ∞ with identity as its scaling matrix. Let $2k \geq 0$ be an even integer, and let f be aw eight $2k$ meromorphic form which is Γ-invariant and with q-expansion at ∞ normalized so its constant term is equal to one. Then, we can express $\log\left(\|f\|(z)\right)$ in terms of parabolic Eisenstein series and Green's functions, as*

$$-2k + 2\pi \sum_{w\in\mathcal{F}_\Gamma} \frac{\mathrm{ord}_w(f)}{\mathrm{ord}(w)} \lim_{s\to 1}\left(G_s^{\Gamma}(z,w) + \mathcal{E}_{\Gamma,\infty}^{\mathrm{par}(z,s)}\right) + \cdots.$$

Here

$$\mathcal{E}_\infty^{\mathrm{par}}(z,s) = \sum_{\Gamma_\infty\backslash\Gamma} \Im(\gamma z)^s.$$

The Kronecker limit formula says that

$$\mathcal{E}_\infty^{\mathrm{par}}(z,s) = \frac{1}{(s-1)\,\mathrm{vol}(\Gamma\backslash\mathbb{H})} + \beta - \frac{1}{\mathrm{vol}(\Gamma\backslash\mathbb{H})}\log\left(|\eta_\infty^4(z)|\Im(z)\right) + \mathrm{O}(s-1).$$

Then we need to define β, and the Green's function $G_s(z,w)$, which is obtained by averaging the kernel $k_s(z,w)$; both functions are eigenfunctions of the Laplacian with eigenvalue $s(1-s)$, and the Green's function has a specified singularity on the

diagonal. The Green's function also has a Laurent series expansion as $s \to 1$ that involves the parabolic Eisenstein series. In particular,

$$\lim_{s\to 1}\left(G_s^\Gamma(z,w)+\mathcal{E}^{\mathrm{par}}_{\Gamma,\infty}(z,w)\right)$$

exists, with a logarithmic singularity on the diagonal.

The Rohrlich–Jensen formula can be understood through the study of the regularized inner product of this last limit with $F_{-n}(\cdot,1)$. Regularization is needed only in the cusp (because the logarithmic singularity is integrable). We can thus write

$$\begin{aligned}&\left\langle F_{-n}(\cdot,1), \overline{\lim_{s\to 1}\left(G_s^\Gamma(z,w)+\mathcal{E}^{\mathrm{par}}_{\Gamma,\infty}(z,w)\right)}\right\rangle\\ &= \lim_{Y\to\infty}\int_{\mathcal{F}(Y)} F_{-n}(z,1)\lim_{s\to 1}\left(G_s(z,w)+\mathcal{E}^{\mathrm{par}}_{\infty}(z,s)\right)\,d\mu(z).\end{aligned}$$

A key observation is that all terms are eigenfunctions of the Laplacian, hence one can seek to compute the inner product in a manner similar to that which yields the Maass–Selberg formula. The key identity is that

$$\begin{aligned}&\int_{\mathcal{F}(Y)} F_{-n}(z,1)\lim_{s\to 1}\left(G_s(z,w)+\mathcal{E}^{\mathrm{par}}_{\infty}(z,s)\right)\,d\mu(z)\\ &= \partial_s\left(-s(1-s)\int_{\mathcal{F}(Y)} F_{-n}(z,1)\left(G_s(z,w)+\mathcal{E}^{\mathrm{par}}_{\infty}(w,s)\,d\mu(z)\right)\right)|_{s=1}.\end{aligned}$$

We can now absorb $-s(1-s)$ into the integrals after applying the hyperbolic Laplacian Δ to each of the two factors in the integrand.

Main results:

Theorem 12. *For any positive integer n and any point $w \in \mathcal{F}$, we have*

$$\left\langle F_{-n}(\cdot,1), \overline{\lim_{s\to 1}\left(G_s(\cdot,w)+\mathcal{E}^{\mathrm{par}}_{\infty}(\cdot,s)\right)}\right\rangle = -\partial_s F_{-n}(w,s)|_{s=1}.$$

Combined with the previous proposition (describing $\log\|f(z)\|$), we get the Rohrlich–Jensen formula:

$$\langle F_{-n}(\cdot,1), \log\|f\|\rangle = -2\pi\sum_{w\in\mathcal{F}}\frac{\mathrm{ord}_w(f)}{\mathrm{ord}(w)}\partial_s F_{-n}(w,s)|_{s=1}.$$

We have further results. For instance, if g is any Γ-invariant analytic function with a pole at ∞, then (Niebur)

$$g(z) = \sum_{n=1}^{K} 2\pi\sqrt{n}a_n F_{-n}(z,1) + c(g)$$

for some constants K, a_n, and $c(g)$ depending only upon g. Then, we have the identity

$$\langle g, \log|f|\rangle = -2\pi\sum_{w\in\mathcal{F}}\frac{\mathrm{ord}_w(f)}{\mathrm{ord}(w)}\left(2\pi\sum_{n=1}^{K}\sqrt{n}a_n\partial_s F_{-n}(w,s)|_{s=1} - \cdots\right).$$

Further found that the generating series for the Niebur Poincaré series at $s=1$ is, in the z-variable (where we sum over q_z), the holomorphic part of the weight two biharmonic Maass form given by differentiating our linear combination of Eisenstein series and Green's function with respect to z.

6. Shenghao Hua, *Joint Value Distribution of Hecke–Maass Forms*

Preprint: [14]

Motivation: semiclassical limit of solutions to Schroedinger equation. Berry's random wave conjecture. QUE, L^p-norms.

Joint Gaussian moments conjecture: statistical indepndence of values of large eigenfunctions (multiplied together and integrated against some test function).

Theorem 13. *Let f and g be normalized Hecke–MAass cusp forms. Then if $2t_f \leq t_g - t_g^\varepsilon$, we have*

$$\int_{\Gamma\backslash\mathbb{H}} \psi f^2 g \ll t_g^{-A},$$

while if $2t_f > t_g - t_g^\varepsilon$, then, assuming GLH, we have

$$\int_{\Gamma\backslash\mathbb{H}} \psi f^2 g = \int_{\Gamma\backslash\mathbb{H}} \psi g + \mathrm{O}(t_f^{-1/4+\varepsilon}(1+|2t_f - t_g|^{-1/4})).$$

Proof uses spectral expansion of ψ into eigenfunctions u, then spectral expansion of $\int (ug)(f^2)$, reducing to the case $g = u$, then further spectral expansion of $\int (g^2)(f^2)$.

Theorem 14. *Assuming GRH and GRC, we have as* $\max(t_f, t_g) \to \infty$

$$\mathbb{E} f^2 g^2 = 1 + \mathrm{O}((\log(t_f + t_g))^{-1/4+\varepsilon}).$$

Proof eventually applies ultimately Soundararajan's method of moments (which requires GRH).

Work in progress: asymptotic formulas for certain averages of L-functions over toroidal families, going beyond [25].

7. Algebraic proof of modular form inequalities for optimal sphere packings, *Seewoo Lee*

Preprint: [17].

Motivating question: finding densest sphere packings.

- $d = 1$
- $d = 2$: Thue 1910, $\Delta_2 = \frac{\pi}{2\sqrt{3}}$
- $d = 3$: Kepler conjecture 1611, Hales 2005, $\Delta_3 = \frac{\pi}{3\sqrt{2}}$, formally verified 2014 in Isabelle/HOL
- (Korkine–Zolotareff, Blichfeldt, Cohn–Kumar): D_4, D_5, D_6, E_7, E_8 and Leech lattice
- Viazovska, 2016 π-day on arXiv: E_8 lattice packing optimal, with $\Delta_8 = \frac{\pi^4}{388}$
- Cohn–Kumar–Miller–Radchenko–Viazovska, March 21st 2016 on arXiv: $\Delta_{24} = \frac{\pi^{12}}{12!}$ Leech lattice (unique even unimodular with nonzero minimal length 2)

Viazovska et al use:

Theorem 15 (Cohn–Elkies, 2013)**.** *Let $r > 0$. Assume there exists a nice function $f : \mathbb{R}^d \to \mathbb{R}$ satisfying*

- $f(0) = \hat{f}(0) > 0$,
- $f(x) \leq 0$ *for all* $\|x\| \geq r$,
- $\hat{f}(y) \geq 0$ *for all* $y \in \mathbb{R}^d$,

then

$$\Delta_d \le \mathrm{vol}(B^d_{r/2}) = \left(\frac{r}{2}\right)^d \frac{\pi^{d/2}}{(d/2)!}.$$

Proof is not hard (fits in one page).

This leads to the hunt for a "magic function" f satisfying the indicated conditions. BAsed on their numerical experiments, Cohn–Elkies conjectured that the optimal sphere packing can be achieved by a magic function when $d = 2, 8, 24$. The magic function f and its Fourier transform $\hat{f}$ should vanish at the lattice points.

Viazovska et al constructed the magic functions for $d = 8, 24$ using modular forms. Decompose into Fourier eigenfunctions $f = f_+ + f_-$ (parity under Fourier transform). Viazovska writes them as

$$f_\pm(x) = \sin^2\left(\frac{\pi\|x\|^2}{2}\right)\int_0^\infty \varphi_\pm(t)e^{-\|x\|^2 t}\,dt,$$

where the $\sin^2$ factor enforces desired roots.

For $d = 8$, we have $\varphi_\pm(t) = t^2\psi_\pm(i/t)$, where $\psi_\pm$ are defined in terms of standard modular forms. We also need (for both $d = 8, 24$) a few non-obvious inequalities involving some of these modular forms. One reason these inequalities are difficult is because they're inhomogeneous with respect to the weights of the forms involved.

Original proofs use bounds of Fourier coefficients of the form

$$|c(n)| \le C_1 e^{C_2\pi\sqrt{n}},$$

following from the Hardy–Ramanujan formula, reducing the question to finite calculations and interval arithmetic.

Ran Romik (2023) gave an alternative and much simpler proof for $d = 8$ that doe snot use any interval arithmetic, but still requires a "calculator" to check inequalities like

$$e^{3\pi}\frac{9\Gamma(1/4)^{16}}{8192\pi^{12}} < 20480.$$

Question 16. Can we prove such inequalities *algebraically*?

The answer is yes, as we now explain. To that end, we'll develop a theory of *(completely) positive quasimodular forms*.

Definition 17. Let $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$. We say that a quasimodular form F is *positive* if

$$F(it) \ge 0$$

for all $t > 0$.

We call it *completely positive* if it has nonnegative q-coefficients at ∞.

Note that "completely positive" implies "positive", but the inclusion is strict in general. For instance, the discriminant form has a product formula that tells you that it is positive, but you can check that it's not completely positive.

Easy facts:

Theorem 18.
- *Anti-derivative preserves positivity.*
- *Derivative preserves complete positivity.*
- *Serre derivative preserves complete positivity.*

"Nontrivial" fact (that we won't use, and which also follows directly from Bernstein's theorem):

Theorem 19. *F is completely positive if and only if all its derivatives are positive.*

"Interesting" fact that we will use:

Theorem 20. *Let $F = \sum_{n \geq n_0} a_n q^n$ with $a_{n_0} > 0$. If some Serre derivative of F is positive, then so is F.*

Examples?

Definition 21 (Kaneko–Koike)**.** For given weight w and depth s, we can speak of *extremal quasimodular forms of weight w and depth s*, which are those whose order of zeros at ∞ is as large as possible.

It's been shown that these have almost all coefficients positive, and conjectured that they're all positive.

Theorem 22. *That conjecture is true for depth 1 extremal forms.*

We also have similar identities proving complete positively of the depth 2 extremal forms of weight $w \leq 14$, but not yet for general w.

This reduces some of the inequalities in the arguments of Viazovska et al to checking some limiting properties of ratios of modular functions. For monotonicity, we check that derivatives are positive, and for this it suffices to check the same for Serre derivatives of derivatives.

Possible future directions:

- Classify the completely positive forms?
- Possible applications in other LP problems?
  - Dual LP, uncertainty principle
  - Any results that are "uniform" in dimensions
- Make a formalization of the proof (e.g., in Lean) easier?

8. Ring of modular forms on certain unitary Shimura variety, *Yuxin Lin*

Recall the definition of modular forms of weight k and level 1, as certain holomorphic functions $f : \mathbb{H} \to \mathbb{C}$.

More generally, let $\mathrm{Sh}(G)$ be a Shimura variety (of level 1), where G is a reductive group over $\mathbb{Q}$. We can understand it as the moduli space of abelian varieties with additional structure. Let $\pi : \mathcal{A} \to \mathrm{Sh}(G)$ be the universal abelian scheme of dimension g.

Let $\underline{\omega} := \pi_* \Omega_{\mathcal{A}/\mathrm{Sh}}$ be the Hodge bundle on $\mathrm{Sh}(G)$, and let $\lambda_{\mathrm{Sh}} := \Lambda^g \underline{\omega}$ be the Hodge line bundle on $\mathrm{Sh}(G)$.

Definition 23. The ring of automorphic forms of scalar-valued weight is the space of global sections of the k-th tensor power of the Hodge line bundle, namely

$$\oplus_k H^0(\mathrm{Sh}, \lambda^{\otimes k}).$$

, 8, 12, 2, 16 An interesting question to ask is, what is the ring structure of such objects? There are some relevant results along these lines.

First, when $G = \mathrm{SL}_2(\mathbb{Z})$, we have that $\mathrm{Sh}(G)$ is the modular curve, and the ring of modular forms is generated by E_4 and E_6 modulo a single relation:

$$\mathcal{M}(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{C}[E_4, E_6]/(E_4^3 - E_6^2).$$

Going one dimension higher, for $G = \mathrm{SL}_2(\mathcal{O}_K)$ and $K = \mathbb{Q}(\sqrt{13})$, we have that $\mathrm{Sh}(G)$ is the Hilbert modular surface with real multiplication by $\mathcal{O}_K$. Van Der Geer and Zagier show that the ring of symmetric Hilbert modular forms of even weight has 4 generators at weights $4, 8, 12, 12$ respectively, with a relation at weight 24, while the ring of Hilbert modular forms of even weight has 5 generators of weights $4, 8, 12, 12, 16$, respectively, with two relations at weight 24 and 32. To that end, they find the minimal model Y of the compactification of $\mathrm{Sh}(G)$, then find the canonical divisor of Y and compute the image S of Y under the canonical embedding, then realize sections of $H^0(\mathrm{Sh}, \lambda^{\otimes 2k})$ as meromorphic sections of $\mathcal{O}_S(k)$ with certain multiplicity at the cusps.

This is the strategy we will imitate, but for certain unitary Shimura varieties rather than Hilbert modular surfaces.

To that end, we need to first introduce the moduli space of curves.

Let $G = \mathbb{Z}/d\mathbb{Z}$ be a finite cyclic group.

Definition 24.
- $\mathcal{M}_G$: the moduli space of admissible stable $\mathbb{Z}/d\mathbb{Z}$ covers of $\mathbb{P}^1$. Objects are pairs $(C/S, \iota : G \to \mathrm{Aut}_S(C))$, i.e., a family of curves over the base scheme together with an embedding of G in their automorphism group.
- $\tilde{\mathcal{M}}_G$, the moduli space of admissible stable $\mathbb{Z}/d\mathbb{Z}$ covers of $\mathbb{P}^1$ with an ordering on the ramification points. The objects are tuples $(C/S, \iota : G \to \mathrm{Aut}_S(C), \sigma_i)$.
- The forgetful morphism $\tilde{\mathcal{M}}_G \to \mathcal{M}_G$ is finite étale.

The irreducible connected components of $\mathcal{M}_G$ are indexed by monodromy datum, which is a triple $(d, r, \underline{a})$, where

- r is the number of branching points on $C/\iota(G)$,
- $\underline{a} = (a(1), \ldots, a(r)) \in G^r$ is the *inertia type*, which records the character of G at the tangent space of the i-th ramification point.

The *Torelli morphism* T associates the moduli of curves with the moduli of principally polarized abelian varieties:

$$\begin{array}{ccc} \mathcal{M}(d, r, \underline{a})_{\mathbb{C}} & \longrightarrow & \mathrm{Sh}(\mathcal{D})_{\mathbb{C}} \\ \downarrow & & \downarrow \\ \mathcal{M}_{g,\mathbb{C}} & \xrightarrow[T]{} & \mathcal{A}_{g,\mathbb{C}}. \end{array}$$

The image of the Torelli morphism is called the *Torelli locus* inside $\mathcal{A}_{g,\mathbb{C}}$. It turns out that the Jacobians coming out of $\mathcal{M}(d, r, \underline{a})$ have larger endomorphisms than generically, and in particular, at least contain the group algebra generated by G. It's thus reasonable to speculate that the image of T on $\mathcal{M}(d, r, a)$ will sit inside a *special* subvariety of $\mathcal{A}_g$, and this special subvariety will be the focus of our study.

More precisely, $\mathrm{Sh}(\mathcal{D})$ is the smallest PEL type Shimura subvariety of $\mathcal{A}_{g,\mathbb{C}}$ containing $T(\mathcal{M}(d, r, \underline{a})_{\mathbb{C}})$. It parametrizes abelian varieties with multiplication by $\mathbb{Z}[G]$ and signature given by $(d, r, \underline{a})$.

We focus on the family $\mathcal{M}[16] = \mathcal{M}(5, 5, (1, 1, 1, 1, 1))$.

- The associated Shimura variety $\mathrm{Sh}(G)$ is of *unitary type*, whose reductive group $\mathcal{G}$ has $\mathbb{R}$ points given by $\mathrm{GU}(3, 0) \times \mathrm{GU}(2, 1)$. Therefore, it is *compact*. The reason is that the signature of this Shimura variety can be computed, and the signature has a zero in one of the components, which means that the

hermitian symmetric domain is compact, and hence this $\mathrm{Sh}(\mathcal{D})$ is compact. In particular, this Shimura variety has no cusp.

- A second reason why this family is so nice is that $\mathrm{Sh}(\mathcal{D})_{\mathbb{C}}$ has dimension 2, which is equal to the dimension of $\mathcal{M}[16]$. Therefore, the Torelli morphism T gives an *isomorphism*

$$T : \mathcal{M}[16] \to \mathrm{Sh}(\mathcal{D}),$$

such that T gives an isomorphism between the corresponding Hodge line bundles. Thus, in order to compute global sections of the Hodge line bundle on $\mathrm{Sh}(\mathcal{D})$, it suffices to do the analogous thing on $\mathcal{M}[16]$, which is a bit simpler, but still tricky, due to nontrivial stabilizers. To that end, we consider:

- The finite étale cover $\tilde{\mathcal{M}[16]}$, which has a natural forgetful morphism

$$q : \tilde{\mathcal{M}[16]} \to \overline{M_{0,5}},$$

that is bijective on field-valued points. This is very nice, because we know the geometry of $\overline{M_{0,5}}$: its Picard group is generated by ten divisors, each isomorphic to $\mathbb{P}^1$.

Proposition 25. *Let Δ be the boundary divisor of $\overline{M_{0,5}}$, and let $\lambda_{\tilde{M}}$ be the Hodge divisor on $\tilde{\mathcal{M}[16]}$. Under the forgetful morphism*

$$q : \tilde{\mathcal{M}[16]} \to \overline{M_{0,5}},$$

we have that $q^*\Delta = 5\lambda_{\tilde{M}}$.

Idea of proof.

- Use Grothendieck–Riemann–Roch to relate $\lambda_{\tilde{M}}$ with the *boundary divisor* of $\tilde{\mathcal{M}}$.
- q exhibits Δ as a multiple of the boundary of $\tilde{\mathcal{M}}[16]$.

□

Our next goal is to fill in the lifts

$$\begin{array}{ccc} \tilde{\mathcal{M}[16]} & \xrightarrow{?} & \mathrm{Sh}_{\mathcal{K}}(\mathcal{D}) \\ \downarrow & & \downarrow{\scriptstyle ?} \\ \mathcal{M}[16] & \xrightarrow[T]{} & \mathrm{Sh}(\mathcal{D}). \end{array}$$

Giving such a cover of T is the same as computing the level subgroup corresponding to this cover. That level subgroup is basically a congruence subgroup:

Proposition 26. *The level subgroup $\mathcal{K}$ at the place p satisfies:*

$$\mathcal{K}_p = \begin{cases} \mathcal{G}(\mathbb{Z}_p) & \text{if } p \neq 5, \\ \mathcal{U}(\mathfrak{m}) & \text{if } p = 5. \end{cases}$$

Putting these parts together via the sequence of morphisms

$$\mathrm{Sh}_{\mathcal{K}}(\mathcal{D}) \xleftarrow{T} \tilde{\mathcal{M}[16]} \xrightarrow{q} \overline{M_{0,5}} \xrightarrow{|-K|} S,$$

we obtain the following:

Proposition 27.
- *The weight m automorphic forms on $\mathrm{Sh}_{\mathcal{K}}(\mathcal{D})$ is*

$$H^0(\mathrm{Sh}_{\mathcal{K}(\mathcal{D}),\lambda_{\mathrm{Sh}}\otimes m}) \cong H^0(S, \mathcal{O}_S(2\lfloor \frac{m}{5} \rfloor)) \cong (\mathbb{C}[x_0,\dots,x_5])/(Q_{12}-Q_{ij}),$$

  with explicit degrees.
- *Similar for the ring of graded automorphic forms.*

From the moduli interpretation, we see that $\mathrm{Sh}(\mathcal{D})$ is the $\mathfrak{S}_5$-invariant of $\mathrm{Sh}_{\mathcal{K}}(\mathcal{D})$. Therefore:

Theorem 28. *The ring of automorphic forms on $\mathrm{Sh}(\mathcal{D})$ is the $\mathfrak{S}_5$-invariant of $\mathrm{Sh}_{\mathcal{K}}(\mathcal{D})$. In particular,*

$$\oplus_m H^0(\mathrm{Sh}(\mathcal{D}), \lambda^{\otimes m}) \cong \mathbb{C}[f_2, f_4, f_6, \varepsilon]/\varepsilon^{10} + \frac{1}{52}(f_2^2 - 4f_4),$$

where the f_{2n} are polynomials of degree $2n$ in x_i, where everything has explicit degrees.

9. Counting modular forms mod p satisfying constraints at p, *Samuele Anni*

Recall the definition of modular forms f of weight k on $\Gamma_0(n)$, with q-expansion $f(z) = \sum a_n q^n$. We also know that there are Hecke operators T_p acting on them. We consider only cuspidal *newforms* ($a_0 = 0$), *normalized* ($a_1 = 1$), which are eigenforms for the Hecke operators and arise from level n and not any smaller level. We use the notation $S_k(n, \mathbb{C})$ and $S_k(n, \mathbb{C})^{\mathrm{new}}$.

Definition 29. The *Hecke algebra* $\mathbb{T}(n,k)$ is the $\mathbb{Z}$-subalgebra of $\mathrm{End}_{\mathbb{C}}(S_k(n,\mathbb{C}))$ generated by Hecke operators T_p for each prime p.

Newforms can be seen as ring homomorphisms $f : \mathbb{T}(n,k) \to \overline{\mathbb{Q}}$.

Theorem 30 (Deligne, Serre, Shimura)**.** *Let n and k be positive integers. Let $\mathbb{F}$ be a finite field of characteristic ℓ, with $\ell \nmid n$, and let $f : \mathbb{T}(n,k) \twoheadrightarrow \mathbb{F}$ be a surjective ring homomorphism. Then there is a unique continuous semisimple representation*

$$\bar{\rho}_f : \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\mathbb{F}),$$

unramified outside $n\ell$, such that for all p not dividing $n\ell$, the trace of Frobenius at p under $\bar{\rho}_f$ is $f(T_p)$, and the determinant is given in terms of the value of f on the diamond operator (central character).

Computing $\bar{\rho}_f$ is "difficult", but theoretically it *can be done in polynomial time* in $n, k, \#\mathbb{F}$:

- Edixhoven, Couveignes, de Jong, Merkl, Bruin, Bosman: $\#\mathbb{F} \leq 32$.
- Mascot, Zeng, Tian: $\#\mathbb{F} \leq 53$.

Fix a prime $p \geq 5$, a level N prime to p, and a weight $k \geq 2$. Let $S_k := S_k(\Gamma_0(Np), \bar{\mathbb{Q}}_p)$ be the space of p-new (i.e., not coming from level N) cuspidal modular forms of level Np and weight k with coefficients in $\bar{\mathbb{Q}}_p$.

We now discuss the **Atkin–Lehner involution**. There exist $x, y, z, t \in \mathbb{Z}$ for which the matrix

$$W_p = \begin{pmatrix} px & y \\ Npz & pt \end{pmatrix}$$

has determinant p.

The matrix W_p normalizes the group $\Gamma_0(Np)$, and for any weight k, it induces a linear operator w_p on the space of cusp forms S_k that commutes with the Hecke operators T_q for all $q \nmid Np$ and acts as its own inverse.

Any cusp form in S_k that is an eigenform for all T_q with $q \nmid N$ is also an eigenform for w_p, withe eigenvalue ± 1. This involution acts on the modular curve.

The Atkin–Lehner involution w_p splits S_k as a sum of plus and minus spaces:

$$S_k = S_k^+ \oplus S_k^-.$$

Since we have dimension formulas for $s_k := \dim S_k$, in order to understand the dimensions

$$s_k^\pm = \dim S_k^\pm$$

of the Atkin–Lehner eigenspaces, it suffices to understand the difference

$$d_k := s_k^+ - s_k^-.$$

We can get some idea from tables of examples (obtained empirically by looking at many more specific examples):

p	d_k
5	± 1
11	± 2
59	± 6
101	± 7

TABLE 1. Values of d_k for different p

Classical result that d_k is constant in absolute value and alternates in sign. Need to introduce modification d_k^* taking into account Eisenstein series of weight two. One can show:

Theorem 31 (Fricke, Yamauchi, Momose, Ogg, Wakatsuki, Helfgott, Martin et al.)**.** *We have*

$$d_k^* = (-1)^{k/2} \frac{\#\mathrm{FP}}{2},$$

where #FP *is the number of fixed points of the Atkin–Lehner involution w_p on $X_0(Np)$.*

The fixed points of w_p on $X_0(Np)$ corresponds to elliptic curves with level structure and CM by $\sqrt{-p}$, in fact the d_k are closely related to class numbers:

$$\#\mathrm{FP} = c_p \cdot h(\sqrt{-p}) \cdot \prod_{q|N,\mathrm{prime}} \left(1 + \left(\frac{-4p}{q}\right)\right).$$

What do we want to do? Systems of mod-p prime-to-Np Hecke eigenvalues correspond to continuous semisimple Galois representations $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\overline{\mathbb{F}}_p)$. Can decompose according to these, and then further via Atkin–Lehner, giving spaces $S_{k,\bar\rho}^\pm$ and their dimensions $s_{k,\bar\rho}^\pm$ and differences $d_{k,\bar\rho}$. As before, $k = 2$ and $\bar\rho$ forces us to make an adjustment, so let

$$d_{k,\bar\rho}^* := \begin{cases} d_{k,\bar\rho} - 1 & \text{if } k = 2 \text{ and } \bar\rho = 1 \oplus \omega, \\ d_{k,\bar\rho} & \text{otherwise,} \end{cases}$$

where ω is the cyclotomic character.

Theorem 32 (Anni, Ghitza, Medvedovsky)**.** *For $k \geq 2$ and any $\Gamma_0(Np)$-modular $\bar{\rho}$, we have*

$$d^*_{k+2,\bar{\rho}[1]} = -d^*_{k,\bar{\rho}}.$$

As an example, for $p = 5$, $N = 23$, we can check that for $k > 2$, we have $d_k = \pm 2$.

Bergdall and Pollack use the Ash–Stevens formula, a fundamentally characteristic p technique for filtering cohomology of modular symbols, to derive their dimension formulas. But Ash–Stevens has nothing to say about Atkin–Lehner, in part because Atkin–Lehner operator requires inverting p. On the other hand, the classical complex methods – trace formulas, Gauss–Bonnet, Riemann–Hurwitz – do not know anything about $\bar{\rho}$.

What we do instead is to combine the *trace formula* (Zagier–Cohen–Osterlé–Cohen–Strömberg and Skoruppa–Zagier–Popa) with an *algebra theorem*, an explicit refinement of Brauer–Nesbitt.

What is this explicit Brauer–Nesbitt? See [2].

Theorem 33 (AGM)**.** *Let M and N be two finite free $\mathbb{Z}_p$-modules of the sam erank d, each with an action of an operator T. Then $\bar{M}^{\mathrm{ss}} \cong \bar{N}^{\mathrm{ss}}$ as $\mathbb{F}_p[T]$-modules if and only if for every n with $1 \leq n \leq d$, we have*

$$\mathrm{trace}(T^n|M) = \mathrm{trace}(T^n|N),$$

(...).

Corollary 34. *If we have modules that injective modulo p (but not necessarily in characteristic zero), then we can check that semisimplifications of certain residual quotients are the same by proving an inequality involving traces.*

Now work with the trace formula. We can deduce statements about dimensions.

10. Integer partitions detect the primes, *Jan-Willem Van Ittersum*

Preprint: [7].

We write s_i for the sizes of the parts, and m_i for the multiplicities of the part, with $s_1 > s_2 > \cdots$.

Definition 35 (MacMahon 1920)**.**

$$M_a(n) := \sum_{\substack{n = m_1 s_1 + \cdots + m_a s_a \\ 0 < s_1 < \cdots < s_a}} m_1 \cdots m_a.$$

Remark 36. This is the sum of multiplicity products of partitions of n with a part sizes.

Consider

$$\psi(n) := (n^2 - 3n + 2)M_1(n) - 8M_2(n).$$

Example 37. $n = 3, a = 1$, in which case $\lambda = (3)$ or $\lambda = (1^3)$, in which case $M_1(3) = 1 + 3 = 4$, or $a = 2$, in which case $\lambda = (2, 1)$, so $M_2(3) = 1 \cdot 1 = 1$, and we compute that $\psi(3) = 2M_1(3) - 8M_2(3) = 0$.

Continuing, take $n = 4, a = 1$, in which case $\lambda = (4)$, $\lambda = (2^2)$ $\lambda = (1^4)$, so $M_1(4) = 1 + 2 + 4 = 7$, or $a = 2$, in which case $\lambda = (3, 1)$ or $\lambda = (2, 1^2)$, so $M_2(4) = 1 \cdot 1 + 1 \cdot 2 = 3$, and so $\psi(4) = 6M_1(4) - 8M_2(4) = 18$.

Continuing, one gets the table, for $n = 2..11$, of $\psi(n) = 0, 0, 18, 0, 120, 0, 270, 192, 504, 0$.

Theorem 38 (Craig–vI–Ono)**.** *For positive integers n, we have*

(1) $(n^2-3n+2)M_1(n)-8M_2(n)\geq 0$,

(2) $(3n^3-13n^2+18n-8)M_1(n)+(12n^2-120n+212)M_2(n)-960M_3(n)\geq 0$

and for $n\geq 2$ these expressions vanish if and only if n is prime.

Remark 39. We call such an expression *prime-detecting*. The sum of two such expressions is likewise prime-detecting, and multiplying $f(n)$ for a polynomial f yields a prime-detecting expression if $f(n)>0$ for all n.

We can get a table of prime-detecting expressions: the first two that appear above, then two more, $(126n^5-\cdots)M_1(n)+\cdots$ and $(300n^8-\cdots)M_1(n)+\cdots$.

Conjecture 40. *These are all such expressions (up to addition and multiplication as before).*

We then checked whether we could get more results of a similar shape, by considering a generalization of the MacMahon function:

Definition 41. For $\ell\in\mathbb{Z}^a_{\geq 0}$, we define the *generalized MacMahon partition function*

$$M_{\underline{\ell}}(n):=\sum_{\substack{n=m_1s_1+\cdots+m_as_a\\ 0<s_1<\cdots<s_a}} m_1^{\ell_1}\cdots m_a^{\ell_a}.$$

You can think of this as a generalization of the divisor function, but for partitions. For this new generalized partition function, much more is possible:

Theorem 42 (Craig–vI–Ono)**.** *Let $d\geq 4$.*

(1) There exist clu$\underline{\ell}\in\mathbb{Z}$ such that $\sum_{|\ell|<d} c_{\underline{\ell}}M_{\underline{\ell}}(n)\geq 0$, where $|\underline{\ell}|=\ell_1+\cdots+\ell_a$.

(2) There are $\gg d^2$ linearly independent such expressions.

Example 43. $63M_{(2,2)}(n)-12M_{(3,0)}(n)-\cdots+12M_{(3,0,0)}(n)=\frac{11}{3}\psi(n)$.

¡++¿

Goal of the talk now is to explain some of the ideas behind these results, which have a lot to do with modular forms.

Prime-detecting quasimodular forms. Let

$$\mathcal{G}_k:=-\frac{B_k}{2k}+\sum_{n\geq 1}\sigma_{k-1}(n)q^n,\qquad D:=q\frac{\partial}{\partial q},\qquad q=e^{2\pi i\tau}.$$

Consider

$$\begin{aligned} f_{k,\ell}&:=(D^\ell+1)\mathcal{G}_{k+1}-(D^k+1)\mathcal{G}_{\ell+1}\\ &=*+\sum_{n\geq 1}\sum_{d|n}\left((n^\ell+1)d^k-(n^k+1)d^\ell\right)q^n.\end{aligned}$$

Note that for $d=1$ one gets $n^\ell+1-n^k-1=n^\ell-n^k$. Calculating similarly with $d=n$, we see that $f_{k,\ell}$ vanishes at prime coefficients. It's also easy to see that its coefficients are positive. But note that $f_{k,\ell}$ is not modular, and not of homogeneous weight. Key features here are quasimodularity and mixed weight.

Theorem 44 (Craig–vI–Ono)**.** *All prime-detecting forms in $\oplus_{k:even}\oplus_{n\geq 0}D^n\mathcal{G}_k$ are linear combinations of D^nH_k, where*

$$H_k=\begin{cases}\frac{1}{6}(D^2-D+1)\mathcal{G}_2-\mathcal{G}_4 & \text{if } k=6,\\ \frac{1}{24}\left(-D^2\mathcal{G}_{k-6}+(D^2+1)\mathcal{G}_{k-4}-\mathcal{G}_{k-2}\right) & \text{if } k=8.\end{cases}$$

That's the first ingredient, but still need to say something about modularity of MacMahon functions. Let

$$\mathcal{G}(\underline{\ell}) = \sum_{n\geq 0} M_{\underline{\ell}}(n)q^n \xrightarrow{q\to 1} \frac{(1-q)^{|\underline{\ell}|+a}}{\prod_i \ell_i!} \sum_{s_1<\cdots<s_a} \frac{1}{s_1^{\ell_1+1}\cdots s_a^{\ell_a+1}} \quad (\ell_a \geq 1,\, \ell_i \geq 0).$$

The algebra $\mathcal{Z}_q := \langle \mathcal{G}(\underline{\ell})\rangle_{\mathbb{Q}}$ was introduced by Bachmann–Kühn. Facts:

- $\mathcal{Z}_q$ is a differential algebra (closed under differentiation and multiplication)
- $\tilde{M} := \mathbb{Q}[\mathcal{G}_2, \mathcal{G}_4, \mathcal{G}_6] \subseteq \mathcal{Z}_q$ (i.e., it contains the space of quasimodular forms).
- (Hoffman–Ihara) We have

$$\sum_{j\geq 0} \mathcal{G}(\underbrace{1,1,\ldots,1}_{j})x^j = \exp\left(\sum_{n\geq 1}\frac{(-1)^{n+1}}{n}x^n \underbrace{\sum_{m}\frac{q^m}{(1-q^m)^{2n}}}_{\sum_{m,s\geq 1}\binom{s+n-1}{s-n}q^{ms}\in\tilde{M}}\right),$$

  which is a linear combination of quasimodular Eisenstein series, hence $M_a(n)$ are coefficients of a quasimodular form.

11. Supersingular abelian surfaces and where to find them, *Gabriele Bogo*

Characteristic zero: $\mathrm{End}(E) \cong \mathbb{Z}$ or $\mathcal{O}_D$, order in an imaginary quadratic field.

Characteristic $p > 0$ (Hasse, 1936), can also be $\mathcal{O}_{\infty,p}$, order in a quaternion algebra over $\mathbb{Q}$. In that case, we call it *supersingular*.

Deuring 1941: for $p \geq 5$, the number of supersingular elliptic curves over $\mathbb{F}_{p^2}$ is

$$\lfloor \frac{p-1}{12} \rfloor + \delta + \varepsilon.$$

Another interpretation: writing $\mathcal{M}_{1,1} \cong \mathbb{H}/\operatorname{SL}_2(\mathbb{Z})$ for the moduli space of elliptic curves over $\mathbb{C}$, there are three special points $\infty, i = \sqrt{-1}, \rho = (-1+\sqrt{-3})/2$, and we can think of Deuring's result as

$$\#(\mathrm{ss}_p) = \lfloor \frac{p-1}{12} \rfloor + \delta + \varepsilon \leq \lfloor \frac{-\chi(\mathbb{H}/\operatorname{SL}_2(\mathbb{Z}))\cdot(p-1)}{2} \rfloor + 2.$$

Let's generalize to higher-dimensional varieties. Let $\mathbb{Q}(\sqrt{D})$ be a real quadratic field, with ring of integers $\mathcal{O}_D$. We consider a family $\mathcal{X} \to \mathcal{C}$ of (p.p.) abelian surfaces, over a curve $\mathcal{C}$, such that:

- the fibers X_c are defined over a number field k, and
- there exists an inclusion of rings $\mathcal{O}_D \hookrightarrow \mathrm{End}_k(X_c)$.

An abelian surface over a field of characteristic p is *supersingular* (resp. *superspecial*) if it is isogenous (resp. isomorphic) to the product of two supersingular elliptic curves.

There are embeddings

$$\mathbb{H}/\Gamma = \mathcal{C} \hookrightarrow X_D(\mathbb{C}) = \mathbb{H}^2/\operatorname{SL}_2(\mathcal{O}_D).$$

For example, if $\Gamma = \operatorname{SL}_2(\mathbb{Z})$, this is the inclusion of elliptic curves into the space of abelian surfaces with multiplication. There are also modular curves $\Gamma_0(N)$. But one can have more interesting examples, related to non-arithmetic curves. To give

an explicit example, consider the triangle group $\Gamma = \Delta(2,5,\infty) \hookrightarrow X_5$, with explicit equation

$$y^2 = \begin{cases} x^5 - 5x^3 + 5x - 2t & \text{if } t \neq \infty, \\ x^5 - 1 & \text{if } t = \infty. \end{cases}$$

We now take the reductions of the above inclusion modulo p, to a map

$$\overline{\mathcal{C}} \to X_D(\mathbb{F}).$$

We want to study the fibers of this family that, after reduction modulo p, are supersingular.

To that end, consider a smooth algebraic curve in a Hilbert modular surface

$$\mathcal{C} \hookrightarrow X_D$$

with second Lyapunov exponent $\lambda_2 \in \mathbb{Q} \cap (0,1]$. For example, for $\mathrm{SL}_2(\mathbb{Z})$, we have $\lambda_2 = 1$, and for modular curves, $\lambda_2 = 1$, while for $\Delta(2,5,\infty)$ we have $\lambda_2 = 1/3$, but for Teichmuller curves, you can have $1/3, 1/5, 1/7$.

Theorem 45 (B–Li, 2024)**.** *The supersingular locus of $\mathcal{C}$ modulo p has cardinality described by*

$$\lfloor \frac{-\chi(\mathcal{C})(p-\lambda_2)}{2} \rfloor \leq \#\mathrm{ss}_p^{\mathcal{C}} \leq \lfloor \frac{-\chi(\mathcal{C})(p-1)(\lambda_2+1)}{2} \rfloor + r, \quad \text{if } \left(\frac{D}{p}\right) = -1,$$

while

$$\#\mathrm{ss}_p^{\mathcal{C}} \leq \lfloor \frac{-\chi(\mathcal{C})(p-1)\lambda_2}{2} \rfloor + r, \quad \text{if } \left(\frac{D}{p}\right) = 1.$$

How to find supersingular abelian surfaces? If $\mathcal{C} = \mathbb{H}/\Gamma$ is of genus zero with Hauptmodul j, then the supersingular locus can be described by a polynomial in j:

$$\mathrm{ss}_p^{\mathcal{C}}(j) := \prod_{\substack{c \in \mathcal{C} \\ \overline{X_c} \text{ supersingular}}} (j - j(c)).$$

Theorem 46 (B.–Li, 2024)**.** *There are two families $\{A_{0,n}(j)\}_n$ and $\{A_{1,n}(j)\}_n$ of orthogonal polynomials (Atkin's polynomials) such that for every p of good reduction,*

$$\mathrm{ss}_p^{\mathcal{C}}(j) \equiv \begin{cases} \mathrm{lcm}(A_{0,n_p}, A_{1,\tilde{n}_p}) & \text{if } \left(\frac{D}{p}\right) = -1, \\ \gcd(A_{0,n_p}, A_{1,\tilde{n}_p}) & \text{if } \left(\frac{D}{p}\right) = 1 \end{cases}$$

for explicit indices n_p and $\tilde{n}_p$.

For example, consider $\Delta(2,5,\infty)$. Let j be a Hauptmodul, with pole at ∞. The scalar products are defined on $\mathbb{R}[j]$ by

$$\langle f, g \rangle_0 := \int_{\pi/5}^{\pi/2} f(e^{i\theta}) g(e^{i\theta})\, dth,$$

$$\langle f, g \rangle_1 := \int_{2\pi/5}^{\pi/2} f(e^{i\theta}) g(e^{i\theta})\, d\theta.$$

The first polynomials $A_{0,n}(j)$ are: $A_{0,0}(j) = 1$, $A_{0,1}(j) = j - 9/20$, etc. Taking $p = 13$, one finds that

$$\mathrm{ss}_{13}(j) \equiv A_{0,3}(j) \equiv j(j+4)(j-1) \pmod{13}.$$

Next, real multiplication splits $H^1_{\mathrm{dR}}(X_c)$ into two eigenspaces. The eigendifferentials induce two second order differnetial equations, called Picard–Fuchs differential equations. TRuncation of the holomorphic solutions can be related to the supersingular locus of $\mathcal{C}$.

Example 47. For $\mathcal{C} = \mathbb{H}/\Delta(2,5,\infty)$,

$$j^n \cdot {}_2F_1\left(\frac{7}{20}, \frac{3}{20}; 1; \frac{1}{j}\right) = U_{0,n}(j) + \mathrm{O}(j^{-1}),$$

etc. Leads to a formula for $\mathrm{ss}_p^{\mathcal{C}}(j)$.

The partial Hasse invariants h_1 and h_2 are characteristic p Hilbert modular forms of non-parallel weight. They have the following properties:

- The divisor of h_i is the component D_i of the non-ordinary locus of $X_D(\mathbb{F})$.
- Their q-expansion is constant, equal to 1, at every cusp.

In particular, they do not lift to Hilbert modular forms in characteristic zeor.

Theorem 48 (B–Li, 2024)**.** *Let $\mathcal{C} \cong \mathbb{H}/\Gamma \hookrightarrow X_D$ be a smooth algebraic curve with good reduction at p. Then the partial Hasse invariants lift to (twisted) modular forms on Γ in characteristic zero.*

12. Congruences for the number of 3- and 6-regular partitions and quadratic forms, *Cristina Ballantine*

Definition 49. A *partition* of a positive number n is a non-increasings equence of positive integers, named *parts*, that add up to n.

We denote by $p(n)$ the number of partitions of n.

Example 50. $p(4) = 5$.

Ramanujan congruences:

$$p(5n+4) \equiv 0 \pmod 5,$$
$$p(7n+5) \equiv 0 \pmod 7,$$
$$p(11n+6) \equiv 0 \pmod{11}.$$

Ramanujan: it appears there are no equally simple properties for any moduli involving primes other than these.

Ahlgren and Ono (2001): there are such Ramanujan congruences modulo every prime $\ell \geq 5$ (actually mod ℓ^m).

How about $\ell = 2, 3$? Radu (2012, formerly Subbarao Conjecture): for $\ell \in \{2,3\}$, there is no arithmetic progression $An + B$ such that $p(An+B) \equiv 0 \pmod \ell$ for all n.

Conjecture (Parkin and Shank): approximately half the values of $p(n)$ are even.

Bellaiche and Nicolas 2016: the number of $n \leq x$ with $p(n)$ even is at least

$$0.069\sqrt{x}\log\log x,$$

while the number that are odd is at least a constant multiple of $\sqrt{x}/\log x$.

The reason we can do this using modular forms is that

$$\sum_{n=0}^{\infty} p(n)q^n = \prod_{i=1}^{\infty} \frac{1}{1-q^i} = \frac{q^{1/24}}{\eta(q)}.$$

Let's write $Q(n)$ for the number of partitions with distinct parts. Then

$$\sum_{n=0} Q(n)q^n = \prod_{i=1}^{\infty}(1+q^i) \equiv \prod_{i=1}^{\infty}(1-q^i) \pmod 2 = \sum_{j\in\mathbb{Z}}(-1)^j q^{j(3j-1)/2},$$

so $Q(n)$ is even asymptotically 100% of the time.

Definition 51. An *r-regular partition* is a partition in which no parts are divisible by r.

Notation 52. $b_r(n) :=$ the number of r-regular partitions of n.

$b_2(n)$ is even asymptotically 100% of the time.

Theorem 53 (Keith–Zanello 2022)**.** *If $p \equiv 13, 17, .19, 23 \pmod{24}$ is prime, then*

$$b_3\left(2(p^2n + pk - 24^{-1})\right) \equiv 0 \pmod 2$$

for $1 \le k \le p-1$, where the inverse is taken modulo p.

Theorem 54 (Yao 2022)**.** *Let $p \ge 5$ for a prime. Then we can construct more arithmetic progressions where this happens, different from the above.*

(1) If $b_3\left(\frac{p^2-1}{12}\right) \equiv 1 \pmod 2$, then we get a whole arithmetic progression.
(2) (...)

Experimentally, we found a bunch of arithmetic progressions on which b_3 was even. For instance, for $\alpha \in \{1, 51, 76, 101\}$,

$$b_3\left(2(5^3n + \alpha)\right) \equiv 0 \pmod 2.$$

Conjectured that one can do this for every prime $p \ge 5$.

The bad news is that the congruences we found for $p = 5, 7, 11$ are special cases of Yao's theorem. My coauthor did more experience and found other primes $p \in \{29, 59, \ldots, 683\}$ where we have

$$b_3\left(2(p^2n + p\alpha - 24^{-1})\right) \equiv 0 \pmod 2,$$

where $0 \le \alpha < p$, $\alpha \ne \lfloor p/24 \rfloor$, and 24^{-1} taken modulo p, that do not fit into Yao's theorem.

We proved the congruences algorithmically, using a result of Radu involving modular forms and Sturm's bound.

Can we describe these primes? What is the general theorem?

Let $\mathcal{P}$ be the set of primes p such that for some $j \in \{1, 4, 8\}$, the equation $x^2 + 24 \cdot 9y^2 = jp$ has primitive solutions.

13. Exact formulae for ranks of partitions, *Qihang Sun*

Preprint: [22].

$p(n)$: partition function. Ramanujan's congruence mod 5 and 7.

Dyson's rank: for each partition $\Lambda = \{\Lambda_1 \ge \Lambda_2 \ge \cdots \ge \Lambda_k\}$, define $\text{rank}(\Lambda) := \Lambda_1 - \kappa$ to be the largest part minus the number of parts. Rank generating function $N(a, b, n)$. Explains Ramanujan congruences, e.g.,

$$N(a, 5, 5n + 4) = \frac{1}{5}p(5n + 4),$$

$$N(a, 7, 7n + 5) = \frac{1}{7}p(7n + 5).$$

These conjectures were all proved by Swinnerton-Dyer in 1954.

We'll focus on the analytic side of those rank functions. For $q = e^{2\pi iz}$ and $w = \zeta_b^a$,

$$\mathcal{R}(w,q) = 1+\sum_{n=1}^{\infty}\sum_{m\in\mathbb{Z}} N(m,n)w^m q^n = 1+\sum_{n=1}^{\infty}\frac{q^{n^2}}{(wq,q)_n(w^{-1}q,q)_n} =: 1+\sum_{n=1}^{\infty} A\left(\frac{a}{b},n\right)q^n.$$

Examples:

- $R(1,q) = 1+\sum p(n)q^n = q^{1/24}/\eta(z)$
- $R(-1,q) = f(q)$ (Ramanujan's 3rd other mock theta function)
- $R(\zeta_3,q) = R(\zeta_3^2,q) = \gamma(q)$ (another mock theta function)

Hardy–Ramanujan (1918):

$$p(n) \sim \frac{1}{4n\sqrt{3}}\exp\left(\pi\sqrt{\frac{2n}{3}}\right),$$

together with more precise result. Rademacher (1937): exact formula.

$A(\frac{1}{2},n) = N(0,2,n)-N(1,2,n)$. Ramanujan claimed, Dragonette (1952) proved:

$$A(\tfrac{1}{2},n) = \frac{(-1)^{n-1}e^{\pi\sqrt{\tilde{n}/6}}}{2\sqrt{\tilde{n}}}+\cdots.$$

Andrews (1966).

Conjectures by Andrews and Lewis, proved by Bringmann (2009).

$$A(\tfrac{1}{3},n) = \frac{4\sqrt{3}i}{(24n-1)^{1/2}}\sum_{3|k}^{\lfloor\sqrt{n}\rfloor}\cdots.$$

Should be an exact formula.

More generally, Bringmann gave asymptotics for $A(\frac{\ell}{u},n)$.

Proofs use Maass Poincaré series.

14. The rational torsion subgroups of the Drinfeld modular Jacobians for prime-power levels, *Sheng-Yang Kevin Ho*

Preprint: [13].

$X_0(N)$: modular curve,

$$J_0(N)(\mathbb{Q}) \cong \mathbb{Z}^r \oplus \mathcal{T}(N).$$

Can we compute the rational torsion subgroup?

Let $\mathcal{C}(N)$ denote the rational cuspidal divisor class group for $X_0(N)$. By a theorem of Manin and Drinfeld, we have

$$\mathcal{C}(N) \subseteq \mathcal{T}(N).$$

Conjecture (generalized Ogg's conjecture): for any positive integer N, we have $\mathcal{C}(N) = \mathcal{T}(N)$.

Strategy:

(1) Compute $\mathcal{C}(N)$ explicitly to give a lower bound for $\mathcal{T}(N)$.
(2) Study the Eisenstein ideal of the Hecke algebra of level N to give an upper bound for $\mathcal{T}(N)$.

Pass to the function field setting.

15. A variation of a theme after Dirichlet, *Chung Hang Kwan*

$$\int_0^1 \int_0^\infty \Phi \left[\begin{pmatrix} y_0 & & \\ & y_0 & \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & u & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right] d^\times y_0 e(-u)\, du,$$

$$\int_0^\infty \int_0^1 \Phi \left[\begin{pmatrix} 1 & u & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} y_0 & 0 & 0 \\ 0 & y_0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right] e(-u)\, du\, d^\times y_0.$$

16. On Hecke eigenvalues of Ikeda lifts, *Nagarjuna Chary Addanki*

Preprint: [1].

$S_k(\Gamma_2)$ decomposes into a Maass space and its complement. The Maass space is generated by Saito–Kurokawa lifts. These lifts are characterized by the signs of their eigenvalues: F is a Saito–Kurokawa lift if and only if all the eigenvalues are positive. The question is whether there is a similar space in higher dimensions. Ikeda lifts are generalizations of these Saito–Kurokawa lifts, so we can: are the eigenvalues of such lifts positive?

We're going to talk about Siegel modular forms, which are defined on the Siegel upper half-space

$$\mathbb{H}_n := \left\{ Z \in M_n(\mathbb{C}),\, {}^tZ = Z,\, \Im(Z) > 0 \right\}.$$

The symplectic group $\mathrm{GSp}_{2n}(R) \leq \mathrm{GL}_{2n}(R)$ is given by the condition ${}^tgJg = \mu(g)J$, where

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

We set $\Gamma_n := \mathrm{Sp}_{2n}(\mathbb{Z})$. A Siegel modular form F of weight k over Γ_n is defined in the usual way. The space of such forms, $M_k(\Gamma_n)$, is finite-dimensional. Elements F admit a Fourier expansion

$$F(Z) = \sum_{\substack{T=T^t,\\ T\geq 0,\\ \text{half-integral}}} A(T) e^{2\pi i \,\mathrm{trace}(TZ)}.$$

We can define a notion of cusp form ($A(T) = 0$ unless $T > 0$) and Petersson inner product.

For each $g \in G(\mathbb{Q})^+ \cap M_{2n}(\mathbb{Z})$, there is an associated Hecke operator $T(g)$ on $M_k(\Gamma_n)$. The Hecke algebra, $\mathcal{H}_n$ is generated by $T(g)$.

- $\mathcal{H}_n$ is commutative.
- To each $m \in \mathbb{N}$ we may associate $T(m) \in \mathcal{H}_n$ such that $T(mn) = T(m)T(n)$ when $(m, n) = 1$.
- Each $T(g)$ is self-adjoint with respect to the Petersson inner product.

Andrianov 1973: basis of simultaneous eigenfunctions.

Eigenvalues may be given in terms of the Satake parameters. Let F be a Hecke eigenform and $\lambda_F(g)$ the eigenvalue corresponding to the operator $T(g)$. For any g with $\mu(g) = p^r$, depending on F, there are $n + 1$ complex numbers

$$\left(a_{0,p}^{(F)}, a_{1,p}^{(F)}, \ldots, a_{n,p}^{(F)} \right)$$

satisfying, with

$$\Gamma_n g \Gamma_n = \sqcup_i \Gamma_n g_i, \quad g_i = \begin{pmatrix} A_i & B_i \\ 0 & D_i \end{pmatrix}, \quad D_i = \begin{pmatrix} p^{d_{i1}} & \cdots & * \\ 0 & \cdots & * \\ 0 & \cdots & p^{d_{in}} \end{pmatrix},$$

we can explicitly describe the Hecke eigenvalue by

$$\lambda(g) = a_{0,p}^r \sum_i \prod_{j=1}^n (a_{j,p}^{-j})^{d_{ij}}.$$

.

Andrianov, 1974: if $F \in S_k(\Gamma_n)$ is a Hecke eigenform, then

$$\sum_{r=0}^{\infty} \lambda_F(p^r) p^{-rs} = \frac{P_{F,p}(p^{-s})}{Q_{F,p}(p^{-s})},$$

$$Q_{F,p}(p^{-s}) = \cdots .$$

Let $f \in S_k(\Gamma_1)$ be a Hecke eigenform.

Theorem 55 (Ikeda, 2001)**.** *Assume that $n \equiv k \pmod 2$. Then*

$$F_f(Z) = \sum_{B \in \mathcal{S}_{2n}(\mathbb{Z})^+} A(B) e^{2\pi i \operatorname{trace}(BZ)}, \quad Z \in \mathbb{H}_{2n}$$

is a Hecke eigenform in $S_{k+n}(\Gamma_{2n})$ and the standard L-function of F is equal to

$$\zeta(s) \prod_{i=1}^{2n} L(s+k+n-i, f, \text{spin}).$$

Here $S_{2n}(\mathbb{Z})^+$ is the set of all positive definite half-integral matrices of size $2n$.

To compute the spin L-function of the Ikeda lift, move to the language of automorphic representations. Let $F_f \in S_{k+n}(\Gamma_{2n})$ be the Ikeda lift of $f \in S_{2k}(\Gamma_1)$. Let Π and π be the associated automorphic representations.

Theorem 56 (Schmidt, 2003)**.** *We have*

$$L_p(s, \Pi, \text{spin}) = \prod_{j=0}^{n} \prod_{\substack{r=j(j-2n) \\ Step\ 2}}^{j(2n-j)} L_p(s + \tfrac{r}{2}, \pi, \text{Sym}^{n-j})^{\beta(r,j,n)}.$$

Here $\beta(r,j,n)$ is given explicitly.

Theorem 57 (A, 2024)**.** *For $n \equiv k \pmod 2$, let $F_f \in S_{k+n}(\Gamma_{2n})$ be the Ikeda lift of $f \in S_{2k}(\Gamma_1)$. For all large enough primes p, we have $\lambda_{F_f}(p) \geq 0$.*

Lemma 58. *If $F_f \in S_{k+2}(\Gamma_4)$ is the Ikeda lift of $f \in S_{2k}(\Gamma_1)$, then $\lambda_{F_f}(p)$ is positive for all primes p.*

For $\lambda_F(p^r)$ with $r > 1$, we use

$$\sum_{r=0}^{\infty} \lambda_F(p^r) p^{-rs} = \frac{P_{F,p}(p^{-s})}{Q_{F,p}(p^{-s})},$$

and we need knowledge of both numerator and denominator. The denominator can be computed using Schmidt's formluaf or the L-function, but the numerator is known only for genus at most 4.

Let $T(p^4)$ be Hecke operators in $\mathcal{H}_4$. Vankov (2011) found polynomials P_p and Q_p over the Hecke algebra such that

$$\sum_{r\geq 0} T(p^r)x^r = \frac{P_p(x)}{Q_p(x)}.$$

$P_{p,F}(x)$ can be retrieved from $P_p(x)$.
$\lambda_F(p^r)$ for genus 4 Ikeda lifts? We do the following.

- Express $\frac{1}{Q_{p,F}(p^{-s})}$ as partial fractions.
- Write each partial fraction as a power series in p^{-s}.
- Multiply by $P_{p,F}(p^{-s})$.
- Compare the coefficients of p^{-rs} on both sides.

This yields a formula for $\lambda_{F_f}(p^r)$ in terms of some $c_i \in \mathbb{Z}[a, a^{-1}]$ that are bounded, uniformly in r. This lead sto:

Theorem 59 (A, 2024)**.** *Let F_f be a Hecke eigenform in $S_{k+2}(\Gamma_4)$, the Ikeda lift of $f \in S_{2k}(\Gamma_1)$. FOr a fixed r, the number $\lambda_{F_f}(p^r)$ is positive for all large enough primes p.*

17. Abelian covers of P1 of p-ordinary Ekedahl–Oort type, *Deepesh Singhal*

Preprint: [18].

Fix a prime p. Let A be a g-dimensional principally polarized abelian variety defined over $\bar{\mathbb{F}}_p$. There are three invariants attached to A:

- p-rank: $f(A) = \dim_{\bar{\mathbb{F}}_p}(\mu_p, A)$. Note that
  - $\#A[p](\bar{\mathbb{F}}_p) = p^f$,
  - $0 \leq f \leq g$.
- Newton polygon: the isogeny class of the p-divisible group $A[p^\infty]$, which can be presented as a sequence of $2g$ rational numbers $NP(A) := (\nu_1, \ldots, \nu_{2g})$.
- Ekedahl–Oort type: the isomorphism class of the p-kernel $A[p]$, which can be presented as an element $w(A) \in \mathfrak{S}_{2g}$.

When $g = 1$, A is an elliptic curve:

- $f(A) = 0$ iff $NP(A) = (\frac{1}{2}, \frac{1}{2})$ iff $w(A) = \mathrm{id}$ iff A is supersingular.
- $f(A) = 1$ iff $NP(A) = (0, 1)$ iff $w(A) = (21)$ iff A is ordinary.

When $g \geq 2$, these are distinct invariants with some relation among each other:

- p-rank can be computed from either the Newton polygon or the EO-type.
- $f(A) = \#\{i : v_i = 0\}$.

One has the *Torelli morphism* from the moduli of curves to the moduli of abelian varieties:

$$\text{Hurwitz space} \xrightarrow{T} \text{Shimura variety}$$

$$C \xrightarrow{T} J(C).$$

- Fact: $\dim(\mathrm{im}(T)) < \dim(\text{ambient Shimura variety})$.
- For $S = \mathcal{A}_{g,1}$, it is known which p-rank, Newton polygons and EO-types give *nonempty* stratum.

Question 60. Which stratum in $\mathcal{S}$ has nonempty intersection with the Torelli locus?

We turn to the Hurwitz space of cyclic covers of $\mathbb{P}^1$. Fix m such that $p \nmid m$. $\mathcal{M}_{\mu_m}$ is the moduli space of μ_m covers of $\mathbb{P}^1$.

- $\mathcal{M}_{\mu_m} = \cup \mathcal{M}(m, r, \underline{a})$ is the decomposition of irreducible components.
- r is the number of branching points.
- $\underline{a} = (a(1), a(2), \ldots, a(r))$ is the monodromy data of each branching point.
- Looking at one of the irreducible components, we have have the restricted Torelli morphism

$$\mathcal{M}(m, r, \underline{a}) \xrightarrow{T} \mathrm{Sh}(H, \mu).$$

Here

- $\mathrm{Sh}(H, \mu)$ is the smallest PEL type Shimura variety containing the image of T, with μ determined by the data $\underline{a}$.
- The set of the Newton polygons occurring is known and denoted by $B(H_{\mathbb{Q}_p}, \mu)$. However, very little is known about which of these Newton polygons have a stratum that intersects the Torelli locus.
- The set $B(H_{\mathbb{Q}_p}, \mu)$ has a partial ordering, and $\nu(H, \mu)$ is the unique maximal element in this set. It is called the *μ-ordinary* Newton polygon.

Example 61. Consider the moduli space of μ_7-covers of $\mathbb{P}^1$, with $r = 4$ branching points, and monodromy datum $(3, 2, 1, 1)$. If $p \equiv 6 \pmod 7$, then there are three Newton polygons:

- the μ-ordinary $\mu = (0, 1)^4 \oplus (\frac{1}{2}, \frac{1}{2})^2$, with p-rank 4.
- $\nu = (0, 1)^2 \oplus (\frac{1}{2}, \frac{1}{2})^4$, with p-rank 2.
- the basic $\beta = (\frac{1}{2}, \frac{1}{2})^6$, with p-rank 0.

We refine Bouw's result and show the sharpness of the upper bound on Newton polygon given by $B(\mathbb{H}_{\mathbb{Q}_p}, \mu)$:

Theorem 62 (Y. Lin, E. Mantovan, S., 2023)**.** *Fix a prime p and let $m \in \mathbb{N}$ such that $p \nmid m$ and $p > m(r-2)$. Then for any $\mathcal{M}(m, r, \underline{a})$ such that $r \leq 5$, we have*

$$T(\mathcal{M}(G, r, \underline{a}))[\nu(H, \mu)] \neq \emptyset.$$

In addition, we also show (...).

Comments:

- Same conclusion holds for abelian covers.
- The statement of the theorem likely remains true without the assumptions that $p > m(r-2)$ and $r \leq 5$. However, these conditions are needed for our technique to work.
- (...)

Proof strategy:

Theorem 63 (Moonen, 2004)**.** *For a PEL-type Shimura variety $\mathrm{Sh}(H, \mu)$, we have*

$$\nu(H, \mu) \textit{ stratum } = \textit{ maximal EO stratum } \subseteq \textit{ maximal } p\textit{-rank stratum.}$$

To prove the non-emptiness of the intersection with maximal EO stratum, we use the equivalence of categories between

- p-kernel of polarized abelian variety,
- polarized mod p Dieudonné module,
- Hasse–Witt triples.

We translate the condition of having maximal EO type to linear algebra conditions on the Hasse–Witt triple. By explicitly computing them, we are able to verify the rank conditions.

18. Hyperbolic Counting Problems, *Marius Voskou*

Preprint: [23].

We start with an interlude on hyperbolic geometry. Work with $\mathrm{PSL}_2(\mathbb{R})$ (orientation-preserving isometries), lattice points $\gamma \cdot z$, where $\gamma \in \Gamma$, a discrete cofinite subgroup of $\mathrm{PSL}_2(\mathbb{R})$, and $z \in \mathbb{H}$.

Consider a hyperbolic circle of center w, radius R.

This looks like a Euclidean circle; the only difference is that the center is slightly lower than what you'd expect, which has to do with the fact that distances blow up as you get closer to the x-axis.

We're now ready to formulate our first hyperbolic counting problem: take

$$N(z, w, R) := \#\left\{\gamma z : \gamma \in \Gamma, \mathrm{dist}(w, \gamma z) \leq R\right\}.$$

The same method as in the Euclidean counting problem does, in principle work. We get a main term comparable to the area, an error term comparable to the circumference. The problem is that in hyperbolic geometry, these are comparable – both are comparable to e^R – so the error term is not smaller than the main term. Instead, we use the spectral theory of automorphic forms.

To that end, the first step is to write our count in terms of the automorphic kernel:

$$N(z, w, R) = K(z, w) = \sum_{\gamma} \mathbf{1}(\mathrm{dist}(w, \gamma z) < R).$$

We know that automorphic kernels have nice spectral expansions in terms of eigenfunctions of the hyperbolic Laplacian. Since it's a 20 minute talk, we'll just take spectral theory as a black box and not say much about it. The thing we want to note is that the hyperbolic Laplacian has two kinds of eigenvalues:

- large eigenvalues: $\lambda \geq 1/4$, $\lambda = s(1-s), s = 1/2 + it$.
- small eigenvalues: $\lambda < 1/4, \lambda = s(1-s), s \in (1/2, 1]$, of which there are finitely many.

Using this spectral expansion, Selberg in the 70's proved an asymptotic formula for $N(z, w, R)$:

Theorem 64 (Selberg 1970, Gunther 1980, Good 1983)**.** *We have*

$$N(z, w, \log X) = \frac{2\pi}{\mathrm{area}(\mathbb{H}/\Gamma)} X + \sum_{s_j} c_j(z, w) X^{s_j} + \mathrm{O}(X^{2/3}),$$

where the sum is over finitely many real numbers $s_j \in (1/2, 1)$, corresponding to the small eigenvalues.

Now, let's move to a slightly more complicated problem: let's ask what happens if instead of points, we have geodesics. For fixed (closed) geodesics ℓ_1, ℓ_2, how many geodesics $\gamma \cdot \ell_1$ have distance $\leq R$ from ℓ_2? For simplicity, take $\ell_1 = \ell_2 = \ell$.

[Picture of a pair of closed geodeics intersecting.]

The distance between closed geodesics is the length of the common perpendicular. We want to count how many closed geodesics are at given distance from one closed geodesic.

Our strategy is again to use the spectral theory of automorphic forms. We can write the count as the integral of an automorphic kernel:

$$N(\ell_1, \ell_2, R) = \int_{\ell_1} \int_{\ell_2} K(z, w)\, ds(z)\, ds(w).$$

This leads to an asymptotic formula:

Theorem 65 (Good 1983, Lekkas–Petridis 2024)**.** *We have*

$$N(\ell, \log X) = \frac{2(\operatorname{len} \ell)^2}{\pi \operatorname{area}(\mathbb{H}/\Gamma)} X + \sum_{s_j} (\ell) X^{s_j} + E(X),$$

where $E(X) = \mathrm{O}(X^{2/3})$.

Let's say a few words about why it's necessary to have a second proof of this. The first proof by Good was hard to read, and some people still question whether it was valid. In particular, if you want to do refinements of this or apply his methods to obtain related results, it's difficult. Petridis–Lekkas found simpler methods that also apply to other problems.

Conjecture 66. *We have* $E(X) = \mathrm{O}(X^{1/2+\varepsilon})$. *(And similarly for the previous problem, and many similar problems.)*

This has been around for many decades, and we have zero improvements on $X^{2/3}$.

What we're going to try to do now is to motivate this conjecture. To that end, we'll start by saying why we can't do any better:

Theorem 67 (V. 2024)**.** *We have*

$$E(X) = \Omega\left(X^{1/2} (\log\log X)^{1/4-\delta}\right).$$

Theorem 68 (Lekkas–Petridis 2024)**.** *We have*

$$\left(\frac{1}{X} \int_X^{2X} |E(u)|^2\, du\right)^{1/2} \ll X^{1/2} \log X.$$

Now, let's restrict to the case of a primitive hyperbolic element

$$h = \begin{pmatrix} m & 0 \\ 0 & m^{-1} \end{pmatrix}.$$

For $\ell = \mathcal{I}/\langle h \rangle$, we have

$$\cosh \operatorname{dist}(\ell, \gamma\ell) = \max\left(|ad + bc|, 1\right),$$

so our counting problem can be rephrased as follows: how many double cosets $\gamma \in \langle h \rangle \backslash \Gamma / \langle h \rangle$ with $|ad + bc| < X$? We can also ask, are there more γ with $ad + bc > 0$ than $ad + bc < 0$, or less? Equivalently, are there more a, d having the same sign than different, or less? Geometric interpretation: the sign corresponds to the direction of the image $\gamma \cdot \ell$, and whether it lies on the left or the right of ℓ. This behaves differently as we tilde ℓ to some ℓ_θ. Idea: consider

$$\frac{\partial^2}{\partial\theta \partial\phi} N(\ell_\theta, \ell_\phi, R).$$

Let's finish with a nice arithmetic application. Fix a prime p. For Γ an appropriate quaternion order, and $m = \left(1 + \sqrt{2}\right)^2$:

Theorem 69 (V. 2023, $p = 5$; Hejhal 1982)**.** *We have*

$$\sum_{n\le X} \mathcal{N}(n)\mathcal{N}(pn \pm 1) = \frac{p}{p + \left(\frac{2}{p}\right)} \cdot \left(\frac{\log m}{\pi}\right)^2 X + \sum_{1/2<s_j<1} a_j^{\pm} X^{s_j} + \mathrm{O}(X^{2/3}),$$

where $a_j^{\pm}$ is real and $\mathcal{N}(n)$ is the number of $\mathfrak{a} \le \mathbb{Z}[\sqrt{2}]$ with $\|(\|\mathfrak{a}) = n$.

That's all.

19. False indefinite theta functions and partitions separated by parity, *Kathrin Bringmann*

Today we'll first say a bit about modular forms and partitions, but we'll probably skip most of that, since we've already seen it. In particular, we need the Dedekind eta function $\eta(q) = q^{1/24}\prod(1-q^n)$ and the theta function $\Theta(q) = \sum q^{n^2}$, together with the partition generating function $P(q) = \sum p(n)q^n$. By work of Hardy and Ramanujan who developed the circle method, we have asymptotics for $p(n)$, which Rademacher improved to an exact formula: letting

$$A_k(n) := \sum_{h(k)^*} \omega_{h,k} e^{-\frac{2\pi i n h}{k}},$$

and the Bessel function I_α, Rademacher obtained

$$p(n) = \frac{2\pi}{(24n-1)^{3/4}} \sum_{k\ge 1} \frac{A_k(n)}{n} I_{3/2}\left(\frac{4\pi\sqrt{n}}{k}\right).$$

Goal: determine asymptotic behavior of $a(n)$ as $n \to \infty$.

Let $A(q) := \sum_{n\ge 0} a(n)q^n$ denote the generating function. Then we can recover $a(n)$ via Cauchy. The Hardy–Ramanujan asymptotics are obtained using

$$I_\kappa(x) \sim \frac{e^x}{\sqrt{2\pi x}}.$$

Rademacher–Zuckerman obtained exact formulas for Fourier coefficients of modular forms of negative weights, and Bringmann–Ono addressed the case of weight zero.

Ramanujan's lost notebook introduced the q-hypergeometric series $\sigma(q)$, with coefficients $S(n)$.

Theorem 70 (Andrews–Dyson–Hickerson)**.** *We have $S(n) = T(24n+1)$, where $T(m)$ is the number of solutions to a certain Pell's equation.*

Definition 71. We say that a q-series is *lacunary* if its coefficients are almost always zero.

Corollary 72. *$\sigma(q)$ is lacunary. $S(n)$ obtains every integer infinitely often.*

Key identity: an expression for $\sigma(q)$ as a false indefinite theta function (not quite a modular form).

Now, what are the partition statistics that we're interested in? We're interested in partitions separated by parity: even parts larger than odd parts (or reversed). You can also require that parts be or not be distinct,. In particular, we'r einterested in $p_{od}^{eu}(n)$, the number of partitions with odd parts smaller than even parts and odd parts distinct. For instance, for $n = 5$, we get 5, $4+1$, $2+2+1$, so $p_{od}^{eu}(5) = 3$.

The associated generating function $F_{od}^{eu}(q)$ is related to σ via

$$\frac{1}{(q^2,q^2)_\infty}\left(1-\cdots\right),$$

where σ is a simple expression involving σ.

Theorem 73 (B–Craig–Nazaroglu)**.** *As $n\to\infty$, we have*

$$\sim \frac{e^{\pi\sqrt{n/3}}}{2\sqrt{3}n}.$$

Proof. Use that coefficients are monotonic, Tauberian theorem, Euler–Maclaurin. □

Let's now tweak the combinatorics a little bit: take $p_{eu}^{ou}(n)$ denote the number of partitions with even parts smaller than odd parts, then the corresponding generating function is

$$F_{eu}^{ou}(q)=\frac{1}{(1-q)(q^2;q^2)_\infty}.$$

We can also define p_{eu}^{od}, whose generating function is

$$\frac{1}{(q^2;q^2)_\infty}\sum_{n\geq 0}q^{n^2}.$$

Next, we look at p_{ed}^{ou} (even parts smaller than odd parts, parts distinct), where the generating function is

$$\frac{1}{2(q;q^2)_\infty}\left((q;-q)_\infty+1-\sum_{n\geq 0}(1-(-1)^nq^n)(-1)^{\cdots}q^{\cdots}\right).$$

This isn't modular, but you can improve it to obtain some sort of modular by introducing, for $z\in\mathbb{C}$ and $\tau\in\mathbb{H}$,

$$\psi(z,\tau):=i\sum_{n\in\mathbb{Z}}\operatorname{sgn}\left(n+\tfrac{1}{2}\right)(-1)^n\zeta^{n+1/2}q^{\frac{1}{2}(n+\frac{1}{2})^2}.$$

We can recover this as the limit as $t\to\infty$ of a $\hat{\psi}(z;\tau,\tau+it)$, where:

Theorem 74 (B.–Nazaroglu)**.** *$\hat{\psi}$ has modular transformation properties.*

Next, we want to look at p_{ed}^{od}: partitions where even parts are smaller than odd parts, odd parts distinct. The generating function is

$$\frac{(-q;q^2)_\infty}{1-q}-\frac{q(-q^2;q^2)_\infty}{1-q},$$

which is just some modular object, so you can use the circle method.

Similarly for a related case.

Next, let's look at p_{ou}^{ed}: odd parts smaller than even parts. The generating function involves Ramanujan's mock theta function $f(q)$, which Zwegers showed how to understand (by completing by adding a second piece, given by an integral of a theta function of weight 3/2).

Harmonic Maass forms are $\hat{f}$ that transform like a modular form of weight $1/2$, annihilated by hyperbolic Laplacian of weight $1/2$. B.–Mahlburg developed a theory of coefficients of mock modular forms multiplied by a modular form. B.–Craig–Nazaroglu obtained leading asymptotics for all coefficients; can one obtain refinements for all of them?

σ is not enough; one needs also its companion

$$\sigma^*(q) = \sum_{n\geq 1} \frac{(-1)^n q^{n^2}}{(q;q^2)_n}.$$

Let $a(n)$ its coefficients.

Theorem 75 (Cohen)**.** *These are the coefficients of a Maass form.*

Definition 76 (Zagier)**.** $f : \mathbb{Q} \to \mathbb{C}$ is a quantum modular form if $h_\gamma(x) := f(x) - f|_k\gamma(x)$ is nice on some nice subsets of $\mathbb{R}$.

For q a root of unity, we have $\sigma(q) = -\sigma^*(q^{-1})$. We define f in terms of σ and σ^*, and also a function $R_\tau(z)$, and a measure defined by a bracket. Then we can concisely express the transformation law of f in terms of these definitions. In particular, we deduce that f is a quantum modular form. This leads to improved asymptotics for the coefficients. First, we split off components that we understand. Let $r_o(n)$ denote the number of partitions of n into distinct odd parts. Its generating function is $\prod_{n\geq 0}(1+q^{2n+1})$. We can decompose into even and odd parts.

Our main theorem:

Theorem 77 (B.–Craig–Nazaroglu)**.** *Very good asymptotics for $\alpha_j(n)$.*

Work in progress: recover determine asymptotic expansion.

Idea of proof: use Maass forms. Build a vector-valued form out of σ. Circle method.

20. On the local functional equation for archimedean exterior square L-function of GL_n, *Ravi Raghunathan*

The purpose of the talk is to complete the archimedean theory of the exterior square L-function of GL_r via the theory of integral representations due to Jacquet and Shalika.

- F: local, characteristic zero
- $\psi_F : F \to \mathbb{C}$, nontrivial
- π: irreducible admissible representation of $G_r := \mathrm{GL}_r(F)$.

Focus on the case F archimedean, since the results are new only in that case.

The proofs are usually simpler when r is odd, so we'll focus on the case $r = 2n$.

The basic idea is as in Tate's thesis. To each irreducible admissible representation π of G_r, the local Langlands correspondence attaches an n-dimensional representation $\rho(\pi)$ of the Weil group (resp. Weil–Deligne group) when F is archimedean (resp. p-adic). For a character $\chi : F^\times \to \mathbb{C}$, the twisted exterior square L-function and ε-factor of π are defined via the local Langlands correspondence:

$$L(s,\pi,\wedge^2,\chi) := L(s,\wedge^2(\rho(\pi))\otimes\chi),$$
$$\varepsilon(s,\pi,\wedge^2,\chi) := \varepsilon(s,\wedge^2(\rho(\pi))\otimes\chi,\psi_F).$$

The right hand sides are the Artin L-function and ε-factor. When the character χ is trivial, we drop it from the notation.

Example 78. If π_p is an unramified representation of $G_n(\mathbb{Q}_p)$, then $\rho(\pi) = \chi_1 \oplus \cdots \oplus \chi_n$, $\chi_i = |.|_p^{\lambda_i}$, then with $\alpha_i(p) = p^{-\lambda_i}$, we have

$$L_G(s, \pi_p, \wedge^2) = \prod_{i<j} \frac{1}{(1 - \alpha_i(p)\alpha_j(p)p^{-s})}.$$

In fact, the above formula is valid (even for finite extensions of $\mathbb{Q}_p$) as long as $\chi_i\chi_j$ is ramified, whenever χ_i and χ_j are both ramified.

If F is p-adic, we denote by $\mathfrak{o}_F$ the ring of integers, and q the cardinality of the residue field. We let N_r denote the standard maximal unipotent subgroup, and Z_r the center of G_r. Let M_r denote the space of all $r \times r$ matrices, and V_r the subspace of upper-triangular $r \times r$ matrices.

We set

$$w_n = \begin{pmatrix} & & 1 \\ & \cdots & \\ 1 & & \end{pmatrix}, \quad w_{n,n} := \begin{pmatrix} 0 & 1_n \\ 1_n & 0 \end{pmatrix}.$$

We view these as elements of G_n or G_r, respectively.

Okay, one more thing. We will assume our representations are generic, i.e., have a Whittaker model, which means you can embed the representation into a space of functions on G. We extend ψ_F to a character of N_r by setting $\psi(u) := \prod_{j=1}^{r-1} \psi_F(u_{j,j+1})$ for $u \in N_r$. We will assume that ψ is an irreducible generic representation of G_n with respect to ψ as above. We denote the Harish–Chandra Schwartz space of Whittaker functions on π by $\mathcal{W}(\pi, \psi)$, and refer to it in this talk simply as the Whittaker model of π.

When F is archimedean, the space $\mathcal{W}(\pi, \psi)$ consists of smooth functions satisfying a moderate growth condition, transformation property with respect to N_r and ψ, and is a Frechét space.

We turn to the integral representation of Jacquet and Shalika. For each $W \in \mathcal{W}(\pi, \psi)$, we define the function $\tilde{W}$ on G_r by

$$\tilde{W}(g) := W(w_n {}^\iota g), \quad {}^\iota g := {}^t g^{-1}.$$

The Whittaker model of $\tilde{\pi}_v$, $\mathcal{W}(\tilde{\pi}, \bar{\psi})$, consists precisely of the set of functions $\{\tilde{W} : W \in \mathcal{W}(\pi, \psi)\}$. Let $\mathcal{S}(F^n)$ be the space of Schwartz functions. Use self-dual Haar measures. Then the integral is

$$J(s, W, \phi) = \int_{N_n \backslash G_n} \int_{V_n \backslash M_n} W\left(\sigma \begin{pmatrix} 1_n & X \\ 0 & 1_n \end{pmatrix} \begin{pmatrix} g & 0 \\ 0 & g \end{pmatrix}\right) \psi(-\operatorname{trace} X)\, dX \phi(e_n g) |\det g|^s\, dg$$

for $W \in \mathcal{W}(\pi, \psi)$ and $\phi \in \mathcal{S}(F^n)$ and $s \in \mathbb{C}$ and σ the permutation matrix given by

$$\begin{pmatrix} 1 & 2 & \cdots & n & | & n+1 & n+2 & \cdots & 2n \\ 1 & 3 & \cdots & 2n-1 & | & 2 & 4 & \cdots & 2n \end{pmatrix}$$

Proposition 79 (Jacquet–Shalika 1990)**.** *(1) For a generic irreducible representation π of G_n, the integrals $J(s, W, \phi)$ converge absolutely for $\Re(s) > 1 - \eta$ for some $\eta > 0$.*

(2) The integrals admit meromorphic continuation.

(3) The integrals define elements of $\mathbb{C}(q^{-s})$.

We can thus take the fractional ideal of $\mathbb{C}[q^s, q^{-s}]$ generated by these integrals. The generator can be taken (by results of Belt) to be of the form $1/P(q^{-s})$. This

is then defined to be the Jacquet–Shalika exterior square L-value. Are these the same as the Artin L-functions? Yes:

Theorem 80 (Kewat–R, Y. Jo)**.** *Let F be a p-adic field. Then with π as above,*

$$L_{JS}(s, \pi, \wedge^2) = L(s, \pi, \wedge^2).$$

We now want to do the same thing in the archimedean setting, where less is known. Implicit in Jacquet–Shalika is that these integrals admit meromorphic continuation.

First main result:

Theorem 81. *For $r = 2n$, the quotients*

$$\Xi(s, W, \phi) := \frac{J(s, W, \phi)}{L(s, \pi, \wedge^2)}$$

are entire and of finite order in vertical strips. Moreover, if π is "nearly tempered", then for every $s_0 \in \mathbb{C}$, we can choose $W \in \mathcal{W}(\pi, \psi)$ and $\phi \in \mathcal{S}(F^n)$ such that $\Xi(s_0, W, \phi) \neq 0$.

Here "nearly tempered" means that every pole of the L-function is realized by one of the integrals, so that the integrals really do capture the L-function more-or-less completely.

Second main result is the local functional equation:

Theorem 82. *With conditions as above, we have*

$$\frac{J(1-s, \rho(w_{n,n})\tilde{W}, \hat{\phi})}{L(1-s, \tilde{\pi}, \wedge^2)} = \varepsilon_{JS}(s, \pi, \wedge^2, \psi) \frac{J(s, W, \phi)}{L(s, \pi, \wedge^2)},$$

where ε_{JS} is of the form $A_0 B_0^s$.

For F p-adic this was proved by Matringe, and in the odd case by Cogdell–???.

The main new tool we will use in the proofs of our results is the following very nice observation of Beuzart–Plessis (2021), which is deduced from Theorem 2 of Finis–Lapid–Muller (2015).

Theorem 83 (Beuzart–Plessis)**.** *Let k be a number field and let w_0 and w_1 be places of k with w_1 non-archimedean. Let V be any open subset of $\mathrm{Temp}(\mathrm{GL}_n(k_{w_0}))$. There exists a cuspidal automorphic representation $\Pi = \otimes'_w \Pi_v$ of $\mathrm{GL}_n(\mathbb{A}_k)$ such that $\Pi_{w_0} \in V$ and Π_w is unramified for every finite place $w \neq w_1$.*

The nice new part of this theorem is that you can embed archimedean representations as well. Moreover, there is this density that happens – the moment you have an open set, you can find something there that can be pushed inside a global representation. Once you have this, you're in great shape, because we were already (in Kewat–R) able to do this whenever we could push our representations into global ones. Indeed, Beuzart–Plessis used the above theorem to prove analogues of our main results (Theorems 81 and 80) for the Asai L-function.

Let's say a bit about the proof. Directly imitating Beuzart–Plessis allows us to prove a functional equation, but we don't' know that the ε-factor that we get matches the Artin ε-factor. However, using the embedding theorem for different pairs of places (three times) allows us to show that the two ε-factors are in fact the same. What remains to show:

- Nonvanishing of $\Xi(s_0, W_v, \phi_v)$ in general.

- Showing that there exist finitely many Whittaker and Schwartz functions that realize the L-factor.

Humphries and Jo [15] prove the second assertion up to a polynomial factor, using a single test vector.

21. Traces of Poincare series at square discriminants and Fourier coefficients of mock modular forms, *Vaibhav Kalia*

(Joint work with PhD supervisor Balesh Kumar.)
[Oops, started late]
$\varepsilon_\delta = 1$ if $\delta \equiv 1(4)$, and $\varepsilon_d = i$ otherwise.
Fix $d, D \equiv 0, 1(4)$; call them *discriminants* if they are nonzero.
Fix $\tau \in \mathbb{H}$.
Let $k \in \frac{1}{2}\mathbb{Z}$. Assume $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ if $k \in \mathbb{Z}$, else $\Gamma = \Gamma_0(4)$.
We define the slash operator and Kohnen's space $M_k^{!,+}$.
Let j be Klein's j-function with q-expansion

$$j(\tau) = q^{-1} + 744 + 196884q + \cdots .$$

Zagier 2002: we can give a basis for $M_{3/2}^{!,+}$ as $\mathcal{G}_d$, indexed by $0 < d \equiv 0, 1(4)$, where

$$\mathcal{G}_d(\tau) = q^{-d} + \sum_{0>D\equiv 0,1(4)} a(d, D)q^{-D},$$

where

$$a(d, D) = \mathrm{Tr}_{D,d}(j - 744) := \frac{1}{\sqrt{d}} \sum_{Q\in \mathrm{PSL}_2(\mathbb{Z})\backslash \mathcal{Q}_{dD}} \frac{\chi_d(Q)}{|\Gamma_Q|}(j - 744)(\tau_Q).$$

Further, if for $0 > d \equiv 0, 1(4)$ we define

$$\mathcal{F}_d(\tau) := q^d + \sum_{0<D\equiv 0,1(4)} a(D, d)q^D,$$

then Borcherds (1995) basis for $M_{1/2}^{!,+}$ is given by $\mathcal{F}_d$ for $0 > d \equiv 0, 1(4)$.

Question 84. Are there analogues of the functions $\mathrm{Tr}_{D,d}(\bullet)$ when $dD > 0$ for weight zero functions? Are they related to any famiy of modular forms?

Answer 1. Yes: harmonic weak Maass forms and mock modular forms.

We recall the definition of a weight k harmonic weak Maass form on Γ. It's any smooth function $f : \mathbb{H} \to \mathbb{C}$ that satisfies $f|_k A(\tau) = f(\tau)$, annihilated by Δ_k (hyperbolic Laplacian), and has an exponential growth condition at all cusps. We denote by $\mathcal{H}_k^!$ the space of such weight k harmonic Maass forms on Γ.

By a *sesqui* harmonic Maasss form of weight k, we mean the same thing, but without the second condition concerning Δ_k, and instead $\xi_k \circ \Delta_k(f) = 0$, where ξ_k is as in the definition of the shadow.

The Fourier expansion of $f \in \mathcal{H}_k^!$ at ∞ with $k \neq 1$ is given by

$$f(\tau) = \sum_{\substack{n\in\mathbb{Z}\\ n\gg -\infty}} c_f^+(n)q^n + c_f^-(0)y^{1-k} + \sum_{\substack{n\in\mathbb{Z}-\{0\}\\ n\ll\infty}} c_f^-(n)W_k(2\pi ny)q^n.$$

Following Zagier (2007), a *mock modular form* of weight k for Γ is the holomorphic part f^+ of $f \in \mathcal{H}_k^!$. It turns out that $\xi_k(f)$ is a weakly holmorphic of weight $2-k$, called the *shadow* of f.

Duke–Imamoglu–Toth (2011) constructed a basis $\mathcal{F}_d$, indexed by $d<0$, for mock modular forms, where

$$\mathcal{F}_d = \sum_{D>0} a(D,d) q^D$$

with $a(d,D)$ defined in terms of the cycle integrals

$$(2\pi)^{-1} \sum_{Q \in \Gamma \backslash \mathcal{Q}_{dD}} \chi_d(Q) \int_{C_Q} (j-744)(\tau) \frac{d\tau}{Q(\tau,1)}$$

for $d, d>0$ with $dD \neq \square$.

To interpret $a(D,d)$ when $dD = \square$, appeal to BFI2015 and Andersen2015.

JKK2013–2014: basis $\mathcal{G}_d$, indexed by $d>0$, with coefficients $b(d,D)$ given in terms of twisted traces of cycle integrals of $\hat{\mathbb{J}}_1(\tau)$, where $\Delta_0(\hat{\mathbb{J}}_1) = -j+720$ (done in the case where $d, D<0$ and $dD \neq \square$).

What should $b(d,D)$ be when $d, D<0$ and $dD = \square$? We'll see in a moment.

We'll need the notion of Niebur Poincaré series (1973), which we've already seen, defined for $\Re(s)>1$ by

$$G_{-1}(\tau, s) = \sum_{A \in \Gamma_\infty \backslash \Gamma} e\left(-\Re(A\tau)\right) \phi_{-1,s}\left(\Im(A\tau)\right).$$

Here

$$\phi_{-1,s}(y) := 2\pi y^{1/2} I_{s-1/2}(2\pi y).$$

It defines a smooth weight zero function for Γ.

The traces of cycle integrals of these Niebur Poincaré series we'll discuss now.

- If $0<d, D$ with D fundamental and $dD \neq \square$, we have (DIT 2011)

$$\mathrm{Tr}_{d,D}(G_{-1}(\tau,s)) := \frac{1}{2\pi} \sum_{Q \in \Gamma \backslash \mathcal{Q}_{dD}} \chi_D(Q) \int_{C_Q} G_{-1}(\tau,s) \frac{d\tau}{Q(\tau,1)}.$$

- With $d, D<0$ and D fundamental and $dD \neq \square$, due to $\chi_D(-Q) = \mathrm{sgn}(D)\chi_D(Q)$, we can define

$$\mathrm{Tr}_{d,D}(G_{-1}(\tau,s)) = \frac{1}{\pi} \sum_{Q \in \Gamma_\infty \backslash \mathcal{Q}_{dD}^+} \chi_D(Q) \int_{S_Q} e(-\Re(\tau)) \phi_{-1,s}(\Im(\tau)) \frac{d\tau}{Q(\tau,1)}.$$

- Superscripted $+$ means $a>0$.

If $0<d$, $D=\square$ and D fundamental, then the above trace is not defined. Andersen 2015:

$$\mathrm{Tr}_{d,D}(G_{-1}(\tau,s)) := \frac{1}{2\pi} \sum_{Q \in \Gamma \backslash \mathcal{Q}_{dD}} \chi_D(Q) \int_{C_Q} G_{-1,Q}(\cdots).$$

ETC.

Modified traces:

$$\mathrm{Tr}_{d,D}(G_{-1}(\tau,s)) = \cdots,$$

$$\tilde{\mathrm{Tr}}_{d,D}(G_{-1}(\tau,s)) := \frac{1}{\pi} \sum_{Q \in \Gamma_\infty \backslash \mathcal{Q}_{dD}^+} \chi_D(Q) \int_{C_Q} e(-\Re(\tau)) \phi_{-1,s}(\Im\tau) \frac{d\tau}{Q(\tau,1)}.$$

JKK2014: $\hat{\mathbb{J}}_1(\tau,s) := \partial_s G_{-1}(\tau,s)$ and proved that $\hat{\mathbb{J}}_1(\tau) := \hat{\mathbb{J}}_1(\tau,1)$ is a sesquiharmonic Maass form of weight zero for Γ. We define

$$\tilde{\mathrm{Tr}}_{d,D}(\hat{\mathbb{J}}_1(\tau)) := \partial\P\tilde{\mathrm{Tr}}_{d,D}(G_{-1}(\tau,s))|_{s=1}$$

when $d, D < 0$ with $dD = \square$.

Theorem 85 (-, Kumar 2024)**.** *For d, D negative with D fundamental and $dD = \square$, we have*

$$b(d,D) = -8\sqrt{dD}\tilde{\mathrm{Tr}}_{d,D}(\cdots),$$

and also ETC.

Idea of the proof:

- $\tilde{\mathrm{Tr}}_{d,D}(G_{-1}(\tau,s)) = \alpha(s,dD)\sum_{c=1}^{\infty}\frac{S(d,D,4c)}{c^{1/2}}J_{s-1/2}\left(\frac{\pi\sqrt{dD}}{c}\right)$, where

$$S(d,D,4a) := \sum_{\substack{b(4a)\\(b^2-dD)/4a\in\mathbb{Z}}} \chi_D\left(\left[a,b,\frac{b^2-dD}{4a}\right]\right)e\left(\frac{b}{2a}\right).$$

- $\tilde{\mathrm{Tr}}_{d,D}(\hat{\mathbb{J}}_1\cdots)$, etc.

We can further relate these traces to the regularized inner product. Let $f, g \in M^{!,+}_{1/2}$ (resp. $M^!_0$). There exists $G \in \mathcal{H}^{!,+}_{3/2}$ (resp. $\mathcal{H}^!_2$) such that $\xi_-(G) = g$ (BDS2017). Then

$$\langle f, g\rangle = \rho_k \sum_{n\in\mathbb{Z}} c_f(n)\cdots.$$

ETC. Relate to Borcherds.

Idea of the proof:

- Since mock modular forms $\mathcal{G}_d$ with shadow $\mathcal{F}_d$ has

$$b(d,D) = 4\sqrt{dD}c_{|d|}(|D|) + 192\pi H(|d|)H(|D|),$$

we get by a computation

$$\langle \mathcal{F}_d, \mathcal{F}_D\rangle = 2\sqrt{|Dd|}c_{|d|}(|D|) + 96\pi H(\cdots),$$

we get ETC.

Next:

Corollary 86. *Let D be a negative fundamental discriminant. Then we have*

$$\tilde{\mathrm{Tr}}_{D,D}(\hat{\mathbb{J}}_1(\tau)) = -\frac{1}{4\pi\sqrt{|D|}}\sum_{n>0}\left(\frac{D}{n}\right)\frac{\langle j_n + 24\sigma(n), j_1 + 24\rangle}{n}e^{-2\pi n/|D|} - \cdots.$$

Idea of the proof: ANS2021 computed inner products of $\mathcal{F}_D$ in terms of exponential integrals, and then explicitly computed $\tilde{J}$, defined such that $\xi_2(\tilde{J}) = j_1 = j - 744$, to be given by $4\pi h_1 - 8\pi E_2^*$, where $h_1 \in H^!_2$ (DIT2016) satisfies $\xi_2(4\pi h_1) = j_1 + 24$, and

$$E_2^*(\tau) := 1 - 24\sum_{n\geq 1}\sigma(n)q^n.$$

We can thus write this inner product in terms of the Fourier coefficients of the harmonic Maass forms h_1. We then just need to use that $\xi_2(4\pi h_1) = j_1 + 24$ to write $c^-_{4\pi h_1}(n) = 4\pi n c_{j_1+24}(-n)$, then similarly for $+$.

As final remarks:

- For $n<0$, the Fourier coefficients $c_{j_1+24}(-n)$ of j_1+24 is the Rademacher–Petersson formula
$$\frac{2\pi}{\sqrt{-n}}\sum_{c>0}\frac{K(-1,-n,c)}{c}I_1\left(\frac{4\pi\sqrt{-n}}{c}\right).$$
- The regularized inner product $\langle j_n+24\sigma(n), j_1+24\rangle$ can also be interpreted in terms of the Rademacher–ETC.

22. Limiting behaviour and modular completions of MacMahon-like q-series, *Badri Vishal Pandey*

Preprint: [4].

Definition 87 (MacMahon's q-series)**.**
$$\mathcal{A}_a(q):=\sum_{1\leq n_1<\cdots<n_a}\frac{q^{n_1+\cdots+n_a}}{(1-q^{n_1})^{a_1}\cdots}.$$

Example 88.
$$\mathcal{A}_1(q)=\sum_{n\geq 1}\frac{q^n}{(1-q^n)^2}=\frac{1}{24}-\frac{1}{24}E_2(\tau).$$
This is a quasimodular form.

Motivation of our work is the following theorem:

Theorem 89 (Amdeberhan–Ono–Singh)**.** *Let $a\in\mathbb{N}$. We have*
$$q^{-\frac{a(a+1)}{2}}\mathcal{A}_a(q)=\prod_{n\geq 1}\frac{1}{(1-q^n)^3}+\mathrm{O}(\cdots).$$

Example 90. We have
$$\prod_{n\geq 1}\frac{1}{(1-q^n)^3}=1+3q+9q^2+22q^4+\cdots,$$
and these agree to increasing order with $q^{-1}\mathcal{A}_1$, $q^{-3}\mathcal{A}_2$, $q^{-6}\mathcal{A}_3$, $q^{-10}\mathcal{A}_4$, $q^{-15}\mathcal{A}_5$, $q^{-21}\mathcal{A}_6$, etc.

Let $p_k(n)$ count the partitions of n with parts having k-colors. Its generating function is
$$\sum_{n\geq 0}p_k(n)q^n=\prod_{n\geq 1}\frac{1}{(1-q^n)^k}.$$

Question 91 (Ono)**.** Find a family of functions that approximate k-colored partitions.

Definition 92. For $a,r,s\in\mathbb{N}$ and $k\in\mathbb{Z}$, define the q-series
$$\mathcal{A}_{a,k,r}(q):=\sum_{1\leq n_1<\cdots<n_a}\frac{q^{r(n_1+\cdots+n_a)}}{(1-q^{n_1})^k\cdots}=:\sum_{n\geq 0}c_{a,k,r}(n)q^n,$$
$$\mathcal{B}_{a,k,r,s}(q)=\sum_{\cdots}\frac{q^{r(n_1^2+\cdots+n_a^2)+s(n_1+\cdots+n_a)}}{\cdots}=:\sum_{n\geq 0}d_{a,k,r,s}(n)q^n.$$

Theorem 93 (Bringmann, Craig, van-Ittersum, P.). *Let $a, r, s \in \mathbb{N}$ and $k \in \mathbb{Z}$. We have*

$$q^{-\frac{ra(a+1)}{2}}\mathcal{A}_{a,k,r}(q) = \prod_{n\geq 1}\frac{1}{(1-q^{rn})(1-q^n)^k} + \mathrm{O}(q^{a+1}),$$

and similarly for $\mathcal{B}$.

Example 94. We have

$$\frac{1}{(1-q^n)^4} = 1 + 4q + 14q^2 + 40q^3 + \cdots,$$

approximately to increasing accuracy by $q^{-1}\mathcal{A}_{1,3,1}$, $q^{-3}\mathcal{A}_{2,3,1}$, $q^{-6}\mathcal{A}_{3,3,1}$, and so on.

We want to study these functions $\mathcal{A}$ and $\mathcal{B}$.

Theorem 95 (Bringmann, Craig, van-Ittersum, P.). *If $a, r \in \mathbb{N}$, then $\mathcal{A}_{a,2r,r}$ is a quasimodular form of (mixed) weight $2ar$.*

Example 96. Formulas for $\mathcal{A}_{2,4,2}$ and $\mathcal{A}_{2,4,3}$ in terms of $G_4, G_6, G_2, G_3, G_7, G_5$.

Remark 97. The proof uses the quasi-shuffle algebra.

Remark 98. If $k \neq 2r$, then odd Eisenstein series are involved.

Corollary 99 (Bringmann, Craig, van-Ittersum, P.). *Let $a, r, m \in \mathbb{N}$. There are infinitely many non-nested AP $An + B$ with*

$$c_{a,2r,r}(An + B) \equiv 0 \pmod{m}.$$

Conjecture 100 (Bringmann, Craig, van-Ittersum, P.). *Let $a, k, r, s \in \mathbb{N}$. For any prime p, we have*

$$c_{a,k,r}(p^{\alpha+1}n + p^\alpha\beta) \equiv 0 \pmod{p^{\nu_p(k)-\alpha}}.$$

ETC.

We move now to results on $\mathcal{B}$, which turns out to have modular properties.

Definition 101. We call a function $\hat{f}(\tau, \bar{\tau})$ a *completion* of $f(\tau)$ if

$$\lim_{\bar{\tau}\to -i\infty}\hat{f}(\tau, \bar{\tau}) = f(\tau).$$

Theorem 102. *Let $k, r, s \in \mathbb{N}$.*

1. *The function $\mathcal{B}_{a,k,r,s}$ is a polynomial in $\mathcal{B}_{k,r,s} := \mathcal{B}_{1,k,r,s}$ of degree a.*
2. *$\mathcal{B}_{k,r,s} + \mathcal{B}_{k,r,k-s}$ are a mixed weight linear combination of:*
   (a) *functions admitting a modular completion*
   (b) *powers of q times theta functions*
3. *If $k = s + 1$, then* (2b) *in the above linear combination vanishes.*

Define Zwegers' multivariable Appell function for $\ell \in \mathbb{N}$ by

$$A_\ell(z_1, z_2\tau) = \zeta_1^{\ell/2}\sum_{n\in\mathbb{Z}}\frac{(-1)^{\ell n}q^{\frac{\ell n(n+1)}{2}}\cdots}{\cdots}.$$

Define some R. Then $\hat{A}_\ell$ is given in terms of A_ℓ and ϑ and R.
Theorem of Zwegers describes transformation of $\hat{A}_\ell$.

Now we briefly describe the idea of the proof of our result. Taylor expansion where Taylor coefficients turn out to be similar functions:

$$\exp\left(\sum_{n\geq 1}\frac{(-1)^{n+1}}{n}\mathcal{B}_{nk,nr,ns}(q)X^n\right)=\sum_{j\geq 0}\mathcal{B}_{j,k,r,s}(q)X^j.$$

From this we get that the coefficients on the right hand side are indeed polynomials in terms of the coefficients on the left hand side. This gives the first part. To get the other parts, we use that

$$\begin{aligned}\mathcal{B}_{j,r,s-k+j}(q)+\mathcal{B}_{j,r,k-s}(q)&=\sum_{n\in\mathbb{Z}-\{0\}}\frac{q^{rn^2+(s-k+j)n}}{(1-q^n)^j}\\&=-\frac{1}{j!}\left[\frac{\partial^i}{\partial\zeta^j}(1-\zeta)\sum_{n\in\mathbb{Z}}\frac{q^{rn^2+(s+1-k)n}}{1-\zeta q^n}\right]|_{z=0}\\&=-\frac{1}{j!}\left[\frac{\partial^j}{\partial\zeta^j}(1-\zeta)\zeta^{-r}A_{2r}(z,(-k-r+s+1)\tau\tau)\right]|_{z=0}.\end{aligned}$$

We get

$$\mathcal{B}_{k,r,k-1}(q)+\mathcal{B}_{k,r,1}(q)=-\frac{1}{k!}\sum_{\ell=0}^{k}\frac{\alpha_{1/2}(k,\ell)}{(2\pi i)^\ell}f_{r,\ell}(\tau),$$

where

$$f_{r,\ell}(\tau):=\left[\frac{\partial^\ell}{\partial z^\ell}\left(\left(\zeta^{-1/2}-\zeta^{1/2}\right)\zeta^{-r}A_{2r}(z,-r\tau,\tau)\right)\right]_{z=0}$$

and $\alpha_{1/2}(k,\ell)$ is a generalized Stirling number defined by a recurrence. Enough to show that $f_{r,\ell}(\tau)$ can be completed.

Fact: Taylor coefficients of a Jacobi form are quasi-modular.

Example 103 ($k=2=s+1, r=1$)**.** (1) We write $\mathcal{B}_{2,2,1,1}(q)$ as the coefficient of X^2 in

$$\exp\left(\mathcal{B}_{2,1,1}(q)X-\tfrac{1}{2}\mathcal{B}_{4,2,2}(q)X^2+\mathrm{O}(X^3)\right),$$

which we compute to be $\frac{1}{2}\mathcal{B}_{2,1,1}^2(q)-\frac{1}{2}\mathcal{B}_{4,2,2}(q)$.

(2) ETC.

23. Explicit lower bounds on the conductor of abelian varieties with specified bad reduction, *Pierre Tchamitchian*

The first part is about L-functions. Consider a "good" L-function: there exists

- $N>0$, the conductor,
- (d_1,d_2) with $d=d_1+2d_2$, the signature,
- μ_j and ν_k, the spectral parameters ($j\leq d_1, k\leq d_2$)

such that the completed L-function

$$\Lambda(s)=N^{s/2}\prod_{j\leq d_1}\Gamma_{\mathbb{R}}(s+\mu_j)\prod_{k\leq d_2}\Gamma_{\mathbb{C}}(s+\nu_k)L(s)$$

satisfies

$$\Lambda(s)=\varepsilon\tilde{\Lambda}(1-s)$$

for some ε, and $L(s)$ admits an Euler product

$$L(s) = \prod_p F_p(p^{-s})^{-1} = \prod_p \left(1 - \alpha_{1,p} p^{-s}\right)^{-1} \cdots \left(1 - \alpha_{d_p,p} p^{-s}\right)^{-1},$$

absolutely convergent for $\sigma > 1$.

Definition 104. We say that $F : \mathbb{R} \to \mathbb{R}$ is a *Mestre test function* if there exists $\varepsilon > 0$ such that

(i) $F(x) \exp\left((\frac{1}{2} + \varepsilon)|x|\right)$ is integrable
(ii) $F(x) \exp\left((\frac{1}{2} + \varepsilon)x\right)$ has bounded variation, is integrable, and the value at each point is the average of its right and left limits
(iii) $(F(x) - F(0))/x$ has bounded variation.

Then we get the Weil–Mestre explicit formula: for $A = \pi^{-d/2} 2^{-d_2} N^{1/2}$ and $\Phi(s) = \int F(x) e^{(s-1/2)x}\, dx$, we have

$$\sum_\rho \Phi(\rho) - \sum_\mu \Phi(\mu) + \sum_{j \le d_1} I(\tfrac{1}{2}, \mu_j/2) + I(1/2, \overline{\mu_j}/2) + \sum_{k \le d_2} J(1, \nu_k) + J(1, \overline{\nu_k})$$

$$= 2F(0)\log(A) - \sum_{p,i,m \ge 1} \left(\alpha_{i,p}^m F(m \log p) + \overline{\alpha_{i,p}}^m F(-m \log p)\right) \frac{\log p}{p^{m/2}}.$$

Morally, the proof is a Fourier transform, as in the prime number theorem. You integrate the function $\Phi(s)\Lambda'(s)/\Lambda(s)$ along the rectangular contour with real parts $-\varepsilon$ and $1 + \varepsilon$. Relate left and right hadn sides using functional equation. Compute explicitly the right hand side using the Euler product, and use a Fourier transform. Because of this, we use GRH.

Trick: choose $F(x)$ such that $\Re(\Phi(s)) > 0$ for s on the critical line. Equivalently, choose $F(x)$ with a positive Fourier transform. Then we have the *Weil–Mestre explicit inequality*

$$2F(0)\log(A) \ge \sum_{j \le d_1} I(\tfrac{1}{2}, \mu_j/2) + I(\tfrac{1}{2}, \overline{\mu_j}/2) + \sum_{k \le d_2} J(1, \nu_k) + J(1, \overline{\nu_k})$$
$$+ \sum_{p,i,m \ge 1} \left(\alpha_{i,p}^m F(m \log p) + \overline{\alpha_{i,p}}^m F(-m \log p)\right) \frac{\log p}{p^{m/2}}.$$

Theorem 105 (Poitou, Odlyzko, 1976)**.** *Let K be a numebr field of degree n and discriminant d_K, with r_1 real embeddings and r_2 complex embeddings. Then, for n big enough,*

$$|d_K|^{1/n} > (215.3)^{r_1/n} (44.7)^{r_2/n}.$$

Let's talk about bad reduction of elliptic curves. Take E over $\mathbb{Q}$, say

$$y^2 = x^3 + ax + b$$

with $4a^3 + 27b^2 \neq 0$. By reducing the coefficients (of a minimal model at p) modulo p, one gets a curve E_p over $\mathbb{F}_p$. Then we end up in one of the following cases:

- E_p is an elliptic curve over $\mathbb{F}_p$
- E_p is a singular curve, with $E_{p,ns} \cong \mathbb{G}_m$
- E_p is a singular curve, and $E_{p,ns} \cong \mathbb{G}_a$

Set respectively $n_p = 0, 1, 2$ depending upon the reduction (unless $p \in \{2, 3\}$).

The *conductor* of E of the elliptic curve is defined by $N_E := \prod p^{n_p}$.

We define the L-series in terms of the $a_p := p + 1 - \#E_p(\mathbb{F}_p)$, by $L_p(s) = 1 - a_p p^{-s} + p^{1-2s}$ if E has good redcution, omitting the final term if bad reduction. We multiply these together to get $L_E(s)$. Then we have the modularity theorem, saying that these admit an analytic continuation. We also have the Mordell–Weil theorem and BSD conjecture.

If F is even, has positive Fourier transform, and support in $[-1, 1]$, and write $F_\lambda(x) = F(x/\lambda)$ for $\lambda > 0$, then we have the Mestre inequality. It involves

$$M_{\lambda,F} = 2\log(2\pi) + 2I_{F_\lambda,2} = 2\,(\cdots)\,.$$

Instead of looking at all elliptic curves, we can only look at the ones with some prescribed reduction.

We can now do the same things for abelian varieties. Looking at the bad reduction, there's a smooth part, torus part and unipotent part. We associate L-functions and Satake parameters. We have a Mestre inequality (1986): the conductor satisfies $N_A \geq 10.323^g$.

LMFDB.

Everything that we described here over $\mathbb{Q}$ also works over number fields.

24. Analytic ranks of modular forms with nontrivial character, *Maarten Derickx*

Link to actual slides: `https://bit.ly/analytic-ranks`.

We write $S_2(\Gamma)^{\text{zero}}$ to consist of those newforms f for which $L(f, 1) = 0$, where $\Gamma = \Gamma_0(N)$ or $\Gamma_1(N)$.

Conjecture 106 (Brumer)**.** *Roughly half the modular forms should have positive analytic rank, and roughly half should have analytic rank zero:*

$$\#S_2(\Gamma_0(p))^{\text{zero}} \sim \frac{1}{2}\#S_2(\Gamma_0(p)).$$

The sign functional equation forces

$$\#S_2(\Gamma_0(p))^{\text{zero}} \geq \left(\tfrac{1}{2} - \varepsilon\right)\#S_2(\Gamma_0(p))^{\text{new}}, \qquad \text{for}$$

¡++¿

[Okay, he's showing some slides of the LMFDB.]

Theorem 107 (Michel–Kowalski)**.** *We have*

$$\#S_2(\Gamma_0(p))^{\text{zero}} \leq \left(\frac{5}{6} + \varepsilon\right)\#S_2(\Gamma_0(p))^{\text{new}}.$$

[More LMFDB. The χ with $r_\chi(p) := \#S_2(p, \chi)^{\text{zero}} > 0$ for the primes $p < 100000$. Pretty rare that there's more than one relevant modular form per character.]

Conjecture 108 (D., Stoll)**.** *If* $\#S_2(p, \chi)^{\text{zero}} > 0$*, then* $\operatorname{ord}\chi \leq 30$*, or at least* $\operatorname{ord}\chi$ *grows super slowly with respect to* p*.*

This is like the equivalent of Brumer's conjecture with nontrivial character.

[More LMFDB discussion. Examples with higher order should come from elliptic curves with higher rank.]

Let's give an application of Conjecture 108. Let p be a prime, K a number field of degree d, E an elliptic curve over K.

Theorem 109 (Oesterle)**.** *Suppose $p \mid E(K)_{\mathrm{tors}}$, then $p < (3^{d/2}+1)^2$.*

Set

$$r_1(p) := \#S_2(\Gamma_1(p))^{\mathrm{zero}} - \#S_2(\Gamma_0(p))^{\mathrm{zero}} = \sum_{\chi \neq \chi_{\mathrm{triv}}} \#S_2(p,\chi)^{\mathrm{zero}}.$$

Theorem 110 (D. Stoll)**.** *Suppose $r_1(p) \leq \log_6(p) - 4$ for p large enough. Then for p large enough, we have $p \mid E(K)_{\mathrm{tors}}$ implies $p \leq 6d+1$.*

(...)

Example 111. Let $p = 99991$. Suppose there exists a number field K of degree d, and an elliptic curve over K, such that if $p \mid E(K)_{\mathrm{tors}}$, then $p \geq 33330$. This is sharp, since complex multiplication gives examples with $p \mid E(K)_{\mathrm{tors}}$ and $d = 33330$.

Proposition 112 (D., Stoll)**.** *For every prime $p = m^2 \pm 108$, there is a genus 2 curve C_p such that*

- $\mathbb{Z}[\zeta_3]$ *is contained in* $\operatorname{End}\operatorname{Jac}(C_p)$ *(i.e.,* $\operatorname{Jac}(C_p)$ *is of* GL_2 *type)*
- $\operatorname{Jac}(C_p)$ *is modular of level p.*
- *If* $\operatorname{Jac}(C_p)$ *is simple, then the associated modular form has character of order 3 or 6.*

Proposition 113. *Every character χ of order 6 in the table satisfies $p = m^2 \pm 108$, and Mordell–Weil rank $\operatorname{Jac}(C_p)(\mathbb{Q}) > 0$.*

Lemma 114 (D., Stoll)*. Assume Bunyakovsky's conjecture. Then there are infinitely many characters $\chi : (\mathbb{Z}/p\mathbb{Z})^* \to \mathbb{C}^\times$ of order 6 such that $\#S_2(p,\chi)^{\mathrm{zero}} \geq 1$.*

We did something similar for characters of order 3, twisting elliptic curves.

Lemma 115 (D., Stoll)*. Let χ mod p be of order 3. Then there exists $f \in S_2(p,\chi)^{\mathrm{zero}}$ with coefficients in $\mathbb{Q}(\zeta_3)$ iff there is an elliptic curve E over $\mathbb{Q}$ such that*

(1) E has conductor p^2,
(2) (...)

Let χ be of order n. Let $\mathbb{S}_2(p,\chi)$ denote the space of cuspidal modular symbols with coefficients in $\mathbb{Q}(\zeta_n)$, and $\mathbb{T}_\chi \subset \operatorname{End} S_2(p,\chi)$ the Hecke algebra.

Lemma 116 (The winding element e)*. There is an explicit $e \in \mathbb{S}_2(p,\chi)$ such that*

$$\dim_{\mathbb{Q}(\zeta_n)}(\mathbb{T}_\chi \otimes \mathbb{Q})e + \#S_2(p,\chi)^{\mathrm{zero}} = \#S_2(p,\chi)^{\mathrm{new}} = \dim S_2(p,\chi).$$

- $\dim_{\mathbb{Q}(\zeta_n)}(\mathbb{T}_\chi \otimes \mathbb{Q})e$ is computable with magma in reasonable time when n is small enough and $p < 100000$.
- If $n \asymp p$, then $\dim_{\mathbb{Q}} \mathbb{Q}(\zeta_n) \asymp p$ which makes the computation unfeasible for $p \asymp 100000$.
- For n large, work over $\mathbb{F}_p$ for a p that splits completely in $\mathbb{Q}(\zeta_n)$.

25. Sharp conditional bounds for moments of products of automorphic L-functions, *Markus Valås Hagen*

Want to study

$$I_k(T) := \int_T^{2T} \left|\zeta(\tfrac{1}{2}+it)\right|^{2k} dt.$$

Gives us a lot of information:

- Taking k large, we amplify the peaks, giving a way to study extreme values.
- Showing that $I_k(T) \ll T^{1+\varepsilon}$ for each $\varepsilon > 0$ and $k \in \mathbb{N}$ is equivalent to knowing the Lindelöf hypothesis, i.e., that $\zeta(\frac{1}{2} + it) \ll t^{\varepsilon}$ for each $\varepsilon > 0$.

Conjecture 117. *$I_k(T) \sim C_k T(\log T)^{k^2}$, for any real $k > 0$, where C_k is predicted precisely via random matrix theory.*

Moments are closely related to the distribution of ζ. Selberg central limit theorem: for fixed $V \in \mathbb{R}$ and large T, we have

$$\Phi_T(V) := \frac{1}{T} \operatorname{vol}\left(t \in [T, 2T] : \frac{\log|\zeta(\frac{1}{2} + it)|}{\sqrt{\frac{1}{2}\log\log T}} \geq V \right) \sim \frac{1}{\sqrt{2\pi}} \int_V^\infty e^{-x^2/2}\, dx.$$

In other words, $\log|\zeta(\frac{1}{2} + it)|$, sampled for $t \in [T, 2T]$, is approximately a normal variable with mean zero and variance $\frac{1}{2}\log\log T$

The order of magnitude of $I_k(T)$ is closely related to $\Phi_T(V)$ for $V \asymp k\sqrt{\log\log T}$.

What do we know about $I_k(T)$?

- Asymptotics for $k = 1, 2$
- $I_k(T) \asymp T(\log T)^{k^2}$ for $0 < k \leq 2$
- Assuming RH, $I_k(T) \ll T(\log T)^{k^2}$ for all $k > 0$.

L-functions in general are attached to some object X, and are used to study that object. X can be an elliptic curve, a modular form, a number field, etc. Also has pleasant analytic properties. It is believed that all L-functions should arise from automorphic L-functions, or products of them. This gives, at least for the speaker, a reason to be interested in automorphic L-functions. GRH (Grand Riemann Hypothesis) asserts the Riemann Hypothesis for all automorphic L-functions, i.e., all nontrivial zeros ρ satisfy $\Re(\rho) = 1/2$.

Automorphic L-functions also have Dirichlet series and Euler product for $\Re(s) > 1$, say

$$L(s) = \sum_{n \geq 1} \frac{a_L(n)}{n^s}.$$

In our proof, we need control over these coefficients. If $\Lambda_L(n)$ denotes the coefficients of the Dirichlet series of $L'/L(s)$, then the Generalized Ramanujan Conjecture gives the pointwise bound

$$|\Lambda_L(n)| \leq m_L \Lambda(n),$$

where m_L is a fixed constant depending only on L. There are also milder alternatives to GRC, such as the so-called "Hypothesis H" of Rudnick–Sarnak, which may be understood as an average version of GRC. In particular, GRC implies Hypothesis H.

From now on, "L-function" means "automorphic L-function."

It's expected that moments of L-functions behave like those of the zeta function, i.e.,

$$\int_T^{2T} \left|L(\tfrac{1}{2} + it)\right|^{2k} \sim C_{k,L} T(\log T)^{k^2}.$$

We want to study moments, but with several L-functions at once, e.g.,

$$\int_T^{2T} \left|L_1(\tfrac{1}{2} + it)\right|^{2k_1} \left|L_2(\tfrac{1}{2} + it)\right|^{2k_2}\, dt.$$

This sheds light on the independence between them. Given two distinct irreducible L-functions, we expect *independence* between them. In particular,

$$\frac{1}{T}\int_T^{2T} \left|L_1(\tfrac{1}{2}+it)\right|^{2k_1}\left|L_2(\tfrac{1}{2}+it)\right|^{2k_2}\,dt$$
$$\approx \frac{1}{T}\int_T^{2T} \left|L_1(\tfrac{1}{2}+it)\right|^{2k_1}\,dt \times \frac{1}{T}\int_T^{2T} \left|L_2(\tfrac{1}{2}+it)\right|^{2k_2}\,dt \asymp (\log T)^{k_1^2+k_2^2}.$$

Milinovich–Turnage-Butterbaugh (2014): Let $k_1,\dots,k_r > 0$ be real numbers. Let $L_1,\dots,L_r$ be distinct irreducible cuspidal automorphic L-functions with unitary central characters. Assume the Generalized Riemann Hypothesis and Hypothesis H. Then, for each $\varepsilon > 0$, we have the almost sharp bound

$$\int_T^{2T} \prod_{i=1}^{r} \left|L_i(\tfrac{1}{2}+it)\right|^{2k_i}\,dt \ll T(\log T)^{k_1^2+\cdots+k_r^2+\varepsilon}.$$

Theorem 118 (H. 2024+)**.** *Keep the assumptions above, but assume GRC instead of Hypothesis H. Then*

$$T(\log T)^{k_1^2+\cdots+k_r^2} \ll \int_T^{2T} \prod_{i=1}^{r} \left|L_i(\tfrac{1}{2}+it)\right|^{2k_i} \ll T(\log T)^{k_1^2+\cdots+k_r^2}.$$

Let's discuss some applications. For $\Re(s) > 1$ and $0 < \alpha \le 1$, the Hurwitz zeta function in defined by

$$\zeta(s,\alpha) = \sum_{n\ge 0} \frac{1}{(n+\alpha)^s}.$$

Admits an analytic continuation to $\mathbb{C}-\{1\}$. These are interesting functions because the analogue of the RH fails when $\alpha \neq 1/2, 1$, but $\zeta(s,\alpha)$ has many similarities with ζ either way. From now on, let $\alpha = a/q$ ($\gcd(a,q) = 1, 1 \le a \le q$). Then it's conjectured that

$$\int_T^{2T} \left|\zeta(\tfrac{1}{2}+it,\alpha)\right|^{2k}\,dt \sim c_{k,\alpha} T(\log T)^{k^2}.$$

- Proven for $k = 1, 2$ unconditionally (Andersson, Rane, Sahay).
- Assuming RH for every Dirichlet L-function modulo q, Sahay (2023) proved for every positive integer k and $\varepsilon > 0$, one has

$$T(\log T)^{k^2} \ll \int_T^{2T} \left|\zeta(\tfrac{1}{2}+it,\alpha)\right|^{2k}\,dt \ll T(\log T)^{k^2+\varepsilon}.$$

Theorem 119 (H. 2024+)**.** *Let $\alpha = a/q$, where $\gcd(a,q) = 1$, $1 \le a \le q$. Assume RH for all Dirichlet L-functions modulo q. Then for all $k > 1/2$, we have*

$$T(\log T)^{k^2} \ll \int_T^{2T} \left|\zeta(\tfrac{1}{2}+it,\alpha)\right|^{2k}\,dt \ll T(\log T)^{k^2}.$$

Proof idea for upper bound, k an integer. By orthogonality of Dirichlet characters, we have

$$\zeta(\tfrac{1}{2}+it,\tfrac{a}{q}) = \frac{q^{1/2+it}}{\varphi(q)} \sum_{\chi(q)} \overline{\chi(a)} L(\tfrac{1}{2}+it,\chi).$$

Taking k-th power and using the multnomial theorem, we get a sum over different products of Dirichlet L-functions with different powers. Taking the second moment,

we have to compute (up to order) many terms of the form (which we know from our main theorem)

$$\prod_{i=1}^{r} \left|L_i(\tfrac{1}{2}+it)\right|^{2k_i}.$$

□

Another application concerns large deviations. Selberg CLT for several L-functions at once:

$$\frac{1}{T}\operatorname{vol}\left(t\in[T,2T]:\frac{\log\left|L_j(\frac{1}{2}+it)\right|}{\sqrt{\frac{1}{2}\log\log T}}\geq V_j, j=1,\dots,r\right)\sim\prod_{j=1}^{r}\int_{V_j}^{\infty}e^{-x^2/2}\,dx$$

for fixed V_j. Controls typical fluctuations. Conditional result (Selberg + Bombieri–Hejhal), requiring a strong zero density estimate.

Theorem 120 (H. 2024+)**.** *Assume GRH and GRC. Let $V_j \asymp \sqrt{\log\log T}$. Then*

$$\frac{1}{T}\log\left(t\in[T,2T]:\frac{\log|L_j(\frac{1}{2}+it)|}{\sqrt{\log\log T}}\geq V_j, 1\leq j\leq r\right)\asymp\exp\left(-(1+o(1))\frac{V_1^2+\cdots+V_r^2}{2}\right).$$

Sketch of proof. The large deviations $V_j \asymp \sqrt{\log\log T}$ are controlled by the moments. Let $V_j = C_j\sqrt{\log\log T}$. Then

$$\frac{1}{T}\operatorname{vol}\left(t\in[T,2T]:\frac{\log|L_j(\frac{1}{2}+it)|}{\sqrt{\frac{1}{2}\log\log T}}\geq V_j, 1\leq j\leq r\right)=\frac{1}{T}\operatorname{vol}\Big(t\in[T,2T]:|L_j(\tfrac{1}{2}+it)|\geq(\log T)^{\sqrt{C_j^2/2}}, 1\leq j\leq$$

Now apply Chebyshev, then our knowledge of moments, then choose $k_j = C_j/\sqrt{2}$ to get the desired bound. □

[Slide that says "Do I have more time?".]

Let's now discuss the proof strategy for moments. The lower bound proof strategy is due to Heath–Brown. For ζ, the argument starts with

$$\int_T^{2T}\left|\zeta(\tfrac{1}{2}+it)\right|^{2k}\,dt\geq\int_T^{2T}\left|\sum_{n\leq T}\frac{d_k(n)}{n^{1/2+it}}\right|^2\,dt-\int_T^{2T}\left|\zeta(\tfrac{1}{2}+it)-\sum_{n\leq T}\frac{d_k(n)}{n^{1/2+it}}\right|^2\,dt.$$

First term on the RHS can be computed by Montgomery–Vaughan mean value theorem. Second term is dealt with by means of a version of Hadamard three lines theorem, moving the line of integration to $\Re(s) = 5/4$, where the corresponding integral is small. For the upper bound, lots of ideas are combined:

- Soundararajan and Chandee: on GRH, bounding L-functions on the critical line by just an Euler product (with no information from zeros)
- Radziwill: integrating over certain subsets where our L-functions behave "typically", instead of the whole of $[T, 2T]$.
- Harper/Radziwill–Soundararajan: splitting up Dirichlet polynomial into many parts with progressively smaller variance.

26. Sub-Weyl bound for GL(2) L-functions, *Prahlad Sharma*

26.1. **Background.** Hardy–Littlewood (1921) (written down by Landau):

$$\zeta(\tfrac{1}{2}+it) \ll_\varepsilon t^{1/6+\varepsilon}, \quad t \geq 1.$$

Uses Weyl differencing technique to estimate $\sum_{n\sim N} n^{it}$. Since then, several improvements have been done to this bound, leading to many important techniques for estimating exponential sums. We'll only mention some of them. One is Bombieri–Iwaniec's bound, refined by Huxley (2005), giving

$$\zeta(\tfrac{1}{2}+it) \ll_\varepsilon t^{1/6-13/1230+\varepsilon}, \quad t \geq 1.$$

The state of the art is due to Bourgain (2017), who obtained

$$\zeta(\tfrac{1}{2}+it) \ll_\varepsilon t^{1/6-1/84+\varepsilon}, \qquad t \geq 1,$$

using decoupling method from harmonic analysis.

For general automorphic L-functions, one has the convexity bound (Phragmen–Lindelöf convexity principle plus functional equation)

$$L(\tfrac{1}{2}+it, F) \ll_\varepsilon \mathfrak{q}(t,F)^{1/4+\varepsilon},$$

while the analogue of the Weyl bound is

$$L(\tfrac{1}{2}+it, F) \ll_\varepsilon \mathfrak{q}(t,F)^{1/6+\varepsilon}.$$

A "sub–Weyl" bound is an improvement on the 1/6 exponent.

There are important applications of subconvexity bounds.

- A. Good (1982): Weyl bound for f holomorphic of level 1,

$$L(\tfrac{1}{2}+it, f) \ll_\varepsilon (t^2)^{1/6+\varepsilon}.$$

- K. Aggarwal and S. Singh (2017): Weyl bound for GL(2) t-aspect, using the "GL(2) delta method".
- K. Aggarwal (2018): Weyl bound for GL(2) t-aspect using the "trivial delta method".
- V. Blomer, S. Jana, P. Nelson (2021): Weyl bound for GL(2) × GL(2) × GL(2) using the period integral approach.

In particular, we haven't yet seen a sub-Weyl bound for GL(2).

26.2. **Main result.** Using a further refinement of the trivial delta method, we obtain:

Theorem 121 (R. Holowinsky, R. Munshi, S., J. Streipel)**.** *Let f be a $\mathrm{SL}_2(\mathbb{Z})$ form (cuspidal or Eisenstein). We have*

$$L(\tfrac{1}{2}+it, f) \ll_{\varepsilon,f} (t^2)^{1/6-1/96+\varepsilon}.$$

Some remarks:

- Taking f to be Eisenstein, we get a sub-Weyl bound for zeta, between Bourgain's and Bombieri–Iwaniec's:

$$\frac{1}{6}-\frac{1}{84} < \frac{1}{6}-\frac{1}{96} < \frac{1}{6}-\frac{1}{168}.$$

- The limiting bound of the method is $\zeta(\frac{1}{2}+it) \ll_\varepsilon t^{1/8+\varepsilon}$. Speaker thinks that with more effort, they could improve upon Bourgain's bound.
- The method is soft, using no exponent pair estimates.

26.3. **Proof sketch.**

26.3.1. *Setup.* By the approximate functional equation,

$$L(\tfrac{1}{2}+it,f) \approx t^{-1/2}\sum_{n\asymp t}\lambda(n)n^{-it}.$$

Trivial estimation gives the convexity bound

$$L(\tfrac{1}{2}+it,f) \ll t^{1/2}.$$

Need to save $t^{1/6}$ plus something more in

$$S := \sum_{n\asymp t}\lambda(n)n^{-it}$$

to get sub–Weyl.

26.3.2. *Trivial delta.* Let's now discuss the δ-symbol. Let $\delta : \mathbb{Z} \to \{0,1\}$ denote the characteristic function of 0:

$$\delta(k) = \begin{cases} 1 & \text{if } k = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$S = \sum_{m\asymp t}\sum_{n\asymp t}\lambda(m)\delta(m-n)n^{-it}.$$

The *trivial delta symbol* comes from observing that for $p > t$, we have

$$\delta(m-n) = \frac{1}{p}\sum_{b(p)} e\left(\frac{b(m-n)}{p}\right).$$

To optimize the delta symbol, we employ "conductor lowering". We first use

$$m-n \ll t/K \rightsquigarrow \frac{1}{K}\int_{v\sim K}(m/n)^{iv}\,dv$$

for some $K < t$, and then detect the smaller equation

$$m-n = 0.$$

This reduces the number of variables we are introducing (saving from the v-integral) and reduces the conductor size arising from the delta symbol expansion.

We are led to proceed with

$$S = \frac{1}{K}\sum_{m\asymp t}\sum_{n\asymp t}\lambda(m)\delta(m-n)n^{-it}\int_{v\asymp K}(m/n)^{iv}\,dv.$$

We choose a set of primes $\mathcal{P}$ of size P such that

$$P > t/K.$$

Then

$$\delta(m-n) = \frac{1}{|\mathcal{P}|}\sum_{p\in\mathcal{P}}\frac{1}{p}\sum_{b(p)} e\left(\frac{b(m-n)}{p}\right).$$

Substituting, we arrive at

$$S = \frac{1}{K}\int_{v\asymp K}\frac{1}{P}\sum_{p\in\mathcal{P}}\frac{1}{p}\sum_{b(p)}\sum_{m\asymp t}\lambda(m)e\left(\frac{bm}{p}\right)m^{iv}\sum_{n\asymp t}e\left(\frac{-bn}{p}\right)n^{-i(t+v)}\,dv.$$

Dualizing (Voronoi and Poisson), get

$$S = \frac{t^{3/2}}{K^{3/2}P^3} \sum_{p\in\mathcal{P}} \sum_{m\asymp p^2K^2/t} \sum_{n\asymp p} \lambda(m) e\left(\frac{\bar{n}m}{p}\right) (n/p)^{it} e^{i\phi(m,n,p)}.$$

Trivially,

$$S \ll t^{1/2}K^{1/2}P = t\cdot \frac{K^{1/2}P}{t^{1/2}}.$$

We are $\frac{K^{1/2}P}{t^{1/2}}$ away from subconvexity.

Everything so far has been as in previous works.

$$\left(\sum_{m\asymp P^2K^2/t} |\lambda(m)|^2\right)^{1/2} \left(\sum_{m\asymp P^2K^2/t} \left|\sum_{p\in\mathcal{P}} \sum_{n\asymp P} e\left(\frac{\bar{n}m}{p}\right) (n/p)^{it} e^{i\phi(m,n,p)}\right|^2\right)^{1/2}.$$

- Diagonal contributes $P^2\cdot P^2K^2/t$.
- Off-diagonal contributes $P^4\cdot\sqrt{K}$.
- Therefore

$$S \ll t^{1/2}K^{1/2} + \frac{t}{K^{1/4}}.$$

With the optimal choice $K = t^{2/3}$, we get

$$S \ll t^{1-1/6}.$$

26.3.3. *Going beyond Weyl.* We go back to the step just before, and use Diophantine approximation. Write

$$S = \frac{t^{3/2}}{K^{3/2}P^3} \sum_{p\in\mathcal{P}} \sum_{n\asymp p} \left(\frac{n}{p}\right)^{it} \sum_{m\asymp p^2K^2/t} \lambda(m) e\left(\frac{\bar{n}m}{p}\right) e^{i\phi(m,n,p)}.$$

Fix $Q \geq 1$. For each (n,p), choose (a,q), with $q\leq Q$, such that

$$\frac{\bar{n}}{p} = \frac{a}{q} + \beta, \quad \beta \leq \frac{1}{qQ}.$$

Then

$$e\left(\frac{\bar{n}m}{p}\right) = e\left(\frac{am}{q}\right) e(m\beta).$$

Choose Q such that

$$m\beta \asymp K \implies Q \asymp P\sqrt{\frac{K}{t}},$$

which is smaller than P, although the analytic conductor remains the same. The GL(2) sum now becomes

$$\sum_m \lambda(m) e\left(\frac{\bar{n}m}{p}\right) e^{i\phi(m,n,p)} = \sum_m \lambda(m) e\left(\frac{am}{q}\right) e(m\beta) e^{i\phi(m,n,p)},$$

where the conductors are $P^2\cdot K^2$ for the LHS and $Q^2\cdot K^2$ for the RHS. After Voronoi once more, we obtain

$$S = \frac{t}{KP^2} \sum_{p\in\mathcal{P}} \sum_{n\asymp p} (n/p)^{it} \sum_{m\asymp K} \lambda(m) e\left(\frac{\bar{a}m}{q}\right) \mathcal{J}(m,n,p).$$

Trivially at this stage, we get

$$S \ll t,$$

which gives convexity.

We observe that

$$e\left(\frac{\bar{a}m}{q}\right)$$

is still additive with respect to m.

We now apply Cauchty–Schwarz followed by Poisson summation, giving

$$\left(\sum_{m\asymp K}|\lambda(m)|^2\right)^{1/2}\left(\sum_{m\asymp K}\left|\sum_{p\in\mathcal{P}}\sum_{n\asymp P}e\left(\frac{\bar{a}m}{q}\right)(n/p)^{it}\mathcal{J}(m,n,p)\right|^2\right)^{1/2}.$$

The diagonal contributes $P^2\cdot K$. The off-diagonal gives $P^4\cdot\sqrt{K}$. Therefore

$$S\ll\frac{t}{P}+\frac{t}{K^{1/4}}.$$

Here the diagonal is negligibly small, while the off-diagonal is the same as earlier.

With the best possible choice $K=t^{1-\varepsilon}$, we get

$$S\ll t^{1-1/4},$$

which gives

$$L(\tfrac{1}{2}+it,f)\ll t^{1/4+\varepsilon},$$
$$\zeta(\tfrac{1}{2}+it)\ll t^{1/8+\varepsilon}.$$

Some degeneracies arise in the non-generic cases $q\leq Qt^{-\delta}$, $\beta\ll\frac{t^{-\delta}}{Q^2}$. These are relatively simpler when $K\ll t^{3/4}$. Thus, for now, we obtain

$$S\ll\frac{t}{(t^{3/4})^{1/4}}=t^{1-3/16},$$

that is to say,

$$L(\tfrac{1}{2}+it,f)\ll t^{1/3-1/48+\varepsilon},$$
$$\zeta(\tfrac{1}{2}+it,f)\ll t^{1/6-1/96+\varepsilon}.$$

27. Hybrid subconvex bounds for selfdual GL_3 L-functions via $\mathrm{GL}_3\times\mathrm{GL}_2$ vs. $\mathrm{GL}_4\times\mathrm{GL}_1$ spectral reciprocity, *Soumendra Ganguly*

Let's start by discussing Motohashi's formula (1997). It concerns

$$\sum_{f\in\mathcal{F}_1}\frac{L(\frac{1}{2},f)^3}{L(1,\operatorname{ad}f)}h(t_f)+\cdots=\text{main term }+\int_{\mathbb{R}}\left|\zeta(\tfrac{1}{2}+it)\right|^4\mathcal{H}(t)\,dt.$$

Here f runs over an orthonormal basis for the space of Hecke–Maass forms for $\mathrm{SL}(2,\mathbb{Z})$, h is some nice enough function, and $\mathcal{H}$ is some integral transform of h.

[Ed: Motohashi's formula goes the other way]

Kwan (2021), Humphries–Khan (2022), Wu (2023): various formulas of the shape

$$\sum_{f}\frac{L(\frac{1}{2},F\otimes f)}{L(1,\operatorname{ad}f)}h(t_f)+\cdots=\text{main term }+\int_{\mathbb{R}}L(\tfrac{1}{2}+it,F)\zeta(\tfrac{1}{2}-it)\mathcal{H}(t)\,dt.$$

These recover Motohashi's formula by specializing to F an Eisenstein series.

We focus on the approach of Humphries–Khan, which works only F self-dual, but works for a broad range of test functions.

We consider generalizations to higher levels. This is joint work with Humphries and Lin. Preprint: [10]. We consider levels $q = q_1 q_2$, where $(q_1, q_2) = 1$, and take for χ_1 a primitive Dirichlet character modulo q_1.

We consider the family

$$\mathcal{F}_2 = \left\{ g \otimes \chi_1 : g \in \mathcal{B}(q, \bar{\chi}_1^2) \right\}.$$

We show that for F self-dual,

$$\begin{aligned} &\sum_{f \in \mathcal{F}_2} \frac{L(\frac{1}{2}, F \otimes f)}{L(1, \operatorname{ad} f)} h(t_f) + \cdots \\ &= \text{main } + \frac{q_2^{1/2}}{q_1} \sum_{\psi_1 (q_1)} \int_{\mathbb{R}} L(\tfrac{1}{2} + it, F \otimes \psi_1) L(\tfrac{1}{2} - it, \overline{\psi_1}) \mathcal{Z}_{\chi_1}(\psi_1, t) \mathcal{H}(t)\, dt + \cdots . \end{aligned} \tag{1}$$

Here $\cdots$ includes contributions from lower level, holomorphic forms, and the continuous spectrum. Also,

$$\mathcal{Z}_{\chi_1}(\psi_1, t) \approx g(\chi_1, \psi_1),$$

which appears in the work of Conrey–Iwaniec and Petrow–Young.

As an application of this, we show the following subconvexity bounds:

$$L(\tfrac{1}{2} + it, F \otimes \chi) \ll_{F, \varepsilon} (q(|t| + 1))^{3/5 + \varepsilon} \left(1 + \frac{q^{2/5}}{q_1^{1/2} (|t| + 1)^{1/10}} + \frac{q_1^{1/8}}{(q(|t| + 1))^{1/10}} \right),$$

where F is fixed and self-dual, while χ is a primitive Dirichlet character of conductor q (any positive integer).

Also, for $f \in \mathcal{B}^*(q^2, \chi_0)$ such that $f \otimes \bar{\chi}$ has level dividing q, we have

$$L(\tfrac{1}{2}, F \otimes f) \ll_{F, \varepsilon} (q(|t_f| + 1))^{6/5 + \varepsilon} \left(1 + \frac{q^4}{q_1 (|t_f| + 1)^{1/5}} + \frac{q_1^{1/4}}{(q(|t| + 1))^{1/5}} \right).$$

These bounds are hybrid in the aspects of q, t_f and t. Taking $q_1 = q$, we get a bound of

$$q^{5/8 + \varepsilon} (|t| + 1)^{3/5 + \varepsilon}.$$

If we further assume

$$q^{4/5} (|t| + 1)^{-1/5} \ll q_1 \ll q^{4/5} (|t| + 1)^{4/5},$$

then the bound becomes

$$(q(|t| + 1))^{3/5 + \varepsilon}.$$

For the proof of the spectral reciprocity formula 1, we replace $1/2$ with s of large enough real part, then open the Dirichlet series and apply Kuznetsov. The diagonal term gives the main term. For the off-diagonal term, we open the Kloosterman sums and apply Voronoi. After some manipulations, we find two Dirichlet series, which eventually yields the L-factors on the right hand side, together with the integral transform.

28. Subconvex bounds for $\mathrm{SO}(n+1) \times \mathrm{SO}(n)$, *Blanca Gil Rosell*

Aim is to find an analogue of subconvex bounds for unitary groups $\mathrm{U}_{n+1} \times \mathrm{U}_n$, but for orthogonal groups $\mathrm{SO}_{n+1} \times \mathrm{SO}_n$.

Branching coefficients: for automorphic forms $\varphi \in \pi$ on G and $\varphi_H \in \pi_H$ on H, then

$$\left| \int_{H(\mathbb{Q})\backslash H(\mathbb{A})} \varphi \cdot \varphi_H \right|^2 = \mathcal{L}(\pi \times \pi_H, \tfrac{1}{2}) \times (\text{local integrals}).$$

Conjecture of Ichino–Ikeda is that $\mathcal{L}(\pi \times \pi_H, \frac{1}{2})$ is essentially $L(\pi \times \pi_H, \frac{1}{2})$. Essentially known for unitary groups, widely open for orthogonal groups.

For $f \in G(\mathbb{A})$,

$$\sum_{\pi} \sum_{\varphi \in \mathcal{B}(\pi)} \rho(f)\varphi(x)\overline{\varphi(y)} = \sum_{\gamma \in G(\mathbb{Q})} f(x^{-1}\gamma y).$$

Choose f and φ_H "microlocalized" (Nelson–Venkatesh) and use the "amplification method" (Duke–Friedlander–Iwaniec, Iwaniec–Sarnak),

$$\sum_{\pi} L(\pi \times \pi_H, \tfrac{1}{2}) w_{f,\varphi_H}(\pi) = \int_{x,y \in H(\mathbb{Q})\backslash H(\mathbb{A})} \varphi_H(x)\overline{\varphi_H(y)} \sum_{\gamma \in G(\mathbb{Q})} f(x^{-1}\gamma y)\, dx\, dy.$$

Reduces to a transversality problem that we now describe. Let F be af ield, $V = V_H \oplus F_e$ a quadratic space.

- $(G, H) = (\mathrm{SO}_{n+1}, \mathrm{SO}_n)$,
- $\mathfrak{g}, \mathfrak{h}$: Lie algebras,
- $\tau \in \mathfrak{g} \rightsquigarrow \tau_H \in \mathfrak{h}$: upper left-block, i.e.,

$$\tau = \begin{pmatrix} \tau_H & * \\ * & * \end{pmatrix}.$$

  Nelson–Venkatesh: choose τ (resp. τ_H) to be microlocal parameters attached to π (resp. π_H). Roughly speaking, their eigenvalues are the parameters of the representations.

Call τ *stable* if $\gcd(P_\tau, P_{\tau_H}) = 1$.

Bounds for geometric side of relative trace formula boil reduce, via Cauchy–Schwarz, to solving the *transversality problem*, which is to show that for $a \in G - HZ$, there exists $z \in H_{\tau_H}$ such that $az \notin HG_\tau$.

- Marshall: depth aspect, τ and τ_H semisimple
- Nelson: depth aspect
- Nelson–Hu: horizontal aspects

The Lie algebra problem is to show that for each $x \in \mathfrak{g}_\tau - \mathfrak{z}$, there exists $z \in \mathfrak{h}_{\tau_H}$ such that $[x, [z, \tau]] \notin [\mathfrak{h}, \tau]$.

Conjecture 122 (Nelson 2024, private communication)**.** *The Lie algebra problem can probably be solved for* $(\mathrm{SO}_{n+1}, \mathrm{SO}_n)$*, and would be a good first PhD project.*

Example 123. The conjecture is false for $(\mathrm{SO}_6, \mathrm{SO}_5)$, by explicit numerical computation.

So for the orthogonal case, we need to address the transversality problem more directly. Experiments with this thus far, some experiments with Gröbner bases.

29. Subconvexity for Rankin–Selberg L-functions, *Sumit Kumar*

Let's start with the setup.

- f: Hecke cusp form for $\Gamma_0(p)$ and nebentypus χ.
- g: Hecke cusp form for $\mathrm{SL}_2(\mathbb{Z})$ and trivial nebentypus.
- Consider the Dirichlet series

$$L(s, f \otimes g) = L(2s, \chi) \sum_{n=1}^{\infty} \frac{\lambda_f(n)\lambda_g(n)}{n^s}, \quad \Re(s) > 1,$$

  where the $\lambda(n)$ are the normalized Fourier coefficients. It satisfies a functional equation and has meromorphic continuation to $\mathbb{C}$.

We want to estimate $L(\frac{1}{2}, f \otimes g)$ as $p \to \infty$ with g fixed.

Functional equation yields the following approximateion by a finite Dirichlet polynomial, of length roughly the square-root of the conductor:

$$L(\tfrac{1}{2}, f \otimes g) \approx \sum_{n \ll p^{1+\varepsilon}} \frac{\lambda_f(n)\lambda_g(n)}{n^{1/2}} W(n).$$

Here W is a smooth function satisfying some nice decay conditions. Using this approximation, one can show that

$$L(\tfrac{1}{2}, f \otimes g) \ll_{g,\varepsilon} p^{1/2+\varepsilon} = (p^2)^{1/4+\varepsilon},$$

the convexity bound. Our aim is to show that

$$L(\tfrac{1}{2}, f \otimes g) \ll_g (p^2)^{1/4-\delta} \text{ for some } \delta > 0. \tag{2}$$

- The first estimate of the form (2) was shown by Kowalski–Michel–Vanderkam [16] when f is holomorphic and χ has conductor $\ll p^{1/2}$.
- Michel [19] removed the the latter condition and obtained (2) with $\delta = 1/2114$ for g holomorphic and χ non-trivial.
- Harcos–Michel [12] proved (2) with $\delta = 1/2826$ for g holomorphic/Maass cusp forms and non-trivial χ.
- Michel–Venkatesh [20] generalized all the above results (uniform in all aspects of f).

Michel and Michel–Harcos use spectral theory of automorphic forms. Michel–Venkatesh use period integral approach (geometric arguments). In this talk, we will use Munshi's delta symbol approach to prove our theorem.

Theorem 124 (K–Aggarwal–Kwan–Leung–Li–Young)**.** *Let f be a Hecke cusp form of prime level p and nebentypus χ, and let g be a fixed* GL(2) *form. Then we have*

$$L(\tfrac{1}{2}, f \otimes g) \ll_{g,\varepsilon} (p^2)^{1/4-1/1100+\varepsilon}.$$

For χ trivial, we have

$$L(\tfrac{1}{2}, f \otimes g) \ll_{g,\varepsilon} \cdots .$$

We use the δ-method along with amplification. Key input is DFI estimates of bilinear sum

$$\sum_n a(n) \sum_m b(m) e\left(\frac{c\bar{m}}{n}\right).$$

Let's now say something about the δ-method. For $n \in \mathbb{Z}$, $\delta(n) = 1$ if $n = 0$ and $\delta(n) = 0$ otherwise. We think $\delta(n) = \int_0^1 e(nx)\,dx$. Delta method is a variant

of classical ricle method with no "minor arc" contributions. We use the Fourier expansion (DFI) for $|n| \leq N$,

$$\delta(n) = \frac{1}{Q} \sum_{q \leq Q} \frac{1}{q} \sum_{a(q)^*} e\left(\frac{an}{q}\right) \int_{\mathbb{R}} g(q,x) e\left(\frac{nx}{qQ}\right) dx,$$

where $g(q,x)$ is a smooth function and $Q \leq \sqrt{N}$.

Let's now give a sketch of the proof. On applying the approximate functional equation, we see that

$$L(\tfrac{1}{2}, f \otimes g) \ll \sup_{N \ll p} \frac{S(N)}{\sqrt{N}},$$

where

$$S(N) = \sum_n \lambda_f(n) \lambda_g(n) W(n/N).$$

Here W is a fixed bump function, taking the value one on some interval. The goal is to show that

$$S(N) \ll N^{1-\delta} \qquad \text{for } N \approx p, \quad \text{some } \delta > 0.$$

To that end, we write

$$\begin{aligned} S(N) &= \sum_n \lambda_f(n) \lambda_g(n) W(n/N) \\ &= \sum_n \sum_m \lambda_f(n) \lambda_g(m) W(n/N) W_1(m/N) \delta(m-n) \\ &= \frac{1}{Q} \sum_q \int_{\mathbb{R}} \frac{g(q,x)}{q} \sum_{a(q)^*} S_1(a,q,x) S_2(a,q,x)\, dx. \end{aligned}$$

Here $Q = \sqrt{N}$. (To be precise, we have suppressed the amplifier from our sketch of the argument, but that is used to treat the diagonal.) Also

$$S_1(a,q,x) = \sum_m \lambda_f(m) e\left(\frac{am}{q}\right) W_1(m,q,x),$$

$$S_2(a,q,x) = \sum_n \lambda_g(n) e\left(\frac{-an}{q}\right) W_2(n,q,x).$$

So what we have done is to separate out the coefficients λ_f and λ_g. The next step is to analyze S_1 and S_2 using Voronoi summation. In the generic case $q \sim Q = \sqrt{p}$, we see that

$$S_1(\cdots) \rightsquigarrow \frac{N \chi(-q)}{Q\sqrt{p}} \sum_{m \asymp p} \lambda_f(m) e\left(\frac{-\overline{ap}m}{q}\right).$$

(This step doesn't lose or gain anything.) Similarly,

$$S_2(\cdots) \rightsquigarrow \frac{N}{Q} \sum_{m \asymp 1} \lambda_f(n) e\left(\frac{\bar{a}m}{q}\right).$$

(Here we have gained a lot.) Combining everything together so far, we arrive at

$$S(N) \rightsquigarrow \frac{1}{\sqrt{p}} \sum_{m \asymp p} \lambda_f(m) \sum_q \chi(-q) \sum_{n \asymp 1} \lambda_f(n) c_q(pn - m).$$

Here

$$c_q(n) = \sideset{}{^*}\sum_{a(q)} e\left(\frac{an}{q}\right)$$

is the Ramanujan sum. Thus far, we have "shown" that

$$S(N) \ll \sqrt{p}Q\sqrt{Q} = Np^{1/4},$$

assuming square-root savings in $c_q(\cdots)$. (Of course, we get more saving than that for c_q, but we're doing a bunch of averaging.) Next, we apply Cuachy to the m-sum to obtain

$$|S(N)|^2 \ll \sum_m \left|\sum_q \chi(-q) \sum_{n \asymp 1} \lambda_f(n) c_q(pn-m)\right|^2.$$

We now open the square and apply Poisson summation to the m-sum:

$$\sum_{m \asymp p} c_{q_1}(pn_1 - m)\overline{c_{q_2}(pn_2 - m)}$$

$$\rightsquigarrow p\sqrt{q_1 q_2}\delta(q_1 = q_2, n_1 = n_2) + \sum_{m \asymp 1} e\left(\frac{\overline{q_2}mpn_1}{q_1} + \frac{\overline{q_1}mpn_2}{q_2}\right).$$

The remaining sum over q_1 and q_2 is given by

$$B = \sum_{q_1 \asymp Q} \sum_{q_2 \asymp Q} \chi(q_1)\overline{\chi}(q_2) e\left(\frac{a\overline{q_1}}{q_2}\right), \quad a = pm(n_2 - n_1).$$

Here we apply DFI estimate $B \ll Q^{2-1/48}$ [8]. The proof follows.

30. Archimedean Distinguished Representations and Exceptional Poles, *Akash Yadav*

Preprint: [24]

Let F be a non-archimedean local field, E either $F \times F$ (split case) or a quadratic extension (inert case). π: irreducible generic representation of $\mathrm{GL}_n(E)$.

Definition 125. A representation π is said to be $\mathrm{GL}_n(F)$-distinguished if $\mathrm{Hom}_{\mathrm{GL}_n(F)}(\pi, 1) \neq 0$.

This is related to Rankin–Selberg and Asai L-functions according as we are in the split or inert case.

- $\psi : E \to S^1$ fixed additive character
- $\mathcal{W}(\pi, \psi)$ Whittaker model of π
- $\mathcal{S} = \mathcal{S}(F^n)$: Schwartz space
- $e_n = (0, 0, \ldots, 1) \in F^n$
- $N_n \leq \mathrm{GL}_n$: subgroup of unipotent upper-triangular matrices

We consider the family of Rankin–Selberg integrals in the split case (resp. Flicker integrals in the inert case)

$$\mathcal{J}(\pi) = \{I(s, W, \Phi) : W \in \mathcal{W}(\pi, \psi), \Phi \in \mathcal{S}\},$$

$$I(s, W, \Phi) := \int_{N_n(F)\backslash \mathrm{GL}_n(F)} W(g)\Phi(e_n g)|\det g|_F^s \, dg.$$

Jacquet, Piatetski-Shapiro and Shalika (resp. Beuzart–Plessis)showed that these integrals converge for $\Re(s)$ large enough and are holomorphic multiples of the

Rankin–Selberg (resp. Asai) L-functions. We have a filtration $\mathcal{S} = \mathcal{S}^0 \supset \cdots \supset \mathcal{S}^m$, where $\mathcal{S}^m$ is the set of vanishing functions to order at least m at 0. At a pole s_0 of maximal order for $\mathcal{J}(\pi)$, each integral as above has a Laurent expansion

$$\frac{B_{s_0}(W,\Phi)}{(s-s_0)^d} + \cdots, \tag{3}$$

where B_{s_0} is bilinear. "Exceptional pole of level d at s_0" means $B_{s_0} \neq 0$ [I think; slides were pretty fast].

Theorem 126 (Y., 2024)**.** *Let π be an irreducible generic representation of $\mathrm{GL}_n(E)$, assumed to be* nearly tempered *in the inert case. Then π is distinguished if and only if the family of integrals $\mathcal{J}(\pi)$ has an exceptional pole of level zero at 0.*

Here we say that π is *nearly tempered* if

$$\pi \cong (\pi_1 \otimes |\det|_E^{r_1}) \times \cdots \times (\pi_k \otimes |\det|_E^{r_k}),$$

where each $\gamma_i \in \mathbb{R}$ or $|\gamma_i| < 1/4$.

In 2010, Matringe established for E/F quadratic non-archimedean that π is distinguished iff its Asai L-function has san exceptional pole at 0. In the non-archimedean setting, the Asai L-function serves as the greatest common divisor of the family of Flicker integrals, ensuring that the quotient of these integrals and the Asai L-function is nonvanishing. But in the archimedean case, the nonvanishing of this quotient is known only for nearly tempered representations (in the inert case).

We'll first prove the "if" direction of this theorem. Suppose $s = s_0$ is a pole of maximal order d for $\mathcal{J}(\pi)$, then we have an expansion as in (3), where

$$B_{s_0}(g \cdot W, g \cdot \Phi) = |\det|^{s_0} B_{s_0}(W,\Phi).$$

Note that $\mathcal{S}^1$ is a codimension 1 subspace of $\mathcal{S}$, being the kernel of the evaluation at zero map $\mathcal{S} \to \mathbb{C}$. Therefore, if $s = s_0$ is an exceptional pole of level 0 for the family $\mathcal{J}(\pi)$, then B_{s_0} is of the form

$$B_{s_0}(W,\Phi) = \lambda_{s_0}(W)\Phi(0),$$

where λ_{s_0} is nonzero and satisfies

$$\lambda_{s_0}(g \cdot W) = |\det g|^{-s_0} \lambda_{s_0}(W).$$

Taking s_0, we see that $\lambda_0 \neq 0$, and so π is $\mathrm{GL}_n(F)$-distinguished.

In the remainder of the talk, we focus on proving the "only if" direction. We focus on the split case; the inert case will follow from a similar argument.

Let W_F be the Weil group of F, either $\mathbb{C}^\times$ if $F = \mathbb{C}$ or $\mathbb{C}^\times \sqcup j\mathbb{C}^\times$ for $F = \mathbb{R}$, where $j^2 = -1$ and $jwj^{-1} = \bar{w}$. To any continuous finite-dimensional $\phi : W_F \to \mathrm{GL}(V)$, Tate associates $L(s,\phi)$ and $\varepsilon(s,\phi,\psi')$. Let $\pi = \pi_1 \otimes \pi_2$: irreducible representation of $\mathrm{GL}_n(E) = \mathrm{GL}_n(F) \times \mathrm{GL}_n(F)$. Let ϕ_1, ϕ_2 denote the associated Langlands parameter, so that $\phi_1 \otimes \phi_2$ is the parameter giving rise to the Rankin–Selberg L-function. In 1983 and 2010, Jacquet, Piatetski-Shapiro and Shalika resolved the analytic properties of Rankin–Selberg integrals.

Theorem 127 (JPSS 1983, Jacquet 2010)**.** *For $\pi = \pi_1 \otimes \pi_@$, W and Φ as above:*

1. *The local integrals $I(s,W,\Phi)$ converges for $\Re(s)$ large enough and extend meromorphically.*
2. *We have the local functional equation.*

(3) The normalized local integrals are holomorphic, and for any $s_0 \in \mathbb{C}$, do not vanish identically.

We turn to *continuity of integrals*. Let

$$P_n = \begin{pmatrix} * & * & \cdots & * \\ \cdots & \cdots & \cdots & * \\ * & * & * & * \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

denote the mirabolic subgroup of GL_n. For $W \in \mathcal{W}(\pi, \psi_n)$, consider the local Rankin–Selberg integrals

$$I(s, W, \Phi).$$

Proposition 128 (Cogdell 2005, Chai 2015)**.** *The normalized local Rankin–Selberg integrals are continuous.*

$$\frac{I(s, W, \Phi)}{L(s, \pi, \mathrm{RS})}.$$

Proposition 129 (Beuzart-Plessis, 2001)**.** *The normalized Rankin–Selberg integrals don't vanish identically.*

Proposition 130 (Ehud Moshe Baruch, 2003)**.** *For a distinguished representation π, any $P_n(F)$-invariant linear form on π is actually $\mathrm{GL}_n(F)$-invariant.*

We now give the proof of Theorem 126 Let $\pi = \pi_1 \otimes \pi_2$, with irreducible generic representations of $\mathrm{GL}_n(F)$. We need to prove that if π is distingiushed, then $s = 0$ is an exceptional pole of level 0 for the family $\mathcal{J}(\pi)$. For $\Re(s)$ sufficiently negative, consider the integral

$$I(1-s, \tilde{W}, \tilde{\Phi}) = \int_{N_n(F)\backslash \mathrm{GL}_n(F)} \tilde{W}(g)\hat{\Phi}(e_n g)|\det g|^{1-s}\, dg.$$

We assume that W is a pure tensor of K_n-finite components. Then we can write

$$\tilde{W}_1(gk) = \sum_l h_l(k)\tilde{W_l}'(g),$$

and similarly for W_2, with g_j in place of h_l.

Now we set $f_i = h_l g j$, and form $\tilde{W}(gk) = \sum_i f_i(k)\tilde{W}_i(g)$. Since π is distingiushed, its central character is distinguished.

Using the Iwasawa decomposition, we get

$$I(1-s, \tilde{W}, \hat{\Phi}) = \cdots,$$

then divide to get:

$$\begin{aligned} \frac{I(1-s, \tilde{W}, \hat{\Phi})}{L(1-s, \tilde{\pi}, \mathrm{RS})} &= \int_{K_n} \frac{\Psi(1-s, \tilde{\pi}(k)\tilde{W})}{L(1-s, \tilde{\pi}, \mathrm{RS})} \int_{F^\times} \hat{\Phi}(e_n ak)|a|^{n(1-s)}\, d^\times a\, dk \\ &= \cdots. \end{aligned}$$

This leads to

$$\frac{I(1, \tilde{W}, \hat{\Phi})}{L(1, \tilde{\pi}, \mathrm{RS})} = \int_{K_n} \frac{\Psi(1, \tilde{\pi}(k), \tilde{W})}{L(1, \tilde{\pi}, \mathrm{RS})} \int_{F^\times} \hat{\Phi}(e_n ak)|a|^{n}\, d^\times a\, dk.$$

As π is distinguished, so is $\tilde{\pi}$. Now pass from the $P_n(F)$-invariance to $\mathrm{GL}_n(F)$-invariance to eventually get the equality

$$\frac{I(1,\tilde{W},\hat{\Phi})}{L(1,\tilde{\pi},\mathrm{RS})} = \frac{\Psi(1,\tilde{W})}{L(1,\tilde{\pi},\mathrm{RS})}\Phi(0),$$

which we now know miust hold for zll W and Φ. Using the functional equation at $s=0$, we get

$$\frac{I(1,\tilde{W},\hat{\Phi})}{L(1,\tilde{\pi},\mathrm{RS})} = \alpha\frac{I(0,W,\Phi)}{L(0,\pi,\mathrm{RS})}$$

for some $\alpha \in \mathbb{C}^\times$. From the earlier calculation, the left hand side of the above equation vanishes for any $\Phi \in S^1$, thus the ratio on the right hand side of the above vanishes. But then $s=0$ is a pole for $L(s,,\pi,\mathrm{RS})$, say of order d. Thus $s=0$ must be a pole for the family $\mathcal{J}(\pi)$ of maximal order d. Observe that

$$B_0(W,\Phi) = \left(\lim_{s\to 0} s^d L(s,\pi,\mathrm{RS})\right)\frac{I(0,W,\Phi)}{L(0,\pi,\mathrm{RS})}$$

for all W and Φ. By previous discussion, the bilinear form B_0 is nontrivial and vanishes on S^1. Therefore, $s=0$ must be an exceptional pole of level 0 for the family $\mathcal{J}(\pi)$. This completes the proof of the theorem. Thank you.

31. Unconditional computation of the class groups of real quadratic fields *Andrei Seymour-howell*

Joint work with Ce Bian, Andrew Booker, Austin Docherty, Michael Jacobson Jr.

Let's start off by saying how I use class number computations. Imaginary class numbers are actually used to compute holomorphic modular forms using a trace formula.

Real quadratic class numbers are used to compute Maass forms, again using a trace formula. Thus

more class numbers = more automorphic forms.

Numerical verification of the Selberg eigenvalue conjecture up to level 880 also used to compute Maass forms by Strombergsson et al.

Previous computatinos:

Source	**Bound**	**Invariants**
Gauss	$\Delta < 1000$	h_Δ
Saito and Wada 1988	$\Delta < 10^6$	Cl_Δ and R_Δ
Jacobson 1988	$\Delta < 10^9$	Cl_Δ and R_Δ
te Riele and Williams 2003	$\Delta < 10^9$	h_Δ and R_Δ

Table 2. Comparison of bounds and invariants for different sources

How fast?

- Hafner and McCurley $\mathrm{O}(\exp((\log\Delta)^{1/2+\varepsilon}))$ (runtime heurstic)
- Buchmann: $\mathrm{O}(\Delta^{1/4+\varepsilon})$
- Lenstra/Shanks $\mathrm{O}(\Delta^{1/5+\varepsilon})$

Unfortunately, these algorithms all rely on GRH.

Now let's talk about imaginary quadratic fields. Ramachandran, Jacobson and Williams used modular forms to unconditionally compute and verify class numbers of imaginary quadratic fields.

- Compute a list of class numbers for imaginary quadratic fields up to discriminant of size say X. With Buchmann's algorithm, you would get a lower bound to the size of the class numbers.
- Use the trace formula for holomorphic forms of fixed weight and level to derive an upper bound and compare the two.

The nice thing about modular forms is that everything is finite . In fact, in their algorithm, no modular forms are needed, since we can choose a space of fixed weight and level that is empty and still derive a nontrivial expression for the class numbers. With this, they managed to compute several class numbers for discriminants of size 10^{12}.

In this talk, we mimic the above for the real quadratic case.

We recall that a function $f : \mathbb{H} \to \mathbb{C}$ is a *Maass cusp form* on $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ if it is

- automorphic,
- an egienfunction of the Laplacian Δ,
- lies in $L^2(\Gamma\backslash\mathbb{H})$, and
- vanishes at the cusp of $\Gamma\backslash\mathbb{H}$.

For any f and nonzero integer n, we have the Hecke operator T_n that maps Maass forms to Maass forms. Furthermore f has a Fourier expansion of the form

$$\sum_{n\neq 0} a(n)\sqrt{y}K_{ir}(2\pi|yn|)e(nx).$$

Here $\lambda = 1/4 + r^2$.

The Selberg trace formula allows one to consider the whole spectrum of Maass cusp forms. Selberg derived this in the 1950's to prove the existence of Maass cusp forms.

In our case, if we have a Hecke eigenbasis $\{f_j\}$ of level 1 Maass cusp forms with respective Laplace iegenvalues $\lambda_j = 1/4 + r_j^2$ and Hecke eigenvalues $a_j(n)$, the Selberg trace formula allows us to compare

$$(\text{spectral side}) \sum_{j=1}^{\infty} a_j(n)H(\lambda_j) = (\text{geometric side}).$$

Here H is some nice analytic test function, and $n \neq 0$. Important for us is that the geometric side involves class numbers of real quadratic fields. Explicitly, if we take some function h (think of a Gaussian) and let g be its Fourier transform, then the trace formula, in more detail, says

$$\sum_{j=1}^{\infty} a_j(n)h(r_j) = \sum_{\substack{t\in\mathbb{Z} \\ \sqrt{D}=\sqrt{t^2-4n}\notin\mathbb{Q} \\ D>0}} L(1,\psi_D)g\left(\log\left(\frac{(|t|+\sqrt{D})^2}{4n}\right) + \Psi(n)\right),$$

where, with $D = d\ell^2$, we have

$$L(1,\psi_D) = \frac{L(1,\psi_d)}{\ell}\prod_{p|\ell}\left(1 + (1-\psi_d(p))\frac{\cdots}{\cdots}\right).$$

Choosing a specific test function, we can write the trace formula as

$$\sum_{j=1}^{\infty} a_j(n)h(r_j) = \sum_{\substack{t\in\mathbb{Z} \\ \sqrt{D}=\sqrt{t^2-4n}\notin\mathbb{Q} \\ 0<D\leq X}} L(1,\psi_D)\left(1-\frac{D}{X}\right)^k + \Psi(n).$$

Truncating the spectral sum and rearranging, we have

$$\sum_{\substack{t\in\mathbb{Z} \\ \sqrt{D}=\sqrt{t^2-4n}\notin\mathbb{Q} \\ 0<D\leq X}} L(1,\psi_D)\left(1-\frac{D}{X}\right)^k = \sum_{r_j\leq R} a_j(n)h(r_j) + E_n - \Psi(n). \tag{4}$$

The point is that E_n, while we don't know it explicitly, we can bound it to be quite small. What this gives us is an *upper bound* that we can compute, provided that we have some list of Maass forms. We saw on Tuesday that there is a list of Maass forms that have been computed. So, how does the algorithm work?

(1) We compute the class number and regulator for real quadratic fields up to discriminant X, plus a bit extra. Recall that Buchmann's algorithm will give a lower bound to the size of the class number unconditionally. We also need the regulator, but we'll go into that later.
(2) Now we need to compute the spectral side of the trace formula for all $n \leq \sqrt{X}$ and a bound for the error E_n. This will be used as an upper bound on the size of the class numbers.
(3) To verify a given class number, we double its contribution to the left hand side of (4) (what we call "the hyperbolic sum") and hope we get something bigger than what our upper bound says, giving the required contradiction and showing that our class number was in fact correct.

An important point of this is that in step (2), we need to rigorously compute the Laplace eigenvalues and all Hecke eigenvalues up to $\sqrt{X}$ for several Maass cusp forms.

So that's basically the algorithm.

As we said, Buchmann's algorithm computes the class number in $\mathrm{O}(\Delta^{1/4+\varepsilon})$ unconditionally. Due to work of De Haan, Jacobson and Williams, we can unconditionally compute the regulator in $\mathrm{O}(\Delta^{1/6+\varepsilon})$. Thus to compute all class numbers and regulators up to discriminant X takes time $\mathrm{O}(X^{5/4+\varepsilon})$.

Now we need to compute the trace formula for all $n \leq \sqrt{X}$. This computation is dominated by the computation of the hyperbolic terms. Overall, we get $\mathrm{O}(X^{5/4+\varepsilon})$.

We implemented this algorithm and have unconditionally computed all class numbers and regulators up to discriminant $X = 10^{11}$. To do this, we computed the Laplace eigenvalues of the first 2184 Maass forms of level 1, and all Hecke eigenvalues a_n with $n \leq \sqrt{X}$. We are currently running this code to improve this bound to $X = 2^{40}$.

Can we do better than this? For the imaginary quadratic field case, to compute the class number, Jacobson and Mosunov used a relation between the Fourier coefficients of one of the classical Jacobi theta function and the Hurwitz class number. Computing the Fourier coefficients can be done in subexponential time. Indeed, if

you take the (third) Jacobi theta function

$$\vartheta_3(\tau) = \sum_{n\in\mathbb{Z}} q^{n^2}$$

and look at its cube's q-expansion

$$\vartheta_3(\tau)^3 = \sum_{n\in 0}^{\infty} E(n)q^n,$$

then $E(n)$ is essentially the Hurwitz class number. So you just need to compute the Fourier coefficients of this well-known modular form, which can be done in subexponential time.

Something similar *should* be possible for real quadratic fields. Goldfeld and Hoffstein [11] derived a half-integral weight Eisenstein series whose Fourier coefficients are related to $L(1,\chi_\Delta)$ for $\Delta > 0$. Using Hejhal's algorithm, one could copute the Fourier coefficients, and hence the $L(1,\chi_\Delta)$ value in subexponential time. However, to get the class number, we still need the regulator. Hence currently the best speed we could obtain is $\mathrm{O}(X^{7/6+\varepsilon})$, to compute all class numbers up to discriminant X. (The regulator computation remains the slowest part, still requiring $D^{1/6}$.)

[picture of Cohen–Lenstra heuristics]

This is just data showing the ratio of actual vs. expected frequencies of odd class numbers h over $\Delta \le X$. Looks like it follows it pretty well. There are some cases where it looks like it doesn't, but that's just because we haven't computed enough. The line describing occurrences of fields with "class number three" looks the weirdest, but someone has convinced the speaker not to be concerned.

Would the computation be faster if we had more eigenvalues for Maass forms? The speed is dominated by computing the left hand side of (4), so not obviously. But maybe one could combine Hejhal's algorithm with Turing's method? Relevant paper of Smajlovic–Jorgensen–Then.

32. Hilbert Eisenstein series and generalized Rankin–Selberg L-series, *Mingkuan Zhang*

Let F be a real quadratic field, $G = \mathrm{SL}_2$. For a Hecke character χ of F, denote by $I(s,\chi)$ the corresponding induced representation on $\mathrm{Res}_{F/\mathbb{Q}} G(\mathbb{A})$, and $E(h,\Phi,s)$ the corresponding Hilbert Eisenstein series.

Denote by $\chi_{E/F}$ the quadratic Hecke character of F for E/F. There is an $\mathrm{SL}_2(\mathbb{A}_F)$-equivariant map

$$\lambda = \prod \lambda_v : \mathcal{S}(W_{\mathbb{A}_F}) \to I(0,\chi),$$

$$g \mapsto \lambda(\phi)(g) := (\omega(g)\phi)(0),$$

whose images are called *Siegel sections*.

At an archimedean place v of F, and for $k \in \mathbb{Z}$ and χ_v a character of $F_v^\times \cong \mathbb{R}^\times$ such that $\chi_v(-1) = (-1)^k$, we have the standard section $\Phi_\infty^k \in I(s,\chi_v)$ of weight k, satisfying

$$\Phi_\infty^k(g\kappa(\theta),s) = e(k\theta)\Phi_\infty^k(g,s).$$

Let $-D_i < 0$ be coprime fundamental discriminants ($i = 1,2$) and $K_i = \mathbb{Q}(\sqrt{-D_i})$. For

$$\Phi = \lambda_{E/F}\left(\mathrm{char}(\hat{\mathfrak{o}}_E)\right) \otimes \Phi_\infty^{k_1} \otimes \Phi_\infty^{k_2},$$

we define the corresponding Eisenstein series

$$E(h,\Phi,s)=\sum_{\eta\in B(F)\backslash G(F)}\Phi(\eta h,s).$$

It is a non-holomorphic Hilbert modular form with level $\mathrm{SL}_2(\mathfrak{o}_F)$.

Given $\Phi=\lambda_{K_i/\mathbb{Q}}(\mathrm{char}(\hat{\mathfrak{o}}_{K_i}))\otimes\Phi_\infty^k$, write

$$E(h,\Phi,s):=\sum_{\eta\in B(\mathbb{Q})\backslash G(\mathbb{Q})}\Phi(\eta h,s).$$

It is a non-holomorphic elliptic modular form with level $\Gamma_0(d_{K_i})$.

Let's discuss CM cycles. Let (V,Q) be a quadratic space over $\mathbb{Q}$ of signature $(n,2)$, and set $H=\mathrm{GSpin}(V)$ and $\mathcal{D}$ for the Grassmannian of two-dimensional negatively oriented subspaces of V_{Rr}. For a compact open $K\subseteq H(\mathbb{A}_f)$, we obtain the corresponding Shimura variety

$$X_K:=H(\mathbb{Q})\backslash(\mathcal{D}\times H(\mathbb{A}_f))/K.$$

To obtain zero-cycles, let $U\subset V$ be a negative definite 2-dimensional subspace. It defines a two point subset $\{z_U^{\pm}\}\subseteq\mathcal{D}$ given by $U_{\mathbb{R}}$, and we obtain a two-point subset $Z(U)\subseteq X_K$.

Let L be an even lattice of V, and $K\subset H(\mathbb{A}_f)$ a compact open subgroup stabilizing $\hat{L}$ and acting trivially on L'/L. Let $f:\mathbb{H}\to S_L$ be a harmonic MAass form of weight $1-\frac{n}{2}-2j$. For $z\in\mathcal{D}$ we obtain a regularized Green function $\Phi^j(z,h,f)$.

The space U determines two definite lattices $N=L\cap U$, $P=L\cap U^\perp$. For any $g\in S_{2-k+2j,\rho_L}$ with $k=1-\frac{n}{2}$, we define

$$L(g,U,s)=\left([\theta_P(\tau),E_N(\tau,s,1)]_j,g(\tau)\right)_{\mathrm{Pet}}.$$

Theorem 131 (Bruinier et al)**.** ...

The Shimura lift is a lift from half-integral weight modular forms to integral weight modular forms.

Theorem 132. ...

We turn to generalized Rankin–Selberg L-series. Consider the quadratic space

$$V:=\left\{A\in M_2(F):A^t=A'\right\},$$

with the quadratic form det. It has signature $(2,2)$ and $\chi_V(x)=(x,D)_{\mathbb{Q}}=\chi_F(x)$. For a cusp form $g\in S_{2j+2,\rho_L}$, the Rankin–Selberg type integral.

(...)

Question 133. Can we express $L(s,g,L)$ in terms of g similarly as in the case $n=1$.

For a Schwartz function $\varphi\in\mathcal{S}(V(\mathbb{A}))$, we define the theta function $\theta(g,h,\varphi)$ and its integral $I(h,\Xi,\varphi,s)$, which coincides with the Eisenstein series attached to a section $\mathcal{F}(h,s,\Xi,\varphi)\in I(s,\chi)$.

Theorem 134 (M. and Y. '24)**.** *For any Siegel section $\Phi\in I(s,\chi)$ and $n\geq 1$, there exists a standard section $\Xi\in I(s,\chi_2)$ and $\varphi\in\mathcal{S}(V(\mathbb{A}))$ and a function $C(s)$ (real-analytic at $s=0$) such that*

$$\mathcal{F}(h,s,\Xi,\varphi)=C(s)\Phi(h,s)+\mathrm{O}(s^n).$$

This means we can compute the derivative of Hilbert Eisenstein series using theta integrals.

Let's now see some examples of this lifting map. We discuss invariant vectors. For $F = F_\alpha = \mathbb{Q}_p(\varpi)$ the ramified quadratic extension of $\mathbb{Q}_p$ with $p = \alpha \cdot \varpi^2$ odd, $\alpha \in \mathbb{Z}_p^\times$, we get a valuation ring $\mathcal{O}$ and different ideal $\mathfrak{d}$.

We denote by $\Phi_\infty^{(k_1,k_2)} \in I(s, \chi_\infty)$ the standard section. We denote by $\varphi_\infty^{k,l}(a, b, \nu_1, \nu_2) \in \mathcal{S}(V(\mathbb{R}))$ the standard Schwartz function. We define certain local factors φ_p. Then the corresponding Eisenstein series is the classical Doi–Naguyama one.

The Shintani lift corresponds to the quadratic space $V_{(2,1)} = (M_2(\mathbb{Q})^{\mathrm{tr}=0}, D_1 \cdot \det)$. To introduce the twisted Shintani lift, we need the genus character $\psi_{\Delta,r}(\mathfrak{c}_h)$. Then we obtain an intertwining map $\psi_{\Delta,r} : \mathcal{D} \to \mathcal{D}(\Delta)$ between the representations $\tilde{\rho}$ and ρ_Δ. We can then define the twisted theta functions

$$\theta_{\Delta,r,h}(\tau, z) := \sum_{\substack{\delta \in \mathcal{D}(\Delta) \\ \pi(\delta) = rh \\ Q_\Delta(\delta) \equiv \cdots}} \cdots .$$

With this twisted version of Shintani lift, we can express generalized Rankin–Selberg L-series associated to a vector-valued lifting of a scalar-valued modular form:

Theorem 135. *Assume $-D_1$ and $-D_2$ are fundamental discriminants and coprime to squarefree N. If G^{vv} is the vector valued lifting of a scalar valued modular form G with level $\Gamma_0(N)$, then*

$$L(s, G^{vv}, L_N) = \left(\frac{\chi_1(p)}{2} \Gamma_{\mathbb{R}}(s + 2 + j) L(s + 1, \chi_1) |D_1|^{\frac{s+1}{2}} \right)^{-1} \times \frac{H_j}{(4\pi D_2)^{\frac{s+j+1}{2}}} \sum_{n=1}^{\infty} \frac{a_{D_2 n^2}(\mathrm{Sh}_{D_1}^{(0,j+1)}(G))}{n^{s+j+1}}.$$

Furthermore, if G is a normalized eigenform, then we can more explicitly evaluate $L(s, G^{vv}, L_N)$.

Theorem 136 (Ye '22)**.** *Let $\pi_N(\tau)$ be a Hauptmodul of a genus zero modular curve $\Gamma_0(N)\backslash\mathbb{H}$ for $N \geq 2$. Then we can cook up a Borcherds lift.*

Theorem 137 (M. and Y. 24)**.** *Let $\pi_p(\tau)$ be the Hauptmodul of a genus zero modular curve $\Gamma_0(p)\backslash\mathbb{H}$ for prime $p \geq 2$. Assume $-D_1$ and $-D_2$ are fundamnetal discriminants and coprime to p. When D_1 or D_2 is sufficiently large, the algebraic integer*

$$\prod_{(z_1,z_2) \in Z(W)} (\pi_p(z_1) - \pi_p(z_2))$$

is not a unit.

33. Symplectic period is preserved under the Jacquet–Langlands correspondence, *Mahendra Verma*

Let F be a number field, $\mathcal{A}$ its adele ring, G an algebraic group over $\mathbb{Q}$, H a closed subgroup. "Automorphic forms on G" means complex-valued functions on $G(\mathbb{Q})\backslash G(\mathbb{A})$ with nice properties. "Automorphic representation of $G(\mathbb{A})$" means realization of representation of $G(\mathbb{A})$ on the space of automorphic forms. An automorphic form f is a *cusp form* if

$$\int_{U(\mathbb{Q})\backslash U(\mathbb{A})} f(ug)\, du = 0$$

for all unipotent radicals U of all proper parabolic subgroups. Cusp forms generate cuspidal automorphic representations.

Let π be an automorphic representation of G. Consider the period integral

$$\int_{H(\mathbb{Q})\backslash H(\mathbb{A})} f(h)\,dh.$$

For a general automorphic form, the period integral may not converge, and it is of interest to define via regularization. We say that an irreducible automorphic representation π of $G(\mathbb{A})$ is $H(\mathbb{A})$-*distinguished* if the above period integral is not identically zero on the space of π. These period integrals are related to special values of certaind cl-functions, and also appear in the context of Langlands functoriality.

Consider the example

$$E(z,s) = \sum_{(m,n)\neq(0,0)} \frac{\Im(z)^s}{|mz+n|^{2s}}, \quad z \in \mathbb{H}, s \in \mathbb{C}$$

on the upper half-plane, which can be lifted as an automorphic form $E(g,s)$ on $\mathrm{SL}_2(\mathbb{Q})\backslash \mathrm{SL}_2(\mathbb{A})$. Further,

$$\int_{\mathrm{SO}_2(\mathbb{Q})\backslash \mathrm{SO}_2(\mathbb{A})} E(g,s)\,dg = \zeta_{\mathbb{Q}(i)}(s).$$

There's number theory behind such period integrals, which provides the main reason to study them.

Our interest is not in which automorphic forms are distinguished. Specifically, we take $G = \mathrm{GL}_{2n}$ and $H = \mathrm{Sp}_{2n}$. Thus, for an automorphic form f on GL_{2n}, we consider the period integral

$$\int_{\mathrm{Sp}_{2n}(\mathbb{Q})\backslash \mathrm{Sp}_{2n}(\mathbb{A})} f(h)\,dh.$$

We say that an irreducible, automorphic representation π of $\mathrm{GL}_{2n}(\mathbb{A})$ is $\mathrm{Sp}_{2n}(\mathbb{A})$-distinguished.

There is a local analogue of symplectic period, which can be defined at each place v of F. We say that a representation π_v of $\mathrm{GL}_{2n}(F_v)$ has a symplectic period if $\mathrm{Hom}_{\mathrm{Sp}_{2n}}(\pi_v, \mathbb{C}) \neq 0$. Recall that an irreducible, cuspidal automorphic representation π of $G(\mathbb{A})$ decomposes as a restricted tensor product over all places v of F, denoted $\pi = \bigotimes \pi_v$. Global nonvanishing implies local nonvanishing.

We need a parametrization of automorphic representations of $\mathrm{GL}_{2n}(\mathbb{A})$, due to Moeglin–Waldspurger. An irreducible unitary representation of $\mathrm{GL}_{2n}(\mathbb{A})$ is called a *discrete* automorphic representation if it occurs as a discrete summand in the space $L^2(\mathrm{GL}_{2n}(F)\backslash \mathrm{GL}_{2n}(\mathbb{A}))$. There is a bijection between discrete automorphic representations π of $G(\mathbb{A})$ and pairs (r,τ), where r divides $2n$ and τ is an irreducible cuspidal automorphic representation of $\mathrm{GL}_r(\mathbb{A})$.

Theorem 138 (Jacquet–Rallis–Offen)**.** *π has a nonzero symplecitc period if and only if $2n/r$ is even. In particular, let π be a cuspidal automorphic representation of GL_{2n}. Then its symplectic period vanishes.*

If $\pi = \bigotimes \pi_v$ has a nonzero symplectic period, then π_v has nonzero $\mathrm{Sp}_{2n}(F_v)$-invariant linear functional. The π_v's are generic, and so have a Whittaker model. Heumos–Rallis proved that an irreducible representation can not have both. (It is

important to note that some aspects of the local questions need global methods, and on the other hand, some global questions need local results.)

The group $\mathrm{Sp}_n(D)$ can be defined as

$$\mathrm{Sp}_n(D) = \left\{ A \in \mathrm{GL}_n(D) : AJ^t\bar{A} = J \right\},$$

where

$$J = \begin{pmatrix} & & & 1 \\ & & 1 & \\ & \cdots & & \\ 1 & & & \end{pmatrix}.$$

The group $\mathrm{Sp}_n(D)$ is the unique non-split inner form of the group $\mathrm{Sp}_{2n}(\mathbb{Q})$. Clearly $\mathrm{Sp}_{2n}(D) \subseteq \mathrm{GL}_{2n}(D)$, and one can again ask for vanishing or nonvanishing of the period integral for automorphic forms.

We can relate these two setups via the Jacquet–Langlands correspondence.

- DS_{2n}: the set of discrete automorphic representations of $\mathrm{GL}_{2n}(\mathbb{A})$.
- DS'_{2n}: the set of discrete automorphic representations of $\mathrm{GL}_n(D_{\mathbb{A}})$.
- For an automorphic representation π of $\mathrm{GL}_n(D)$, we denote by $\mathrm{JL}(\pi)$ its Jacquet–Langlands lift to GL_{2n}.
- (We say that a discrete automorphic representation π of GL_{2n} is D-compatible if it is so at each place.)

Theorem 139 (Jacquet–Langlands–Badulescu–???)**.** *There is a unique injective map*

$$\mathrm{JL} : \mathrm{DS}'_n \to \mathrm{DS}_{2n}$$

such that for all $\pi' \in \mathrm{DS}'_n$, we have $\mathrm{JL}(\pi')_v = \pi'_v$ for all places $v \notin V$. [Further details the speaker says aren't so important for the talk.]

When $D = F$, cuspidal automorphic representations are not always distinguished.

Proposition 140. *For odd $n \geq 1$, there is a cuspidal automorphic representation π of $\mathrm{GL}_n(D)$ and some $f \in \pi$ such that*

$$\int_{[\mathrm{Sp}_n(D)]} f \neq 0.$$

Theorem 141. *Suppose π is a cuspidal automorphic representation of $\mathrm{GL}_n(D)$ whose Jacquet–Langlands lift is cuspidal. Then the symplectic period of that lift vanishes identically.*

Theorem 142. *If π is an automorphic representation of $\mathrm{GL}_n(D)$ that appears in the discrete spectrum and is distinguished by $\mathrm{Sp}_n(D)$, then its Jacquet–Langlands lift to GL_{2n} is distinguished by Sp_{2n}.*

Conjectured by Dipendra Prasad that the converse is also true. We have some partial results, and in progress for general n.

Conjecture 143. *An automorphic representation in the discrete spectrum of $\mathrm{GL}_n(D)$ is distinguished by $\mathrm{Sp}_n(D)$ if and only if its Jacquet–Langlands lift to GL_{2n} is distinguished by Sp_{2n}*

Let σ be an irreducible cuspidal automorphic representation of $\mathrm{GL}_1(D)$. Then we can construct Eisenstein series $E(g, \varphi, s)$, converging absolutely for $\Re(s)$ large, with meromorphic continuation to all s and a simple pole inside $\Re(s) \geq 0$ and depending on σ, the only possible pole in that region is either at $s = 1$ or $s = 2$. For $s_0 \in \{1, 2\}$, we define the residue $E_{-s_0}(\phi)$ by

$$E_{-s_0}(g, \phi) = \lim_{s \to s_0} (s - s_0) E(g, \phi, s).$$

These are square-integrable automorphic forms on $\mathrm{GL}_2(D)$.

Theorem 144. *The function $E_{-1}(g, \phi)$ is integrable over $[\mathrm{Sp}_2(D)]$ and*

$$\int_{[\mathrm{Sp}_2(D)]} E_{-1}(\phi, \bullet) = \int_{K \cap \mathrm{Sp}_2(D_{\mathbb{A}})} \int_{(M \cap H) \backslash (M \cap H)(D_{\mathbb{A}})^1} \phi(mk)\, dm\, dk.$$

Moreover, there is a choice of ϕ such that the integral is nonzero.

As a consequence of the above theorem, we have the following result:

Theorem 145. *Any nonzero cuspidal automorphic representation in the discrete spectrum of $\mathrm{GL}_2(D)$ is of the form $\pi = J(2, \sigma)$ and is distinguished by $\mathrm{Sp}_2(D)$. Further, its image under the global Jacquet–Langlands correspondence is distinguished by Sp_4.*

34. Minor arc bounds for primes, *Mayank Pandey*

The main object we'll be discussing is the following exponential sum, for X large:

$$S(\alpha) := \sum_{p < X} e(p\alpha), \qquad e(\alpha) := e^{2\pi i \alpha}.$$

This has been studied for the purpose of understanding the additive structure of primes. For example, it was introduced in studying the weak Goldbach conjecture by the circle method, not exactly in this form, but by Hardy–Littlewood, and Vinogradov was able to obtain a sufficiently good understanding of this sum to show that every large enough odd number is a sum of three primes. But there are many problems concerning linear patterns in the primes that can be solved simply by understanding this exponential sum. Currently, we can't hope to understand $S(\alpha)$ for all α, but

- when α is "structured", we try to estimate $S(\alpha)$ (the *major arcs* case), while
- for "generic" α, we try to bound $S(\alpha)$ from above (the *minor arcs* case).

This is often the case for exponential sums, that you try to study them in this manner. Very often, as is the case with the primes, "structured" ends up meaning that α is close to a rational with small denominator. This isn't always the case – for instance, if you have a sum over numbers missing a digit, you won't just have rationals with small denominator, but instead something involving the base – but there's none of that here.

We'll discuss some improvements on our understanding of minor arcs. Before we get into this, we'll talk about what one might expect for $S(\alpha)$ and how that relates to α. Say we have some α, and we're trying to understand $e(n\alpha)$. One thing one might do is approximate α by a rational with small denominator:

$$\alpha = \frac{a}{q} + \beta, \qquad q \leq \sqrt{X}, \quad |\beta| \leq \frac{1}{q\sqrt{X}}.$$

We're guaranteed such an approximation by Dirichlet. What we're looking at is then

$$e\left(\frac{an}{q}\right)e(n\beta).$$

This is somewhat lower in complexity than $e(n\alpha)$, because if you expand the latter into multiplicative characters (archimedean and non-archimedean), then the conductors involved are at most

$$C(\alpha) := q(1+|\beta|X) \ll \sqrt{X}.$$

(We note that $C(\alpha)$ is not quite well-defined due to the possibility of overlap, but to be more precise, you could take an infimum over all such approximations.) We'll also write

$$T := 1 + |\beta|X$$

for the archimedean part. One might expect that

$$S(\alpha) = \mathrm{MT} + \mathrm{O}(X^{1/2}),$$

where

$$\mathrm{MT} \approx \frac{X}{qT}.$$

Under GRH, one gets the error term $\mathrm{O}(X^{3/4})$ – this is how Hardy–Littlewood handled the weak Goldbach conjecture under GRH for large enough n.

Now we'll discuss what exactly we mean by "structured" and "generic". We'll define the major arcs to height Q to be

$$\mathcal{M}_Q := \{\alpha \in \mathbb{R}/\mathbb{Z} : C(\alpha) \le Q\}.$$

This is formulated a bit differently, but equivalent to the major arcs you see in the circle method, where you look at conditions like

$$\left|\alpha - \frac{a}{q}\right| \le \frac{Q}{qX}, \qquad q \le Q.$$

The minor arcs are defined to be the complement:

$$m_Q := (\mathbb{R}/\mathbb{Z}) - \mathcal{M}_Q.$$

We'll assume that the major arcs can be dealt with, maybe after averaging over Q, as is often the case. It often suffices to understand major arcs for Q up to some large power of a logarithm. We'll be talking about upper bounds on the minor arcs. What Vinogradov showed in the 1930's is that on the minor arcs,

$$\sup_{\alpha \in m_Q} |S(\alpha)| \ll \frac{X}{\sqrt{Q}} + X\exp\left(-c\sqrt{\log X}\right).$$

This here is a bit artificial. There's a result of Vaughan (1977) who replaced the error term with $X^{4/5}$ (up to logarithms – we're focusing here on the exponents). So now Q should be thought of as a small power of X, and we'll aim to improve things in that region.

The main theorem that we'll present today is the following joint result with Maksym Radziwill:

Theorem 146 (P., Radziwill, 2024)**.** *We have*

$$\sup_{\alpha\in m_Q}|S(\alpha)|\ll\frac{X}{\sqrt{Q}}+X^{4/5-\delta}$$

for

- $\delta<1/120$ *(smooth cutoff),*
- $\delta<1/160$ *(sharp cutoff).*

We'll now discuss Vaughan's $X^{4/5}$ and how we get around that. Vinogradov noticed that to study a general sum over primes, say

$$\sum_{p\asymp X}a(p),$$

where we're trying to show cancellation in such a sum, it suffices to understand, for arbitrarily sets of scales $N_1,\dots,N_k$ with $N_1\cdots N_k=X$, sums of the sort

$$\sum_{\substack{n_1\asymp N_1\\ \cdots\\ n_k\asymp N_k}}a(n_1\cdots n_k).$$

With this idea, you can show that this is the sum of $(\log X)^{\mathrm{O}(1)}$ many sums of the shape

$$\sum_{d\asymp D}\left|\sum_{m\asymp X/D}a(md)\right|$$

and

$$\sum_{\substack{\sqrt{D}\ll m,n\ll X/\sqrt{D}\\ mn\asymp X}}\alpha(m)\beta(n)a(mn).$$

This is basically what Vaughan does. Specializing to our case of $a(n)=e(n\alpha)$, we obtain the estimates

$$\sum_{d\asymp D}\left|\sum_{m\asymp X/D}a(md)\right|\ll D+\frac{X}{D}+\frac{X}{Q},\tag{5}$$

and

$$\sum_{\substack{\sqrt{D}\ll m,n\ll X/\sqrt{D}\\ mn\asymp X}}\alpha(m)\beta(n)a(mn)\ll\frac{X}{D^{1/4}}+\frac{X}{(\sqrt{X}/\sqrt{D})^{1/4}}+\frac{X}{\sqrt{Q}}.\tag{6}$$

If you optimize, then you end up taking $D^{4/5}$, which gives Vaughan's bound. In the first estimate (5), we're just estimating a geometric series. For the second estimate (6), we apply Cauchy.

There's the case $N_1=\cdots=N_5=X^{1/5}$. It turns out that if you do some combinatorics, then this is the only case that one encounters. Namely, to improve the $X^{4/5}$ bound, it suffices to do so just for the following sort of sum:

$$\sum_{n_1,\dots,n_5\asymp X^{1/5}}e(n_1\cdots n_5\alpha)\ll\frac{X}{\sqrt{Q}}+X^{4/5-\delta}.\tag{7}$$

We'll focus on the generic case, which turns out to be the most difficult, where

$$\alpha=\frac{a}{q},\qquad q\asymp\sqrt{X}.$$

Rather than proceeding as in the Type I or Type II bounds, we pass to characters and put in absolute values to see that the left hand side of (7) is majorized by

$$\sum_{\chi} \left| \sum_{n \asymp X^{1/5}} \chi(n) \right|^5.$$

If you apply Cauchy, then the above is

$$\leq \left(\frac{1}{q} \sum_{\chi} \left| \sum_{N^{1/5}} \chi(n) \right|^4 \right)^2 \left(\sum_{\chi} \left| \sum_{N^{1/5}} \chi(n) \right|^6 \right)^{1/2}. \tag{8}$$

The fourth moment can be understood as a sum of divisor functions:

$$\frac{1}{q} \sum_{\chi} \left| \sum_{n \asymp X^{2/5}} \chi(n) d(n) \right|^2.$$

One can show that you have square root cancellation on average, and that this is sharp.. So there's nothing to do in the first factor in (8) to improve the upper bound. For the second factor, we have

$$\sum_{\chi} \left| \sum_{n \asymp X^{3/5}} \chi(n) d(n) \right|^3 \ll X^{6/5}.$$

Our result then boils down to improving this last estimate in the exponent. This reduces to estimating

$$\# \left\{ \chi : \left| \sum_{n \asymp X^{2/5}} \chi(n) d(n) \right| \geq V \right\}.$$

In this case, the above is at most

$$X^{\frac{2}{5}\frac{3}{4}} = X^{\frac{3}{10}}.$$

Recently, Guth–Maynard finally improved an old estimate of Huxley, and it turns out that their methods readily adapt to a sum over $\chi(n)$ as well. The final result they have is slightly worse because at some point they chop up into intervals, but for our purposes, we can take certain estimates in Guth–Maynard that translate to the setting of $\chi(n)$, requiring no substantial modification to the proof. We can thus get $X^{3/10-\delta}$ via a "Guth–Maynard q-analogue". That's how we succeed in the end.

35. TBA, *Jay Jorgenson*

Any way we can construct automorphic forms is worthwhile. What we'll discuss now is partially completed and ongoing work with Cogdell and Smajlovic. We're motivated by three examples, all falling under the canopy of "Kronecker limit formula":

Example 147 (Parabolic Eisenstein series)**.**

$$E_{\mathrm{par}}(s, z) = \sum_{\gamma \in \Gamma_\infty \backslash \Gamma} (\Im \gamma z)^s, \qquad \Gamma = \mathrm{PSL}_2(\mathbb{Z}).$$

In general, as $s \to 0$, the above is

$$C_0 + C_1 \left(\log \|\Delta(t)\|^2 + C_2 \right) s + \mathrm{O}(s^2).$$

It's well known that this satisfies the differential equation

$$(\Delta + s(s-1))\,\mathcal{E}_{\mathrm{par}}(s,z).$$

Example 148 (Hyperbolic Eisenstein series)**.** Kudla–Millson had what we saw as a real parallel to this. They considered any Fuchsian group of the first kind. They picked some hyperbolic geodesic γ on there, and considered the form-valued Eisenstein series

$$\mathcal{E}_{\mathrm{hyp},\gamma}(s,z) = \sum_{\Gamma_\gamma\backslash\Gamma} \left(\cosh(d(\eta z,\gamma))\right)^{-s}\, d\gamma z.$$

In this case,

$$(\Delta + s(s-1))\,\mathcal{E}_{\mathrm{hyp}}(s,z) = \pm s_2 \mathcal{E}_{\mathrm{hyp}}(s+2,z).$$

That's the first part. The next part they proved, the speaker found particularly striking. They studied the behavior as $s \to 0$, and showed that with the right normalization, it is given by

$$\omega_\gamma + \mathrm{O}(s).$$

Jurg Kramer and I cooked up what we thought was the analogue for the elliptic Eisenstein series:

Example 149. Pick a point w in the Riemann surface, and set

$$\mathcal{E}_{\ell,w}(s,z) = \sum_{\eta\in\Gamma} \left(\sinh(d(\eta z,w))\right)^{-s}.$$

Kramer–von Pippich took $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$ and $w = i$ and obtained the $s \to 0$ limit

$$-\log\left|E_4(z)\Delta(z)^{-1/2}(z)\right| s + \mathrm{O}(s^2)$$

and for $w = \rho$ obtained instead

$$-\log\left|E_4(z)\Delta(z)^{-1/3}(z)\right| s + \mathrm{O}(s^2).$$

von Pippch studied $\mathcal{E}_{\mathrm{ell}}$ in general, for $\Gamma \leq \mathrm{PSL}_2(\mathbb{R})$. Then the present collaboration tried to see to what extent this generalized.

Theorem 150 (CJS)**.** *Let X be any compact complex projective Kähler variety. Let D be any divisor, smooth up to dimension 2. Set*

$$C_\rho(s,z,w) = (\Delta + s(s-\rho))^{-1}.$$

Then

$$\int_D C_p(s,z,w)\mu(w) = C_{-1}(s) + C_0\left(\log\|F_D(z)\|^2 + C_1\right) s + \mathrm{O}(s^2).$$

If X happened to be a compact Riemann surface, then we've recovered the results concerning elliptic Eisenstein series. Next:

(1) For $X = \Gamma\backslash\mathbb{H}$,

$$\sum_{z\in D_f}\sum_{w\in D_g} \implies f(\operatorname{div} g) = g(\operatorname{div} f).$$

Gave another proof of Weil reciprocity.

(2) $X = \mathbb{CP}^n$, $D = (H)$ (hyperplane section) with $\deg(D) = 1$. Then F_D is a linear form. We started fighting the constant C_1. By looking at the spectral expansion, we obtained an evaluation of the Mahler measure of that linear form. What turned out to be extremely interesting for us is that the only instance where the Mahler measure had been evaluated explicitly was when all the coefficients were 1. This turned out to be the one case we couldn't handle, due to sharpness of Cauchy–Schwarz in that case. But we were able to evaluate it. For us, it began to indicate that there was something worthwhile here.

To be precise, our result was valid for $n \geq 3$. For $n = 2$, there was a technical difficulty in the proof, resolved in a later paper by others.

What's next? We should really try to work this out not just for the compact Kähler manifolds, but something more exotic, like finite volume quotients of symmetric spaces. Look for instance at Sp_4. Take the Siegel upper half-space. There is a Siegel modular form χ_{10} that vanishes on the "diagonal divisor". The speaker thinks we should have a nice Eisenstein series involving such divisors for which the Kronecker limit formula gives χ_{10}. This should allow us to have some additional point of view on Mahler measures, not just as integrals of log-norms of equations, but rather something that looks more like integrals of log-norms of hypersections of divisors, and that could be a lot of fun.

[question on the relevance of χ_{10} – yeah, could do others]

[question about whether one could use points other than ρ or i in the $\mathrm{SL}_2(\mathbb{Z})$ result – yeah, could consider other points]

[thanks to the organizers and scientific committee, applause]

[next talk at 11:30 – stay tuned!]

36. The global Gan–Gross–Prasad conjecture for Fourier–Jacobi periods on unitary groups, *Paul Boisseau*

Joint work with Weixiao Lu and Hang Xue.

- F: number field, $\mathbb{A}$: adele ring
- G:reductive group over F, $[G] := G(F)\backslash G(\mathbb{A})$
- $H \leq G$: algebraic subgroup, $[H] := H(F)\backslash H(\mathbb{A})$
- π: irreducible unitary cuspidal automorphic representation of G,
- $r : {}^L G \to \mathrm{GL}(V)$.

The goal is to find relations when $\varphi \in \pi$ varies for

$$\underbrace{\int_{[H]} \varphi(h)\, dh}_{\text{period side}} \longleftrightarrow \underbrace{L(s, \pi, r)}_{L\text{-function side}} .$$

Applications of such statements include

- analytic properties of L-functions, poles,
- functionl equation, and
- computation of special values.

To make this statement more precise, let's give two examples.

Let's start with Tate's thesis, working with GL_1, thus let χ be an automorphic character of $\mathbb{A}^\times$. For $\Phi \in \mathcal{S}(\mathbb{A})$, set

$$\Theta(h, \Phi) := \sum_{\chi \in F^\times} \Phi(hx), \quad h \in \mathbb{A}^\times.$$

For $\Re(s) > 1$, consider

$$Z(\chi, \Phi, s) := \int_{F^\times \backslash \mathbb{A}^\times} \chi(h)\Theta(h, \Phi)|h|^s \, dh.$$

Theorem 151 (Tate)**.** *Assume that $\Phi = \bigotimes \Phi_v$. For S a large enough finite set of places of F, we have*

$$Z(\chi, \Phi, s) = L(s, \chi) \prod_{v \in S} \frac{Z_v(\chi_v, \Phi_v, s)}{L_v(s, \chi_v)}.$$

For our second example, we describe Rankin–Selberg theory. We now take

$$G = \mathrm{GL}_n \times \mathrm{GL}_n \geq H = \mathrm{GL}_n \xrightarrow{\Delta} G.$$

For $\Phi \in \mathcal{S}(\mathbb{A}^n)$, set

$$\Theta(h, \Phi) = \sum_{x \in F^n - \{0\}} \Phi({}^t hx), \qquad h \in H(\mathbb{A}).$$

Let π be a cuspidal automorphic representation of G. For $\varphi \in \pi$, consider

$$\mathcal{P}_H(\varphi, \Phi, s) = \int_{[H]} \varphi(h)\Theta(h, \Phi)|\det h|^s \, dh \qquad (\Re(s) > 1).$$

In this setting, there is the following result:

Theorem 152 (Jacquet, Piatetski-Shapiro, Shalika)**.** *Assume that $\varphi = \bigotimes \varphi_v$ and $\Phi = \bigotimes \Phi_v$. Then*

$$\mathcal{P}_H(\varphi, \Phi, s) = L(s, \pi) \prod_{v \in S} \frac{Z_v(\varphi_v, \Phi_v, s)}{L_v(s, \pi_v)}.$$

Moreover, for generic s, we have

$$\mathcal{P}_H(\cdot, \cdot, s) \neq 0 \iff L(s, \pi) \neq 0.$$

Our goal is to generalize these integral representations to $\mathrm{U}(V) \subset \mathrm{U}(V) \times \mathrm{U}(V)$. Some data:

- E/F: quadratic extension of number fields
- $(V, \langle \bullet, \bullet \rangle)$: Hermitian space over E/F of dimension n.
- $G_V = \mathrm{U}(V) \times \mathrm{U}(V)$, $H_V = \mathrm{U}(V)$.

We first need a period $\mathcal{P}_{H_V}$ and an L-function.

One technical point we need to deal with is the Weil representation. Set:

- ψ: nontrivial unitary character of $F\backslash\mathbb{A}$.
- η: the quadratic character of $F^\times\backslash\mathbb{A}^\times$ associated to E/F by global class field theory.
- μ: a character of $E^\times\backslash\mathbb{A}_E^\times$ such that $\mu|_{F^\times\backslash\mathbb{A}^\times} = \eta$.

By a classical construction of Weil and Kudla, one can associate to (ψ, μ) an automorphic representation $\omega_{\psi,\mu}$ of $\mathrm{U}(V)(\mathbb{A})$. This is the *Weil representation*. It is realized on $\mathcal{S}(\mathbb{A}^n)$. This space will be used to construct our theta series: given

$$\phi \in \omega_{\psi,\mu} = \mathcal{S}(\mathbb{A}^n),$$

we set

$$\theta(h, \phi) := \sum_{x \in F^n} \left(\omega_{\psi,\mu}(h)\phi\right)(x), \qquad h \in H_V(\mathbb{A}).$$

We now do the same thing as in Rankin–Selberg theory. Let π be a cuspidal representation of $G_V(\mathbb{A})$, and $\varphi \in \pi$. The Fourier–Jacobi period is then

$$\mathcal{P}_{H_V}(\varphi, \phi) = \int_{[H_V]} \varphi(h)\theta(h, \phi)\, dh. \tag{9}$$

Remark 153. We explicate the Weil representation and theta series in the split case. If $E = F \times F$, then $V = F^n \times F^n$, $\mathrm{U}(V) = \mathrm{GL}_n$ and

$$\theta(h, \phi) = \mu(h)|\det h|^{1/2} \sum_{x \in F^n} \phi({}^t h x).$$

What do we expect the period (9) to represent in this case? Rankin–Selberg L-function.

In general, there exists $\mathrm{BC} : {}^L\mathrm{U}(V) \to {}^L\mathrm{GL}_{n,E}$. By functoriality, there should be a map from automorphic representations of $\mathrm{U}(V)$ to automorphic representations of $\mathrm{GL}_{n,E}$.

Theorem 154 (Mok, Kaletha–Minguez–Shin–White)**.** *For every cuspidal representation π of $\mathrm{U}(V)$, the base-change $\mathrm{BC}(\pi)$ exists.*

If π is a cuspidal representation of G_V, we have a completed Rankin–Selberg L-function $L(s, \mathrm{BC}(\pi))$. We want to vary V but keep the L-function fixed.

Definition 155. Let V_1, V_2 be two skew-Hermitian spaces. Let π_1 be a cuspidal representation of G_{V_1}, and π_2 a cuspidal representation of G_{V_2}. We say that π_1 and π_2 are in the same *L-packet* if $\mathrm{BC}(\pi_1) = \mathrm{BC}(\pi_2)$. (In particular, they have the same L-function.)

Theorem 156 (B., Lu, Xue)**.** *Let π be an irreducible unitary cuspidal representation of G_V. Assume that $\mathrm{BC}(\pi)$ is generic. The following are equivalent:*

(1) $L(\frac{1}{2}, \mathrm{BC}(\pi) \otimes \mu) \neq 0$.

(2) There exists V' and π' (cuspidal representation of $G_{V'}$) in the same L-packet as π such that

$$\left(\mathcal{P}_{H_{V'}}\right)|_{\pi' \otimes \omega_{\psi,\mu}} \neq 0.$$

Some remarks:

- By local GGP (Gan, Ichino, Xue), the pair (V', π') is unique.
- This theorem was known under some local conditions by works of Xue.
- In the Bessel case, this was proved by Beuzart-Plessis, Chaudouard, Zydor. Their proof was an inspiration for our proof in the Fourier–Jacobi case.

So we have Theorem 156, which is very nice, but what we would really like to have is a formula relating the period to the L-value. For this, we need to define the local Fourier–Jacobi periods. Write $\pi = \bigotimes \pi_v$, $\omega_{\psi,\mu} = \bigotimes \omega_v$, $\varphi = \bigotimes \varphi_v$, $\phi = \bigotimes \phi_v$,

$\langle \bullet, \bullet \rangle_\pi = \prod_v \langle \bullet, \bullet \rangle_v$, $\langle \bullet, \bullet \rangle_\omega = \prod \langle \bullet, \bullet \rangle_v$, $dh = \prod dh_v$. Let v be a place of F. The local Fourier–Jacobi period is

$$\mathcal{P}_{H_V,v}(\varphi_v, \varphi_v) = \int_{H_V(F_v)} \langle \pi_v(h_v)\varphi_v, \varphi_v \rangle_v \langle \omega_v(h_v)\phi_v, \phi_v \rangle_v \, dh_v.$$

- If π_v is tempered, this is absolutely convergent.
- Multiplicity one result (Aizenbud–Gourevitch–Rallis–Schiffmann , Sun–Zhu):

$$\dim \mathrm{Hom}_{H_V(F_v)}(\pi_v \otimes \omega_v, \mathbb{C}) \leq 1.$$

- Non-vanishing (Xue):

$$\mathcal{P}_{H_V,v} \neq 0 \iff \dim \mathrm{Hom}_{H_V(F_v)}(\pi_v \otimes \omega_v, \mathbb{C}) > 0.$$

Theorem 157 (B., Lu, Xue, 2024)**.** *Let π be an irreducible unitary cuspidal representation of G_V. Assume that $\mathrm{BC}(\pi)$ is generic and that for all v, π_v is tempered. Then*

$$|\mathcal{P}_{H_V}(\varphi, \phi)|^2 = 2^{-\beta} \Delta \frac{L(\frac{1}{2}, \mathrm{BC}(\pi) \otimes \mu)}{L(1, \pi, \mathrm{Ad})} \times \prod_{v \in S} \mathcal{P}_{H_V,v}(\varphi_v, \varphi_v) \Delta_v^{-1} \frac{L(1, \pi_v, \mathrm{Ad})}{L(\frac{1}{2}, \mathrm{BC}(\pi_v) \otimes \mu_v)},$$

where

- $\Delta = \prod_{i=1}^n L(i, \eta^i)$, $\Delta_v = \prod_{i=1}^n L(i, \eta_v^i)$,
- $\beta \in \mathbb{N}$ *is the number of isobaric components of* $\mathrm{BC}(\pi)$,
- $L(s, \pi, \mathrm{Ad})$ *is defined using* BC *and the Asai L-function.*

We note that the Ramanujan conjecture predicts that if $\mathrm{BC}(\pi)$ is generic, then π is tempered.

Corollary 158. *Let π be an irreducible unitary cuspidal representation of G_V. Assume that $\mathrm{BC}(\pi)$ is generic and that for all v, π_v is tempered. Then*

$$(\mathcal{P}_{H_V})|_{\pi \otimes \omega_{\psi,\mu}} \neq 0 \iff \begin{cases} L(\frac{1}{2}, \mathrm{BC}(\pi) \otimes \mu) \neq 0, \\ \textit{for all } v, \textit{ we have } \mathrm{Hom}(\cdots) \neq 0 \end{cases}$$

37. Counting higher degree prime ideals in number fields and effective Chebotarev, *Aditya Ghosh*

References:

- Modular forms and CDT, Murty–Murty–Samadha
- Some applications of CDT, Serre.

Motivating question: given a number field K, how many prime ideals are there of degree k up to a certain point. A revised question concerns the number $\pi_k(x, K)$ of degree k prime ideals $\mathfrak{p}$ over rational p such that $p \leq x$.

Theorem 159 (Breuillard–G, 2023)**.** *We have*

$$\pi_k(x, K) \sim \frac{P_k}{k} \mathrm{Li}(x).$$

If $P_k = 0$, then $\pi_k(x, K) = 0$.

Here P_k are the Parker numbers for the associated Galois action.

Let's recall the Chebotarev density theorem. We have the prime number theorem (Hadamard, 1896) and prime ideal theorem (Landau 1903). Chebotarev density

theorem gives equidistribution of Frobenius classes. One also has explicit results, e.g., Lagarias–Odlyzsko (1977): assuming GRH, for every $x > 2$,

$$\left|\pi_C(x, L|K) - \frac{|C|}{|G|}\operatorname{Lii}(x)\right| \le c_1 \left\{\frac{|C|}{|G|} + \cdots\right\} + \cdots .$$

There's also an unconditional version.

Serre 1981: reduced the error a bit: assuming GRH, for every $x > 2$,

$$|\pi_f(x, L|K) - \cdots| \ll \cdots .$$

Consider the case where L is the Galois closure of a number field K. Take $G = \operatorname{Gal}(L/\mathbb{Q})$ and $H = \operatorname{Gal}(L/K)$.

Theorem 160 (Frobenius)**.** *Let p be a rational prime unramified in L. Let U be a prime in $\mathcal{O}_L$ lying over p. Let $\phi := \operatorname{Frob}(U|p) \le G$. ϕ acts by right multiplication on right cosets $H\backslash G$ with permutation cycle $(H\sigma_1, \ldots, H\sigma_1\phi^{m_1-1}\cdots)$. Then ETC.*

We can do the same with degree r prime ideals. To count them, define class function on G,

$$f_r(\sigma) = \#\left\{r\text{-cycles in permutation action of } \sigma \text{ on } H\backslash G\right\}.$$

We need estimates for this.

Theorem 161 (Parker)**.**

$$\frac{P_k}{k} = \frac{1}{|G|}\sum_{g\in G} c_k(g).$$

The case $k = 1$ gives us Burnside's lemma.

We get versions of Chebotarev with less error than before.

Theorem 162. $\cdots$

Also a version with unconditional error:

Theorem 163. $\cdots$.

Idea or proof: decompose into irreducible characters of S_n:

$$z_k = \sum_{\lambda:\text{partition of } n} \langle z_k, \chi^\lambda\rangle \chi^\lambda,$$

where (due to Jacques, 1974):

$$\langle z_k, \chi^\lambda\rangle = \begin{cases} \frac{1}{k} & \text{if } \lambda_1 = n, \\ \frac{1}{k}(-1)^{k-\lambda_2} & \text{if } \lambda_1 = n-k \text{ and } \lambda_3 \le 1, \\ \frac{1}{k}(-1)^{k-\lambda_1} & \text{if } \lambda_2 = n-k+1 \text{ and } \lambda_3 \cdots . \end{cases}$$

ETC.

ETC.

Improving past results and dropping AC.

Serre, 1997. For M_K the product of the $|P_K|$ distinct prime factors of Δ_K, we have

$$\log \Delta_K \le (n_K - 1)\log M_K + n_K|P_K|\log n_K,$$

which we improve by a bit.

ETC.

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